

Smart Migration and Digital Technology in Northern Ethiopia

Diploma Thesis

Supervisor:

doc. Mgr. et Mgr. Robert Stojanov

Author:

Tinsae SimeonWonjelo

Brno 2024

Statutory declaration

I hereby declare that, this thesis entitled: **Smart Migration and Digital Technology in Northern Ethiopia** was written and completed by me. I also declare that all the sources and information used to complete the thesis are included in the list of references. I agree that the thesis could be made public in accordance with Article 47b of Act No. 111/1998 Coll., Higher Education Institutions and on Amendments and Supplements to Some Other Acts (the Higher Education Act), and in accordance with the current Directive on publishing of the final thesis.

I am aware that my thesis is written in accordance to Act. 121/2000 Coll., on Copyright and therefore Mendel University in Brno has the right to conclude licence agreements on the utilization of the thesis as a school work in accordance with Article 60(1) of the Copyright Act.

Before concluding a license agreement on utilization of the work by another person, I will request a written statement from the university that the licence agreement is not in contradiction to legitimate interests of the university, and I will also pay a prospective fee to cover the cost incurred in creating the work to the full amount of such costs.

In Brno on 11th May, 2024

Acknowledgment

I would like to express my sincere gratitude to my supervisor, Doc. Mgr. Robert Stojanov, for his continuous guidance, insightful feedback, and for patiently reading multiple drafts of my thesis.

I am also deeply thankful to my husband, Biruk, and my daughter, Lillian, for their endless emotional support and assistance throughout this journey. I would also like to thank my dad, Mr. Simeon, for his encouraging words, and my mom, and my brother (Belay), for their support. Without their encouragement and understanding, this achievement would not have been possible. Thank you all for your support and contributions.

Table of Contents

1	Introduction.....	1
1.1	Background of the study	1
1.2	Research gap	2
2	Literature Review.....	5
2.1	Migration.....	5
2.2	Conceptual background.....	5
2.3	Theoretical concept	6
2.3.1	Migration theory	6
2.3.2	Framework for Machine Learning.....	8
2.4	Migration and drivers in northern Ethiopia.....	10
2.5	Conflict in northern Ethiopia overview.....	11
2.6	Majority humanitarian crisis	13
2.7	Role of mobile phones and digital technologies in migration.....	13
2.8	Social media and migration.....	15
3	Research Methodology	17
3.1	Data collection method and analysis.....	17
3.3	Ethical considerations	27
4	Analysis and Result.....	28
4.1	The socio-demographic profile of IDPs	28
4.2	Job and income-related characteristics of the IDPs	36
4.3	Smartphone technology usage by IDP	40
4.4	The increased usage of mobile phones after migration and migrants' characteristics.....	44
4.5	The social media usage of the IDPs	46
4.6	The attitude toward the usage of technology among migrants.....	52
4.7	Building the models	52
5	Discussion	64
5.1	Reasons for migrants to use smartphone.....	64
5.2	Most commonly used mobile applications.....	66
5.3	Patterns of smartphone usage.....	68

5.4 Prediction results of the implemented models	70
5.5 The interaction between migrants and smartphones using UML.....	70
6 Conclusion and Recommendation	73
References	76
List of Tables	86
List of Figures	87
Abbreviations	89
Appendix	90

1 Introduction

1.1 Background of the study

Migration is a complex and multifaceted phenomenon that has been shaped by various economic, social, and political factors (Tataru, 2020). With the increasing availability and affordability of digital technologies, the role of technology in shaping migration patterns and outcomes has become an area of growing interest for researchers and policymakers (Hagen-Zanker, 2017). Digital technologies have been transformative in many areas of life in Africa, including communication, commerce, and education. In the context of migration, digital technologies have the potential to facilitate and enhance migration processes, including providing information and connecting migrants with opportunities. However, they can also create new challenges and risks, such as increasing the vulnerability of migrants to exploitation and trafficking (Bernal, 2020).

Ethiopia has a lengthy history of internal and external border migration movements. The country has been a hub of trade and cultural exchange for centuries, with migration playing a significant role in shaping its history and culture (Bundervoet, 2018). Internal migration has been a common phenomenon in Ethiopia, with people moving from rural to urban areas in search of better economic opportunities. This has led to the growth of cities such as Addis Ababa, now one of Africa's largest cities. Ethiopians migrate to other countries in search of better economic opportunities, to escape conflict or political instability, or to reunite with family members.

In the early 20th century, Ethiopian migrants were drawn to cities in the Middle East such as Yemen using irregular ways, where they worked in various industries. Furthermore, many Ethiopians migrated to the United States and other Western countries for educational and employment opportunities (ILO, 2020).

The UN defines internally displaced people (IDP) as "individuals or groups of individuals who have been compelled to flee or to leave their homes or places of habitual residence, particularly due to or in order to avoid the effects of armed conflict, situations of generalized violence, violations of human rights, or natural or man-made disasters, and who have not crossed an internationally recognized state border" (IDMC, 2022b).

According to the IDMC global report, 45% of the world's internally displaced people reside in Sub-Saharan Africa. As a result, Ethiopia's 3.85 million internally displaced people (IDP) are the second highest in Africa, trailing only the Democratic Republic of the Congo's over 5.6 million IDP (IDMC, 2023).

In Ethiopia, approximately 1.7 million people were displaced due to the recent conflict in the Tigray region. This conflict has a strong impact on displacement and movements within Ethiopia and within the region in general (IOM, 2021).

More migration is pushing state and non-state entities to look for innovative approaches to migration control. According to a study (Nalbandian & Dreher, 2022), digital technologies are being used more and more in the field of migration to improve procedures and manage migrants. The spread of technical solutions is providing fresh perspectives on the benefits and challenges associated with migration.

Digital technology has been used in migration management procedures increasingly frequently in recent years. Digital technologies are emerging technologies that involve the most recent efforts in digitization and digitalization. It can be defined in other words; it is the process of converting physical objects into a digital format that can be processed by a computer (Molnar, 2019)

The integration of digital technology and smart migration offers opportunities and difficulties in the current crisis. Using digital technologies allows migrants to build networks and get information that will help them guide their journey and provide support as they negotiate the opportunities and challenges of life in their new society (Alencar A, 2023). Sometimes they don't make life better for migrants. As stated by Alencar A (2023), many people who have been forcefully relocated have challenges due to low levels of digital literacy, socioeconomic, language, and cultural limitations, as well as limited access to the internet.

The use of digital technologies has facilitated and enhanced migration processes, providing migrants with access to information, communication tools, and digital platforms that connect them with opportunities and resources. Migrants from Syria in Europe use smartphones to get support, to avoid loneliness and stress (Alencar et al., 2019). They use smartphones to communicate with smugglers and to get information about their travel paths (Gillespie et al., 2016). In addition to this, migrants use their smartphones to look for job opportunities in their destination places (Alencar, 2020; Zijlstra & Liempt, 2017).

1.2 Research gap

Previous studies on digital technology and smart migration have focused more on other countries or global contexts and provide valuable insights into the use of digital technologies in the migration process, there is a need for further research that specifically examines the use of smartphones in the migration process in northern Ethiopia. Studies on how digital technologies are adopted and used in the unique sociocultural and environmental setting of northern Ethiopia are needed; that take into account variables like level of education, gender, age, and language diversity.

It helps to fully understand how these technologies can be used to help host communities and migrants in northern Ethiopia during the migration process. This is particularly important given that smartphones are becoming increasingly affordable and accessible in the region and are likely to play an increasingly important role in the migration process.

Therefore, the study aims to address this gap in the literature by collecting data and analyzing the extracted insights into the reasons for migrants from northern Ethiopia's smartphone usage, and other relevant factors, which applications they prefer to use, and how they use smartphones in the migration process.

Furthermore, this study will build a machine-learning model that predicts the reasons for smartphone usage and most used mobile applications. Compared to other algorithms, machine learning (ML) algorithms are known to have the best usability in data analysis systems because they bring high precision with increased performance capacity in a variety of industries, including agriculture, mining, commercial, and more (Abdul-Jabbar & K. Farhan, 2022).

The following procedures are followed to perform the analysis and build a model for predictions

1. Data collection: This includes collecting data on migration flows, demographic information, economics, and more
2. Data cleaning and preprocessing: To ensure consistency and accuracy, cleaning and preprocessing the data after it has been collected is crucial. This could include organizing data into a useful structure, removing duplication, addressing missing values, and splitting, and dropping irrelevant columns.
3. Exploratory Data Analysis (EDA): To find patterns, relationships, and important insights, exploratory analysis is carried out using the cleaned data. Graphs and plots are made using Python visualization tools like Matplotlib and Seaborn to assist in identifying trends in the data.
4. Feature Engineering: In this step, key features from the data are chosen and transformed so they may be used in machine learning models.
5. Model selection and training: in this phase, appropriate machine learning models will be chosen and trained using the EDA results.
6. Model evaluation: After the training phase, metrics including accuracy, precision, recall, and F1-score will be used to assess the model's performance.
7. Prediction: To forecast the trend of smartphone usage and mostly used applications, predictions will be made on unseen data using the trained model.

The main objective of the study is to explore the reasons for migrants' use of smartphones, the most commonly used mobile applications concerning their situation, and the patterns of mobile device and application use among migrants in the Tigray, and Amhara regions of Ethiopia.

- What are the reasons for migrants' use of smartphones in northern Ethiopia?
- What are the most commonly used mobile applications with respect to their situation in northern Ethiopia?
- What are the patterns of mobile device and application use among migrants in northern Ethiopia?

Additionally, this study will provide insights into not only the research questions but also into the following areas

- The exploratory analysis provides a comprehensive understanding of the socio-demographic profile of the migrants, the challenges faced by migrants for using smartphones, and the correlation between the variables.
- Understanding migrants' remittance preferences
- Building models that will predict the *reason for mobile phone usage* and *preferred applications*.

By doing so, this study can inform the development of policies and programs that promote safe, orderly, and regular migration and protect the rights and well-being of migrants in the context of smartphone use.

Considerable internal and external migration flow occurs in northern Ethiopia. Addressing this topic will help to manage these migration patterns efficiently. The study on the topic of smart migration and digital technology has its importance for concerned organizations that manage migration or/and take care of migrants, for migrants and researchers, in northern Ethiopia. It helps to increase the knowledge and skill scope of the researcher in the way of study.

The geography of the study encompasses both the northern Ethiopia region, from which the internally displaced people (IDPs) originate, and the specific locations where they have relocated. Specifically, the study focuses on IDPs who have moved from the northern Ethiopia region to conflict-free areas within the Amhara region, including Woldia, Debre Birhan, Dessie, and Bahir Dar, as well as the capital city of Ethiopia, Addis Ababa.

2 Literature Review

2.1 Migration

People migration is driven by political, social, demographic, economic, and environmental factors. The study by Telsaç (2022) explained that today, the migration journey persists as individuals and groups alike seek improved opportunities and a better quality of life by relocating from one place to another. The reasons for migration, most of them are motivated by the desire to escape ongoing political unrest and conflict in favor of a better life away from home (Telsaç, 2022).

There are numerous classifications for migration. Based on whether they left their countries' boundaries, people are also categorized as internally or externally displaced. People who are internally displaced leave their homes yet stay in their country of origin. According to the UN convention, they may be given food, medication, and other necessities, but they are not eligible for refugee status (ICRC, 2010).

IOM categorized migration into Forced and voluntary migration (EMM, 2024). Forced migration is one type of migration in which people relocate overseas due to various reasons, such as civil unrest, conflict, persecution, or extreme climate and natural disasters. When people voluntarily choose to leave their home nation and are granted permission by the government to enter another country, this is known as voluntary migration. It can be to look for a job and Job opportunity abroad or family unification (EMM, 2024).

2.2 Conceptual background

Smart Migration and Digital Technology

Smart migration is a modern method of moving people around, in which people use smart technology, often called "smart devices," and make use of creative technological solutions. As Unruh et al. (2017) explain, that smart migration makes life better for displaced populations and encourages action rather than passivity. Smart migration plays a vital role in simplifying the migration process, facilitating access to accurate information, and optimizing the challenges faced by migrants (Alencar A, 2023). As the article by Alencar A, (2023) shows, that digital technologies help migrants secure data storage and ask for humanitarian help.

Digital technology is a digital resource that is used for the creation, storage, and management of data. Digital technology is everywhere, and this influences our unwillingness or readiness to be mindful of our unconsciousness when using digital systems in our daily lives. This is especially true when it comes to the internet, social media, work, family, entertainment, and so on (Glückler & Panitz, 2024). Digital technologies and data-driven approaches facilitate migration processes, that have the potential to enhance social capital and provide opportunities for migrants to cross borders safely and integrate into host societies (Azhar et al., n.d.).

2.3 Theoretical concept

Migration theories aim to comprehend the reasons and incentives behind the decisions made by people and households to relocate, either inside or outside of their country of origin. They also seek to identify the elements that contribute to the persistence of migration patterns over an extended period. Different theories employ different concepts, assumptions, and frames of reference depending on their discipline of origin and the time in which they were formulated (Tristan Claridge, 2018).

The intersection of smart migration and digital technologies presents a dynamic landscape, particularly in regions undergoing rapid socio-economic transitions like northern Ethiopia. As technology continues to evolve, its integration into migration processes reshapes how individuals, communities, and governments perceive and engage with mobility.

This theoretical exploration delves into the foundational concepts that underlie the intricate relationship between smart migration and digital technologies and provides valuable frameworks for understanding the economic drivers, impacts, and processes of migration in the context of northern Ethiopia. Introducing these theories will be able to give a deeper understanding of the drivers, impacts, and possible strategies for a beneficial migration process.

2.3.1 Migration theory

I. Network theory

The network theory or social network theory indicates the advancing role of networks and relationships in people's behavioral patterns and the outcome aspects of an individual. As the theory says, individuals are regarded as parts of social networks, which are connected by links and relationships. These social networks influence human behavior and often lead to a change of mind and a situation (Borgatti & Halgin, 2011).

II. Social capital theory

Social Capital Theory is an economic theory that focuses on social relationships and networks as a tool to facilitate access to resources and opportunities. As the theory states, social networks provide social capital to individuals such as resources, information, and social support that they can access through their social relationships. In the context of migration, social capital theory has been used to explain how social networks can facilitate the integration of migrants into destination societies. According to social capital theory, migrants might use their social networks to access social support, guidance, and information on job opportunities and housing as well as other resources (Tristan Claridge, 2018).

III. Human agency theory

Human agency theory is a theoretical framework that emphasizes the role of individuals in shaping their own migration experiences. The theory suggests that individuals are not simply passive recipients of external forces but are instead active agents who exercise choice and control over their own lives. Human agency theory emphasizes the importance of understanding the subjective experiences of migrants and how they navigate complex social, economic, and political contexts to achieve their goals (Hambali et al., 2019).

IV. Push and pull theory

The idea of push and pull factors states that push factors are those aspects of living in one's current place of residence that led to unrest, such as political repression, poverty, unemployment, fast population expansion, and poor social standing. In contrast, the pull factors that urge people to migrate include political freedom, economic opportunity, well-being, and the welfare state of the destination nation (Urbański, 2022).

V. Conflict-induced displacement theory

Conflict-induced displacement theories emphasize both visible/immediate (proximate) causes and historical root causes, such as political unrest and animosity. These historical factors, combined with current and visible causes such as war, have led people to be forcibly displaced to other places (Lischer, 2007).

VI. Transnationalism theory

Transnationalism theory is a theoretical perspective that focuses on the continuing bonds and links that the migrants keep with their home countries. Pries (2015) explains migrants are not just detached from their home countries, they are continuously involved in transnational practices and relationships that cross national boundaries.

VII. Neoclassical economics theory

Neoclassical economics consider migration as a logical process of decision-making wherein individual actors aim to maximize their economic welfare and utility (Massey et al., 1993). The Massey study describes due to a cost-benefit analysis that makes them assume a good net return from moving, typically financial, people choose to migrate. migrants calculate the expenses and advantages of relocating to different countries abroad and moving to an area where they believe their expected discounted earnings would be highest over a given period (Simon, 1990).

VIII. The new economics of migration theory

The new economics of migration theory continues the principles of neoclassical economics by considering how informal institutions, social networks, and household decision-making influence migration behavior (Blackwell et al., 1991; Massey et al., 1993).

The significance of social connections and community ties in supporting migration decisions is emphasized by the new economics of migration (Massey et al., 1993). The study explained according to this theory, remittances are a major source of support for homes in sending countries, and migrants rely on networks of family and friends to minimize the costs and challenges associated with migrating.

IX. Dual labor market theory

Dual labor market theory posits that migration is driven not only by differences in wages but also by structural factors within the labor market (Massey et al., 1993). Because various worker categories are subject to different labor market logic and management practices, dual labor market theorists offer a critique of standard economic interpretations of labor market inequality based on the decisions of individual agents (Elger, 2015; Massey et al., 1993). This approach relies on the idea that these markets are categorized according to distinct skill sets, job behavior requirements, and regulation of the system (Klimczuk & Klimczuk-Kochańska, 2016).

2.3.2 Framework for Machine Learning

A machine learning framework (workflow) is an organized approach or collection of rules that provides an organized way of developing, employing, and evaluating models and algorithms for machine learning as shown in Figure 1.

Without requiring human input, machine learning (ML) enables software programs to learn from the data and improve their accuracy in forecasting results. Put differently, it enables software applications to learn by doing as they are told (Madakam et al., 2022). Within AI, machine learning techniques become effective technique, these days have been implemented in prediction, time series analysis, and more (Carvalho et al., 2019). High-dimensional and multivariate data can be handled using machine learning techniques, which can also be used to uncover hidden correlations in data in dynamic and complicated contexts (Vandana Kolisetty & Singh Rajput, 2020).

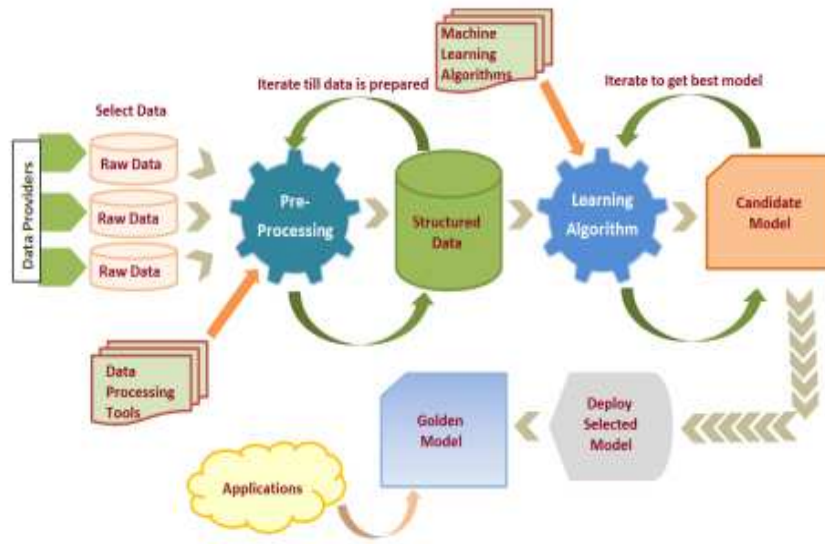


Figure 1 - Machine learning phases

Source: (Vandana Kolisetty & Singh Rajput, 2020)

Machine learning includes both supervised and unsupervised learning. As Madakam et al. (2022) explain, Supervised learning is a labeled dataset, where each input data point is paired with its corresponding output or target variable, whereas unsupervised learning is presented with unlabeled data.

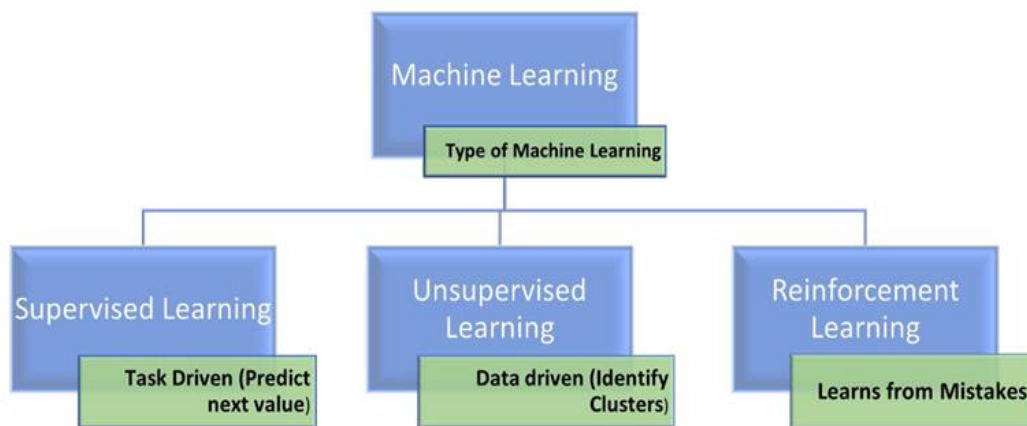


Figure 2 - Machine learning approaches

Source: (Manne, 2021)

The framework for ML includes different phases of the workflow for machine learning. The process includes getting the data from the relevant sources, cleaning the raw data, exploring and understanding the data by doing analysis, building a predictive model, and interpreting the final result (Jeff Saltz, 2024).

2.4 Migration and drivers in northern Ethiopia

In Ethiopia, civilians have migrated from place to place due to diverse factors. They travel for economic reasons, for social obligations, or under pressure from family and relatives due to dangerous civil war, starvation, and drought (Adugna Girmachew, 2021; Dessalegn et al., 2023). In addition to this, many people emigrate to get better educational opportunities in communist nations like the former East Germany, the USSR, and Cuba, to manage their financial status, search for jobs away from home, and send remittances back to their homes (Adugna Girmachew, 2021; ILO, 2018).

Numerous economic, social, and political considerations are reflected in the complex and varied migration patterns and motivations found in northern Ethiopia. Rural-urban migration, where people relocate from rural areas to urban centers in quest of better economic prospects, is one of the main movement patterns in northern Ethiopia (Carter et al., 2016).

International migration from northern Ethiopia is another pattern. Many Ethiopians leave the country in quest of better economic prospects, frequently relocating to the Middle East and Gulf. Another noteworthy trend is forced migration, which is caused by natural disasters, political unrest, and violence. Both domestic and international displacements are included (Dessalegn et al., 2023).

Another factor for migration in Ethiopia is civil war, conflict, and political unrest. These destabilizing aspects can cause displacement of people from their homes in search of safety and security. Political instability presents itself in one way or another such as internal conflicts or ethnic uprisings, and this often results in loss of one's native land. Consequently, many people from Ethiopia have fled to other parts or countries where they think the security and stability to be higher (Adugna Girmachew, 2021).

Ethiopian migration to different countries during the 1974 revolution, which was caused by the downfall of Emperor Haile Selassie and the replacement of the Derg regime, was significant. Before that, Ethiopians who went abroad did so primarily for study purposes and then returned home (Aaron Matteo Terrazas, 2007).

For almost 40 years, the government has promoted resettlement as a means of reducing hunger, drought, and food insecurity as well as enhancing household livelihoods (Tsegay, 2023). According to its collectivization and villagization programs, the Derg regime with socialist ideology encouraged resettlement as a "dual strategy" to address the famine at its source and guarantee household food self-sufficiency in response to the drought and famine of the 1980s (Alula Pankhurst & François Piguet, 2004).

Due to drought in the northern part of Ethiopia, more than 600,000 individuals were moved between 1984 and 1986, and from 600,000 – 700,000 people migrated to the neighboring countries (Vestal, 1985). It failed to provide resettles with significant improvements in their standard of

living while frequently escalating host-community conflict (Alula Pankhurst & François Piguet, 2004).

Comparably, the Ethiopian People's Revolutionary Democratic Front (EPRDF) brought back resettlement in the 2000s with the primary goal of enhancing livelihoods and lowering poverty. Through its two programs (2003–2006), the EPRDF relocated 1.2 million people (Cernea, 2009).

Additionally, in the Afar, Somali, Benishangul-Gumuz, and Gambella regions, a "state-engineered social relocation" involving the sedentarization of pastoralist communities was carried out as part of the "commune" program (Vaughan & Gebremichael, 2011). Long-standing tensions and periodic conflicts have become all too common in Ethiopia after these resettlement programs.

Resettled individuals have been redisplayed in recent years due to several factors, including the Oromo Liberation Army (OLA), Gumuz armed groups, Unidentified Armed Groups (UAGs), and counter-insurgency operations by Ethiopian government troops, as shown by the Benishangul Gumuz and Oromia regional states (Tsegaye Bereket & Gezahegne Kiya, 2023). Other reasons include intercommunal violence, attacks by armed forces, and retaliatory actions taken by host community members against the resettled individuals (Maria Gerth-Niculescu, 2021).

In the Tigray region due to several factors like Economic instability, youth migrated to Europe and the Gulf states. Additionally, pressure from close family and friends makes them look for a better life (Italian Agency for Development Cooperation, 2017).

2.5 Conflict in northern Ethiopia overview

After taking down the Derg regime, EPRDF controlled Ethiopia for 28 years (Opalo & Smith, 2021). TPLF (Tigray People Liberation Front) was the one front from northern Ethiopia (Tigray region) that formed the EPRDF (Ethiopian People's Revolutionary Democratic Front).

With the election of PM Abiy Ahmed in 2018, they began to lose influence, and lately, their power has become limited to the Tigray region. According to (Jima, 2023) findings the conflict between the central government The Ethiopian People's Revolution Democratic Front Coalition's collapse, the central government's eviction of TPLF members, and the postponement of the elections were some of the reasons.

The conflict in Tigray began in November 2020 and turned into a full-scale war due to tensions between the federal government and the Tigray People's Liberation Front (TPLF). More than 2.3 million people were forced to flee their homes and adjacent regional states in 2021 and 2022 because of the conflict's spread into the Afar and Amhara regional states; some of them took up asylum in Sudan (Human Rights Watch, 2024).

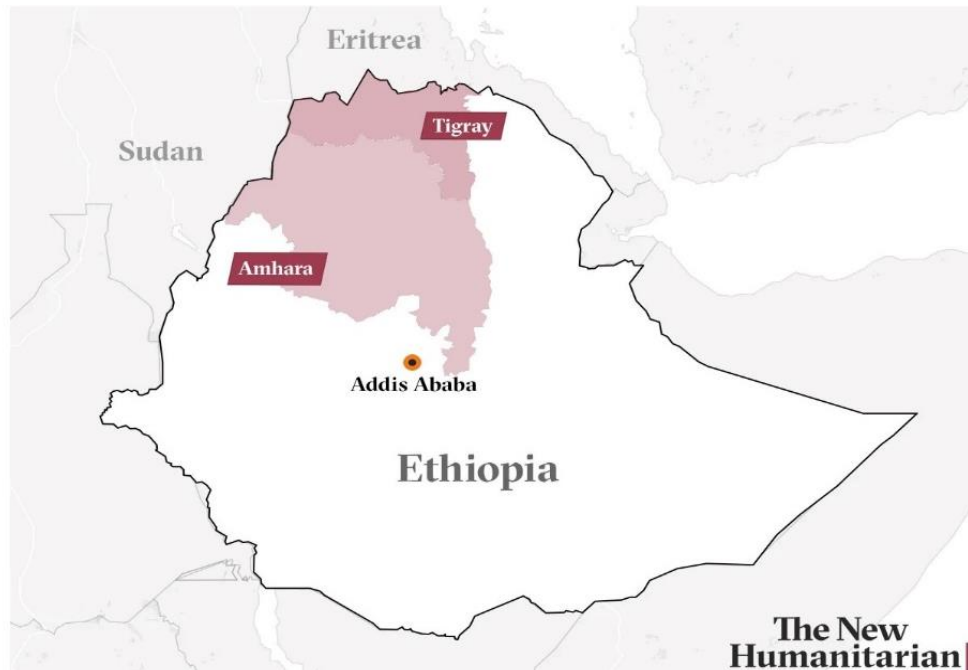


Figure 3 - Conflict area

Source: (Fred Harter, 2023)

According to IDMC (2022a), Ethiopia is the country with the highest number of internally displaced people, with 5.1 million new IDPs in 2021. By the end of 2022, the number of IDPs in Ethiopia become 3.85 million next to DR Congo (IDMC, 2023).



Figure 4 - Sub-Saharan countries with the highest IDPs

Source: (IDMC, 2023)

Due to the Tigray war (northern Ethiopia war), more than 100,000 deaths were recorded in 2022 (Norwegian Center for Humanitarian Studies, 2023). Around 96,223 Eritrean refugees were living in four refugee camps in the Tigray region (Mai Aini, Adi Harush, Shimelba, and Hitsats) receiving shelter before the fighting broke out in the region in November 2020 (USA for UNHCR, 2022).

Immediately after the conflict, they were compelled to relocate to a makeshift shelter in the Afar region and Alemwach, Amhara region (UNHCR, 2022).

2.6 Majority humanitarian crisis

Due to the conflict in northern Ethiopia, civilians faced hardship and several challenges. Since the battle between the Ethiopian Federal Government and the TPLF began in November 2020, around 2 million people have been forced to relocate within Tigray, and approximately 63,000 people have fled to Sudan from western Tigray after Amhara forces overtook the area, according to Pellet, 2021.

The fact that many Internally Displaced People (IDPs) depend on host communities for basic needs like food and water strains already limited resources. The majority of those affected by the fighting have landed in Mekelle, the capital of Tigray; there are food and medicine shortages, overcrowding in living conditions, and a lack of adequate aid (Philippe Pellet, 2021).

Numerous studies confirm that a serious humanitarian catastrophe is developing in Tigray and its surrounding areas. As (Philippe Pellet, 2021) explained, that this crisis and the consequences of Ethiopia's ongoing severe economic crisis may eventually lead to migration outside, including to Europe.

2.7 Role of mobile phones and digital technologies in migration

As Walani (2015) mentioned through network processes and structures, and expression in a shared sense of experience, ICT is being developed and used globally in a way that crosses national boundaries. Recent technological advancements, such as social media, mobile phones, and affordable mobile tariffs, may have a significant impact on the movement of people from place to place. Migration is both positively and negatively facilitated by this (Diminescu, 2008).

History shows there is a connection between technology and migration patterns. Every aspect of life has been greatly impacted by technology, but information and communication technologies (ICTs) have a significant worldwide impact on the migrant (Iqbal et al., 2019). Migrants are dependent on their cell phones when they leave their homes. At every point of their trip, smartphones and tablets have a big influence on their experiences. (Ennaji, 2021). The researcher shows the emotional impact of technology on migrants, especially smartphones and social networks. Considering perspectives, migrants are relying on smartphones to stay connected, access information, and shape their migration process.

Smartphones and social networks allow migrants to bond with their family and their communities. VoIP calls and video calls, online platforms, like Facebook Messenger, and apps such as WhatsApp enable migrants to be psychologically tied to their relatives no matter how far geographically, they are. Despite the hardships of migrating, migrants feel relieved and comfortable thanks to this continuous talk. Additionally, Migrants use smartphones to explore important information and make wise decisions. With the lack of these digital platforms, migrants

might find themselves suffering from emptiness and loneliness since they not only rely on them to get in touch with their loved ones but also reach out for help in times of need.

Using digital technology helps migrants integrate and adapt to their new place while also additionally enabling them to keep in contact with their friends and family, look for humanitarian assistance, and receive news about their home (Dekker et al., 2018; Dekker & Engbersen, 2014; Fedorova & Potemkin, 2020). Fedorova & Potemkin (2020), explained technologies such as the Internet and video cameras have made it easier for migrants to integrate and adapt to their new society while also keeping ties to their homes. Technologies like the Internet and video cameras have significantly facilitated the process of integration and adaptation for migrants. These technologies allow migrants to stay in touch with their families and friends, as well as to document and share their experiences using video cameras and social media.

The research by Jauhiainen et al. (2022) also explained the impact of smartphones on Syrian migrants in Turkey. It shows they maintain transnational ties using smartphones and the internet. Even though they are financially unstable, they still find a way to use a smartphone with the internet.

With the use of digital technologies, migrants can get information before deciding to migrate to reduce risk in their destination place (Iqbal et al., 2019). Smartphones also allow migrants to access information on migration routes, legal rights and protections, and other important information, enabling them to make informed decisions and avoid potential dangers during their journey (Stremlau & Tsalapatanis, 2022). Mobile phones play a significant role in refugees' way of life; They allow people to find jobs and provide services like mobile money services (GSMA, 2017).

The study by Awad & Tossell (2021) shows smartphones become the “life vest” for Syrian refugees, through which they ease and adjust to the opportunities and the cultural context in the Netherlands. They could do those things which are cases of communication with family members and friends, accessing useful information, finding, and using resources and services, searching for job openings, or connecting with local communities and fellow strugglers. These technological devices play indispensable functions in the connection of newcomers with their resettlement process and in the solving of problems bound to happen while they are adapting to a new country and culture.

The focus of globalization concepts is on global technical and economic developments. Mobile phones and online banking platforms considerably simplify the process of getting bank services (Tabit & Moussir, 2017). People use mobile money transfer services for local and international transactions as an alternative to traditional banking systems (Baldé, 2009). The time and location constraints associated with financial operations have been reduced with mobile banking (Mehari, 2019). It allows money transfers between accounts and balance inquiries without the need to visit a branch bank.

But on the other hand, as the researcher Ennaji (2021) shows using a smartphone might come with risks and difficulties, like being the victim of smugglers. However, some migrants prefer to use navigation tools such as Google Maps and GPS instead of relying on smugglers.

2.8 Social media and migration

Widespread communication became easier when the internet developed as social media. Because it makes it possible for people to meet with people they know or don't know (Haythornthwaite, 2016). As McGregor & Siegel (2013), the phrase "social media" is broad and includes social networks like Facebook, blogs, etc., and collaborative projects like Wikipedia and VoIP technologies.

Among other digital technologies like phones, TVs, and radios, social media stands out because it allows simple and affordable real-time transmission of migration information between migrant networks and expected migrants and helps them to keep in touch with their family, relatives, and friends back home (Dekker & Engbersen, 2014; Lara, 2015).

The four major ways that social media can change the lives of migrants mentioned by Dekker & Engbersen Themis (2012) are by establishing connections with friends and family of migrants, giving knowledge/creating awareness about the migration process, creating a connection with people who can help them with the journey, and creating hidden connections to achieve certain objectives.

Social media platforms have become crucial means of sharing information. It has facilitated a deterritorialized social environment that makes communication easier for geographically separated individuals in migrant networks (Dekker & Engbersen, 2014). Its importance as a communication tool is increasing quickly, and different migrants use different online resources and applications. For some migrants, their use has been quite limited, and for people whose countries' internet infrastructure is poorly developed, it is negligible (Fedorova & Potemkin, 2020).

The study by Alencar (2017) illustrates that Syrian refugees in the Netherlands use social media pages like Facebook groups for events and activity organizations. The study also demonstrates how migrants provide each other with emotional support to cope with the loss experienced in their home countries through social media.

The evidence provided by Zijlstra & Liempt (2017) shows that migrant people use and contribute to social media information. For example, social media can influence their decisions on travel routes and modes of transportation (e.g., plane, boat, or on foot). The researcher Zijlstra & Liempt (2017) illustrates the stories of immigrants from Syria, who want to go to the Netherlands, and how they keep in touch and share their experiences of their journey via social media.

In the study by Alencar et al. (2019) the testimony from the migrants from Syria, and Iraq shows how smartphones helped the migrants to get emotional support from their families and additionally, they use it to update their families about their status.

Migrants use social media as a way of their political ideology expression tool after they leave their homes. Sharing pictures and videos via social media has made a big contribution to the Arab Spring (Kassem Chebib & Minatullah Sohail, 2011).

However, the challenge of social media for migration is accuracy and liability. The information that comes through social media might be less or not be correct because people might share rumors and incorrect information (Alpes, 2017; Dekker et al., 2018; Fokkema & de Haas, 2015). As (Dekker et al., 2018) described cost and access to social media are still not the same, and due to this reason, the number of times people use social media to communicate becomes limited.

The study by Alencar A (2023) shows, that because of their bad past experiences on Facebook and WhatsApp groups, Iraqi refugees in France have occasionally been hesitant to trust the information they got from these sites.

Sometimes communicating with their family makes them stressed. Not helping them and not keeping in touch all the time makes them frustrated and feel bad. Awad & Tossell (2021) put the stressful story of a Syrian asylum seeker traumatized by the information from his family about killings, the continuous violence, chaos, and the general peril that characterizes their home country.

The problem of resources for charging their phones and accessing Wi-Fi are some of the challenges faced by migrants (Alencar et al., 2019). During these kinds of situations, migrants use analog methods (Gillespie et al., 2018). Vodafone has launched useful charging equipment that can charge up to 60 phones, which is highly beneficial for migrants in situations like this. According to Alencar A (2023), due to the high cost of mobile connectivity and migrants' limited savings for purchasing phones and SIM cards, they encounter significant challenges.

3 Research Methodology

After reviewing different sources related to the problem under study literature review was classified comprehensively into three parts (themes) include what is the concept behind migration, the most commonly related theory, and the conceptual framework. To identify all relevant publications presented in different sections, databases including PubMed, Google Scholar, Web of Science, Journals, and papers published until 2024 were searched carefully using the title, keyword, and abstract. The citations are imported to documents using Mendeley Reference Manager.

3.1 Data collection method and analysis

Internally Displaced people due to conflict and war in the northern part of Ethiopia are displaced and moved to other parts of the country including Addis Ababa (UNHCR, 2021). The data for the study was collected in Addis Ababa (the capital of Ethiopia), Amhara regional state in Bahir Dar, Debre Birhan town, Dessie, Debre Sina, and Woldia.

To get the answers to the research questions, a one-to-one interview was conducted with 300 respondents in the selected places. The interview includes both Genders (Male and Female) and who are in between the ages of 18 - 65.

For the data collection, the author selected four data collectors to collect the data in the selected places. The data collectors were selected from Addis Ababa, the Amhara region, and the Tigray region. The interviews began in November 2022, and the data collection process was conducted in the Amharic and Tigrigna languages.

The questionnaire was filled out using Google Forms, with the data collectors entering the information directly online. In some locations where internet connection is slow, the questionnaire was printed on paper, and the data collectors filled out the form manually. Later, they transferred the information from the printed paper to the online form.

To choose respondents for the study on the meeting point of smart migration and digital technologies, a chain referral sampling technique is implemented. This approach depends on a series of participant referrals to people who have experienced the phenomenon being studied. Using referrals, The author aims to get access to a wide variety of viewpoints and ideas from networks related to the topic.

The selected respondents are individuals who have migrated from northern Ethiopia due to the conflict. Addis Ababa, as well as the Amhara regional states Bahir Dar, Debre Sina, Debre Birhan town, Dessie, and Woldia, are the locations where the interview is conducted.

The questionnaire is divided into three sections. The first section of the questionnaire was concerned with information about the interview like the location of the interview, the date, and the

language in which the interview is conducted. The second section dealt with the demographic and general data about the respondents, and the third section was about the usage of mobile devices.

The questions were structured in both close-ended types and responses to the questions were measured through a Likert scale of five rating scale disagree (Joshi et al., 2015; UNHCR, 2021). For the third section, a 5-point behavioral scale i.e. strongly agree, agree, neutral, disagree, and strongly is used. The Likert scale was used to make questions clear and permit efficient interpretation of the data.

The Visual Paradigm tool is implemented to draw the Unified Modeling Language (UML) use case diagram, the data analysis process, and machine learning implementation diagrams.

The data analysis (Exploratory data analysis) technique incorporates descriptive statistics such as frequencies and percentages. Additionally, correlation analysis is implemented to identify and quantify the strength and direction of relationships between variables.

As McKinney (2013) explains, that Python programming language is now a competitive choice for data manipulation tasks thanks to its improved library support, like pandas. When combined with its ability in general-purpose programming, Python becomes a great option. The analysis was conducted using Google Colaboratory, an online platform designed for collaborative coding and data processing. The analysis results will be presented with both tables and figures (bar chart, pie chart, and tables).

The unprocessed data that is collected from the respondents via Google Forms, is exported into a CSV (Comma-Separated Values) file. This CSV file typically contains the raw data as entered by the data collectors, and it serves as the starting point for the data preprocessing (cleaning process).

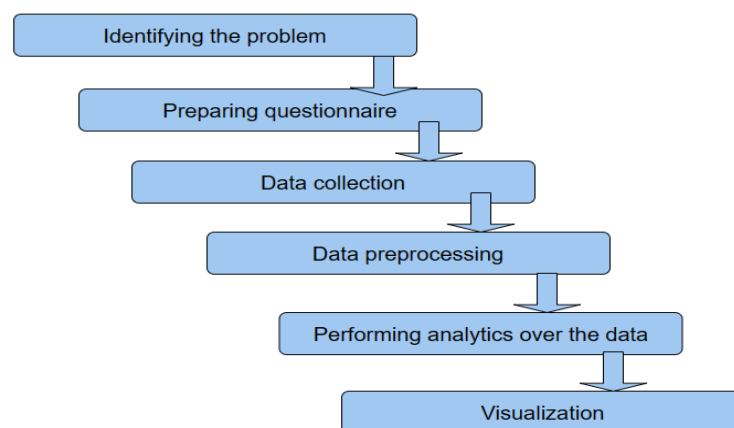
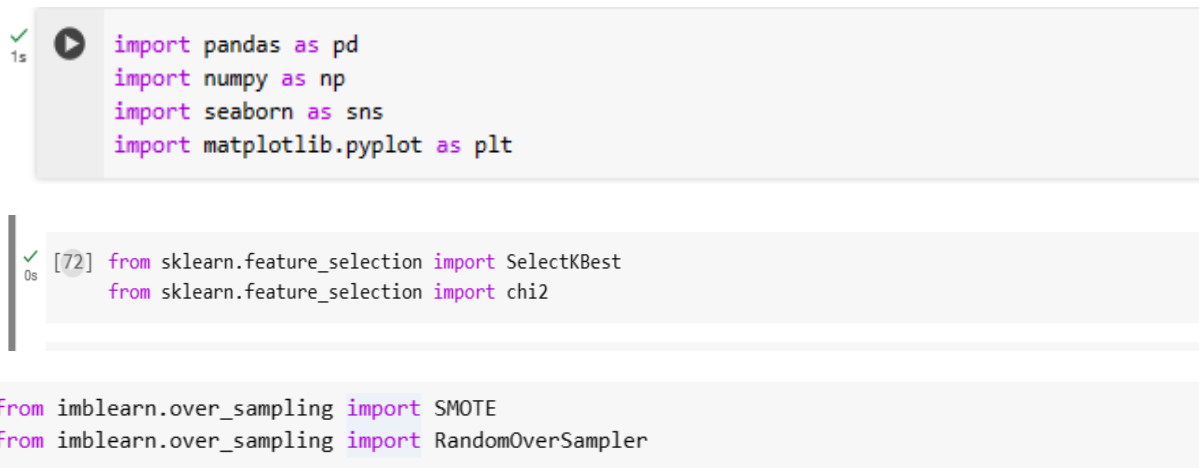


Figure 5 - Data analysis process

3.1.1 Data preprocessing

3.1.1.1 Importing the Necessary Libraries

The first step in the data analysis process is importing the necessary libraries. NumPy provides support for numerical operations, particularly with arrays as shown in (Figure 6). Pandas is used for manipulating and processing data in data frames, and Seaborn and Matplotlib are used for data visualization and plotting.



```
✓ 1s ▶ import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

✓ 0s [72] from sklearn.feature_selection import SelectKBest
from sklearn.feature_selection import chi2

from imblearn.over_sampling import SMOTE
from imblearn.over_sampling import RandomOverSampler
```

Figure 6 - Importing libraries for the data analysis

Pandas: Pandas offer features and data structures for effectively working with structured data. It is used for both data analysis and data manipulation. Pandas integrate relational databases and spreadsheets' data manipulation tools with NumPy's high-performance array computing characteristics (McKinney, 2013).

NumPy (Numerical Python): NumPy is an important library used to perform numerical computation. When it comes to data storage and manipulation, NumPy arrays outperform the other built-in Python data structures (McKinney, 2013).

Matplotlib: is a widely used library in Python for creating visualizations, offering a flexible and powerful toolkit for data visualization tasks. Its popularity stems from its versatility, providing a vast array of tools for constructing a wide variety of plots and charts to effectively communicate insights from data (Barrett et al., 2005).

Seaborn: It gives a high-level and advanced interface on matplotlib. It has good integration with pandas data structures and enables the generation of sophisticated plots and charts (Waskom, 2021).

3.1.1.2 Loading the data

The next step as shown in (Figure 7) involves mounting the dataset, loading it into the environment, and reading the CSV (comma-separated file) file into the data frame to begin analysis.

Using Python's pandas module, the raw CSV data which is exported from Google Forms is loaded into a pandas DataFrame, a two-dimensional data set with a labels structure. The dataset is shown tabularly in this DataFrame, where each row corresponds to an observation or data point and each column to a variable or characteristic.

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
df = pd.read_csv('/content/drive/MyDrive/Smart Migration - Digital Technologies and Migration in the Northern Ethiopia.c
```

Figure 7 - Loading and reading the dataset

3.1.1.3 Data cleaning

Preprocessing data is important to ensure the accuracy of data analysis employing different methods. It is a method for transforming unclean data into a clean data set (Fan et al., 2021). This process makes the feature engineering process which involves modifying the data using different methods, easier (Mahadevan Malamahadevan, 2024)

Renaming the column

Meaningful column names significantly increase the readability of the code, making it simpler to understand for readers. By providing descriptive names to columns, the code becomes more intuitive, reducing the cognitive load required to understand the dataset's structure.

To rename the columns python has the `‘.rename()’` method will do. It takes two parameters. (`‘old_column_name’` : `‘new_column_name’`) as parameters. Figure 8 shows renaming the column names which are directly taken from the dataset (questions) into understandable and simpler names.

```

columns=['Date','Language','Location','Age group','Gender','Education Level','Grade','Level','University','Program',
'year','level','Mention University','ys program','previous job','sector','currently working','current job',
'job sector','current location','sub city','region&town','howlongyou stay','income source','economy sr',
'money from abroad','money transfer way','bank use','married?','children','# of children','live with',
'phone brand','mobile usage reason','phone language','social media?','sm most used','sm not usage reason',
'main reason of social media','mobile usage increase?','money spend for mobile','is MD helpful?','how?',
'MD for moving','MD for moving How','Device affordable?','hortage of Electric power','Internet connection problem'
]

4] dataset = dataset.set_axis(columns, axis=1)

5] dataset.info()

```

Figure 8 - Renaming column

Correcting misspelling

Misspelled data may bias the results of analysis, producing false conclusions and insights. Correcting spelling will increase the credibility of the analysis process. To do so, as shown in (Figure 9) the dictionary that holds the correct values and misspelled values is created. Which will replace the values of misspelled values with the correct ones. As shown in Figure 9 the value `lang_dict` holds the values as a dictionary, Then the next step is assigning the dictionary values to the column that holds the values (Interview language).

```

✓ [8] lang_dict = {
0s      "Amharic ": "Amharic", "Amharic`": "Amharic", "Tigrigna ": "Tigrigna", "Tigrngna": "Tigrigna"
    }

✓ [9] df = df.replace({
0s      "Interview Language": lang_dict
    })

✓ [10] df['Interview Language'].value_counts()
0s
Interview Language
Amharic      225
Tigrigna      75
Name: count, dtype: int64

```

Figure 9 - Correcting misspelling

Splitting the values and making them individual columns

Splitting the values which are collected in one column which are separated by different types of delimiters and making them into individual columns is an important step in the data cleaning process. This process helps to structure and organize the data in a more meaningful way, making it easier to analyze and work with. By splitting them, it is possible to do the analysis easily and do transformations as needed.

Pandas data frame has the method '**str. split()**'. As shown in Figure 10, It is implemented for splitting the values which are separated by (;) delimiter.

```
df['Social Media'] = df['sm most used'].astype(str).replace(['nan', 'NaN'], '')

# Step 2: Split the column, with exception handling for NaN or unexpected data
df['Social Media List'] = df['Social Media'].apply(
    lambda x: x.split(';') if x.strip() != '' else []
)

# Step 3: Find all unique platforms, ensuring correct data types
unique_platforms = set(
    platform for sublist in df['Social Media List'] for platform in sublist
)

# Step 4: Create one-hot encoded columns for each unique platform
for platform in unique_platforms:
    df[platform] = df['Social Media List'].apply(
        lambda platforms: 1 if platform in platforms else 0
    )
```

Figure 10 - Splitting into separate columns

Dropping columns

After splitting the values into a new column, it's often a good practice to drop the original column that holds the non-split values and any temporary columns created during the splitting process. This helps to clean up the data frame and remove redundant or unnecessary columns.

In pandas, the method '**.drop()**' is used to drop the columns. Figure 11 shows the lists that are going to be deleted and dropping columns that have more than 40% null values.

```
df = df.drop(df.loc[:, list((100 * (df.isnull().sum() / len(df.index)) > 40))].columns, axis=1)

df.drop(columns=['Social Media List', 'Social Media', 'sm most used'], inplace=True)
print(df)
```

Figure 11 - Drop columns

Cleaning duplicates

Figure 12 below shows the code that will check if there is a duplicate, it will drop it.

```
df=df.drop_duplicates()
for i in df.duplicated():
    if i==True:
        print( i)
```

Figure 12 - dropping duplicates

Merging columns

In the column values, 2 columns have the same meaning, but they will be considered different due to spelling differences. Figure 13 shows 2 different columns that have the same meanings. To merge them, the '+' operator is implemented.

```
[ ] df['Sm for activism'] = df['Sm for For For Social Activism'] | df['Sm for Activism']
```

```
[ ] df['Sm for activism'] = df['Sm for For For Social Activism'] + df['Sm for Activism']
```

```
[ ] df['Sm for activism'].value_counts()
```

```
Sm for activism
0      264
1       36
Name: count, dtype: int64
```

Figure 13 - Merging 2 column values

Label encoding

Our columns are categorical data (object datatype) and it is necessary to do label encoding. Which is assigning a unique integer for each category.

```
df[df.columns] = df[df.columns].apply(LabelEncoder().fit_transform)
```

Figure 14 - Label encoding

Dealing with class imbalance

imbalanced class can lead to biased information. Balancing techniques such as resampling or algorithmic adjustments help mitigate this issue.

```
77] from imblearn.over_sampling import SMOTE
    from imblearn.over_sampling import RandomOverSampler

78] oversampler = RandomOverSampler(sampling_strategy='auto', random_state=42)
    X_resampled, y_resampled = oversampler.fit_resample(X_new, y)
```

Figure 15 - handling imbalance class

The technique implemented for handling data imbalance is the oversampling technique. Oversampling is a technique used in machine learning and data analysis to address the issue of class imbalance in a dataset (Islam et al., 2022). By generating additional instances or repeating some instances, It is possible to increase the number of minority class instances or samples through oversampling (Mohammed et al., 2020).

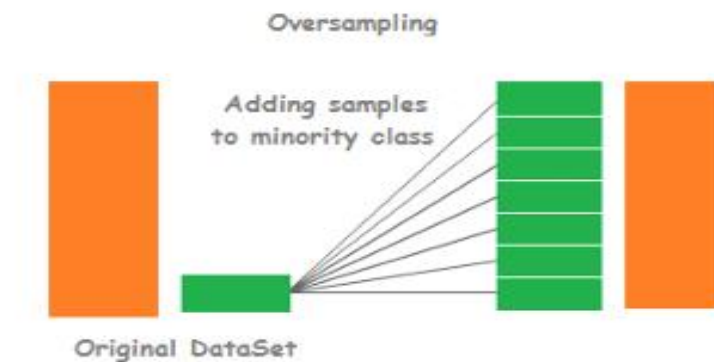


Figure 16 - Oversampling technique

Source: (Mohammed et al., 2020)

Feature Selection

Feature selection is the process of selecting a subset of relevant features (Sabir Rosidin et al., 2021). For the selection of mostly associated features, the chi-square technique is implemented. Since the data is categorical, implementing the chi-square is suitable (ken Stewart, 2023).

$$X^2 = \frac{\sum (O_i - E_i)^2}{E_i}$$

Where,

O_i – frequency of a particular category,

E_i – Expected frequency,

K – the number of possible categories

Equation 1 - Chi-square formula

Implemented Models

The target variables (the reasons for smartphone usage and mostly used applications) in the dataset, have multiple categorical labels (the model predicts outcomes with more than two possible categories), as a result, the predictions of the *reasons for smartphone usage* and *mostly used applications*, the following 2 models (decision tree, and random forest) are implemented.

- **Decision Tree:** The decision tree is composed of nodes that come together to form a rooted tree, which is a directed tree with a node known as the "root" that is devoid of any incoming edges (Rokach & Maimon, 2006).

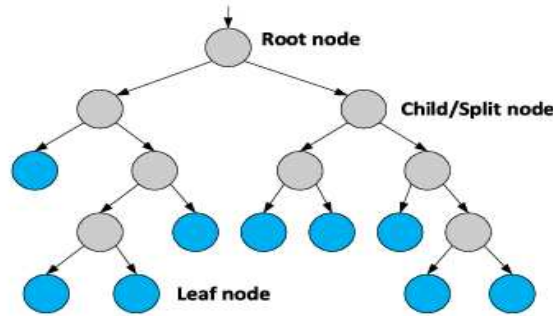


Figure 17 - Decision tree random

Source: (Camana Acosta et al., 2020)

- **Random Forest:** For classification and regression applications, the Random Forest algorithm is a popular learning method. As (Breiman, 2001), A random forest is created by combining different tree predictors so that every tree in the forest is dependent on the values of a random vector that is randomly sampled and has the same distribution for every tree.

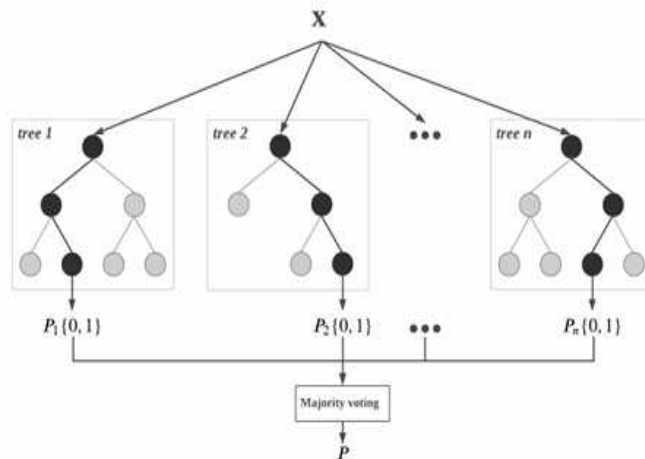


Figure 18 - Random forest classifier

Source: (Shahhosseini & Hu, 2020)

1.2 Machine learning implementation procedure

Figure 19 shows the overall procedure implemented in this study to analyze the data and make predictions by using machine learning techniques.

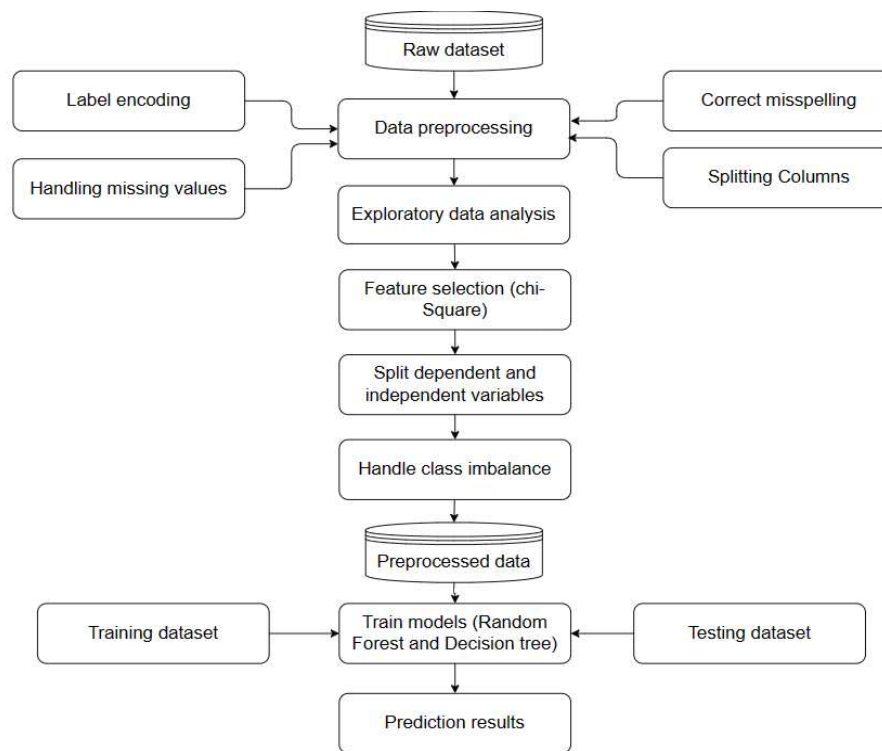


Figure 19 - Machine learning implementation procedure

3.3 Ethical considerations

Respondents must participate voluntarily in the study. The respondents are only between 18 and 65. Individuals below the age of 18 may not have the legal capacity to provide informed consent for participating in research studies. By limiting respondents to adults aged 18 - 65, the researcher minimizes potential risks and ensures that the study adheres to ethical standards regarding the welfare and rights of participants.

The data collected from this study is kept in a locked folder and used only for academic purposes. Additionally, it was expressly stated on the front page of the questionnaire, and respondents were thanked in advance. The researcher is still liable and responsible for any ethics violations. The researcher has received a letter of confirmation from the university's (Mendel University's) ethical committee that confirms the questionnaire is ethically valid.

4 Analysis and Result

After implementing the essential phase of data preprocessing step by step in the previous chapter (Methodology), the preceding steps are performed in this chapter. These steps are crucial phases in the data preprocessing process since they directly affect the dataset's quality.

4.1 The socio-demographic profile of IDPs

Many of the interviews were conducted in the Amharic language, with 225 interviews representing 75%. the proportion of each language by number, along with their percentage distribution used in the interviews.

Figure 20 illustrates the distribution of languages in which the interview was conducted in Barchart, along with the corresponding numerical values displayed within it and the percentage distribution.

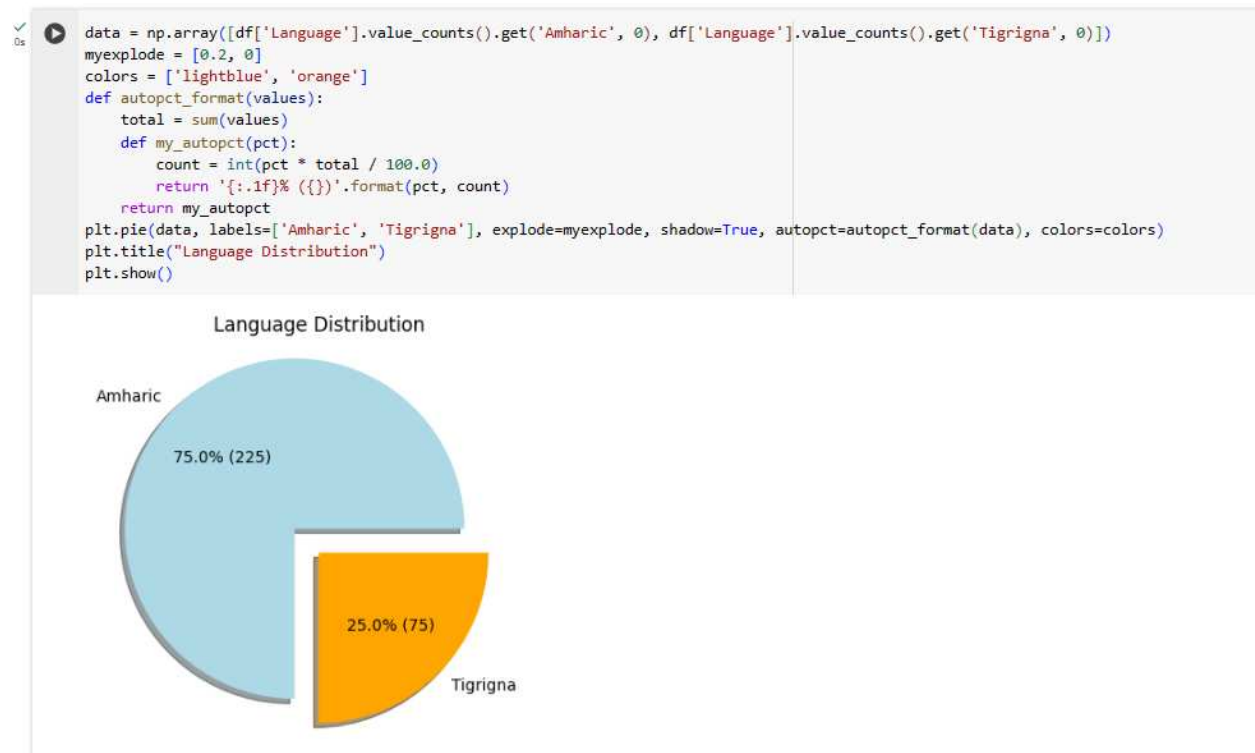


Figure 20 - Interview language distribution in pie and percentage

The age distribution among the interviewees appears to be fairly even across various groups, with the 23 to 30 age group representing a slightly higher proportion at 20.7%.

```

data = np.array([value_counts.get(uv, 0) for uv in unique_values])
total = sum(data)
colors = ['green', 'lightcoral', 'lightblue', 'orange']
plt.bar(unique_values, data, color=colors)
plt.title("Age Distribution")
plt.xlabel("Age Group")
plt.ylabel("Count")
for i, count in enumerate(data):
    pct = (count / total) * 100
    plt.text(i, count + 0.5, f'{count} ({pct:.1f}%)', ha='center', va='bottom')
plt.show()

```

Figure 21 - Generating Age distribution visualization using Google Collaboratory

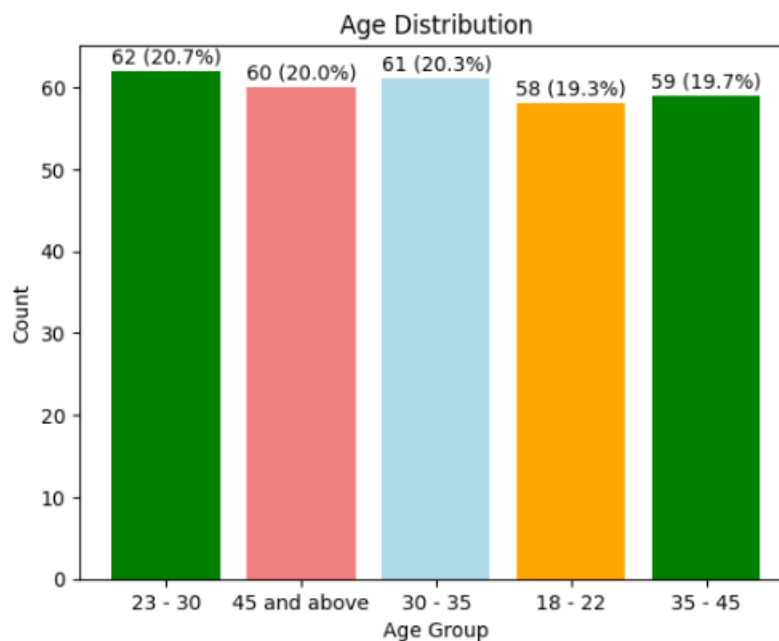


Figure 22 - Age distribution in bar chart and percentage

This age range often encapsulates individuals in their prime working years, highlighting the impact of displacement on a significant segment of the population. The majority of the participants were female 154 (51.33%).



Figure 23 - Gender distribution

The educational levels of the study were majorly shared by never educated 162 (54%). The majority of interviews (80.7%) conducted outside Addis Ababa point to the widespread impact of the conflict, beyond the conflict zone. About 19.3% of the migrants live in Addis Ababa.

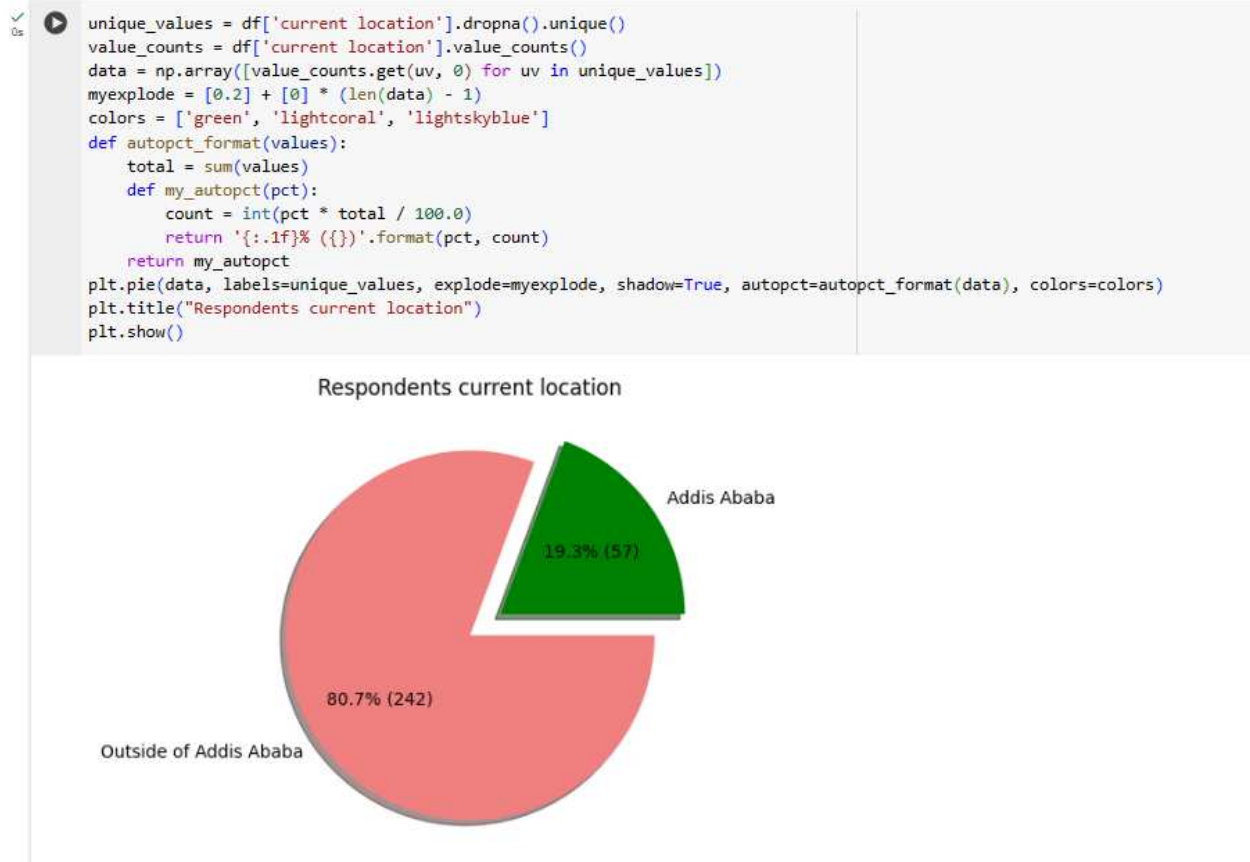


Figure 24 - Migrants' location.

Out of the total participants, 37.67% stayed in their current locations for more than one year. Figure 26 shows the steps and parameters followed to generate a pie chart that shows how long the respondents stayed in their current location.

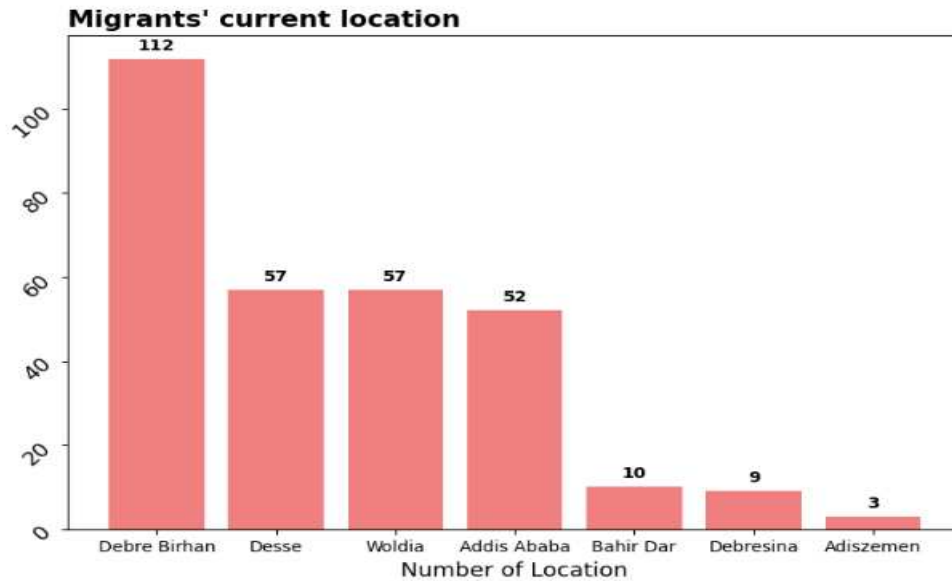


Figure 25 – Migrants' current location

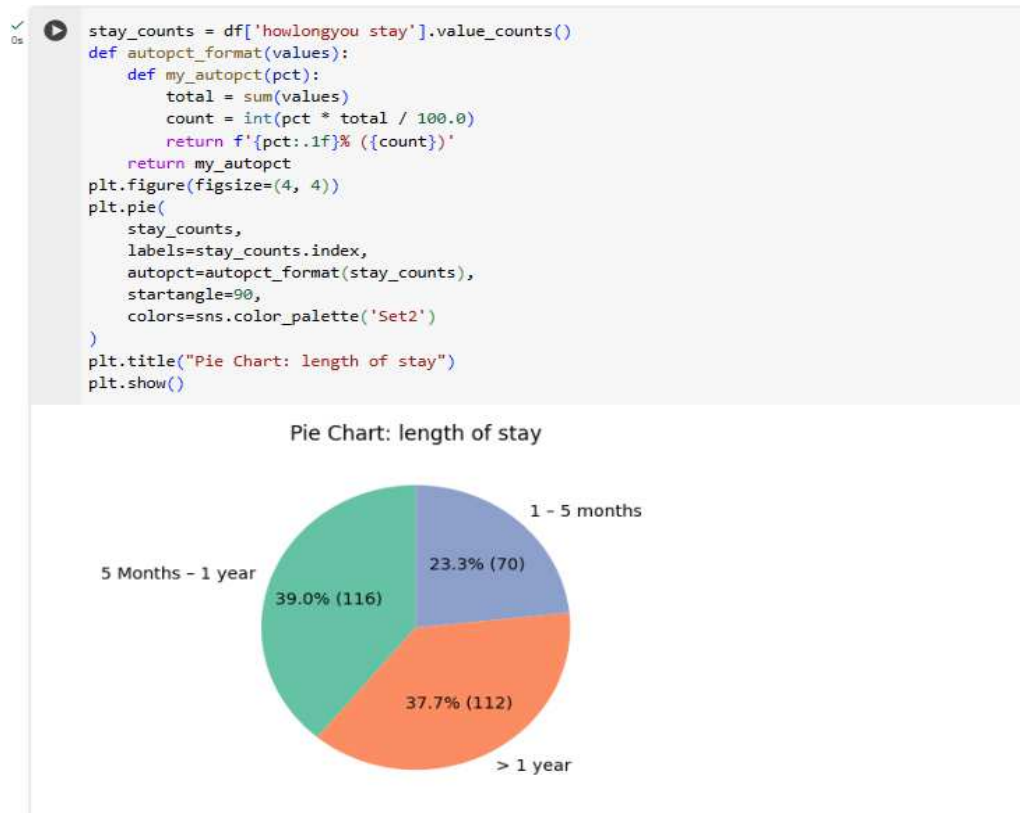


Figure 26 - The proportion of participants residing in the current location

The findings on marital status (48.33% married) and parenthood (45.7% with children) provide insights into the family structures of the displaced population.

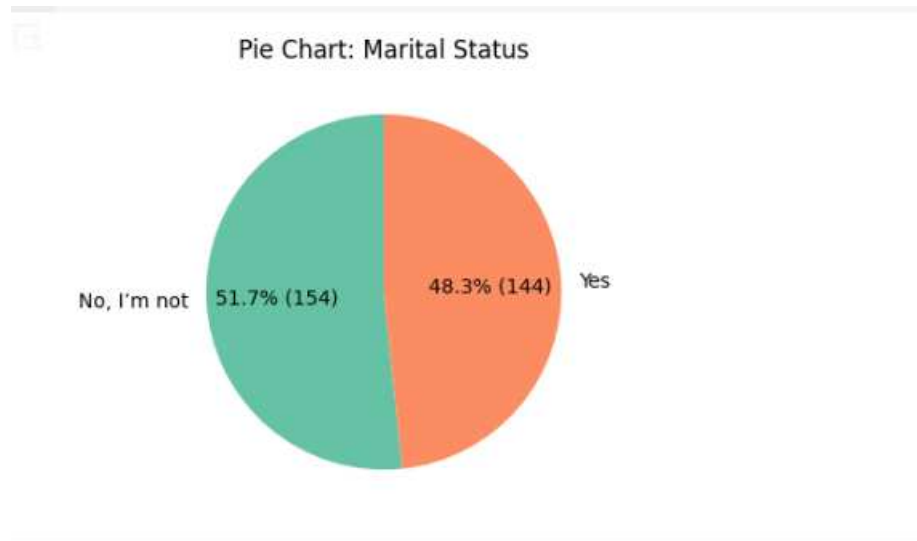


Figure 27 - Marital Status

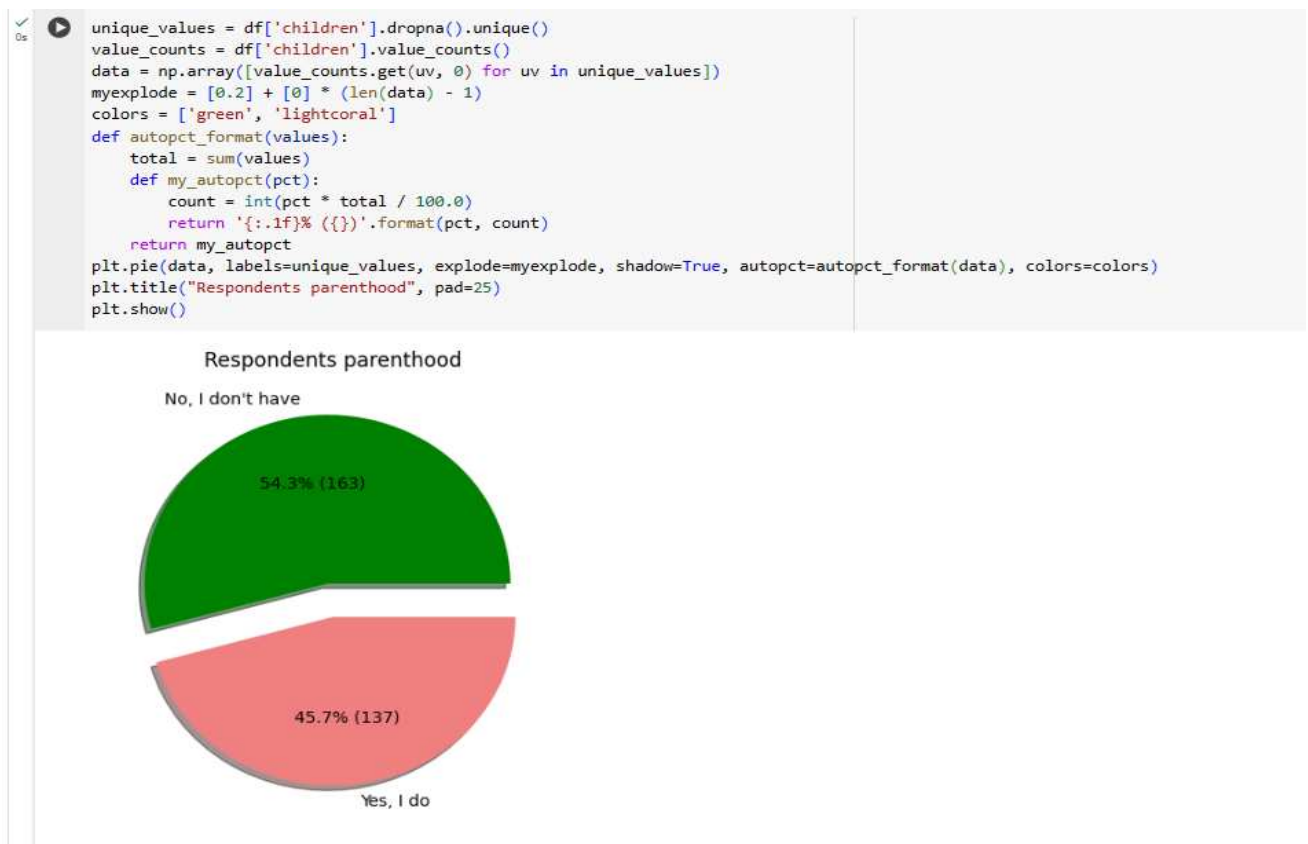


Figure 28 - Participants Parenthood

```

✓ [104] education_counts = df['Education Level'].value_counts()
0s      total_count = education_counts.sum()
      education_table = pd.DataFrame({
          'Education Level': education_counts.index,
          'Frequency': education_counts.values,
          'Percentage': (education_counts.values / total_count) * 100
      })
      education_table['Percentage'] = education_table['Percentage'].round(2)
      print("Education Level Distribution (Frequency and Percentage):")
      print(education_table)

```

Figure 29 - Generating a table of IDPs' education level using Google Colaboratory

Figure 30 shows the distribution of education levels among the respondents, providing a comprehensive view of the varying degrees of educational attainment within the study's participant group.

Education Level Distribution (Frequency and Percentage):

	Education Level	Frequency	Percentage
0	I don't study	162	54.00
1	University Graduate	49	16.33
2	University Student	42	14.00
3	High school	21	7.00
4	TVET Graduate	14	4.67
5	TVET Student	7	2.33
6	Middle School	3	1.00
7	Catering	2	0.67

Figure 30 - Education level of respondents

Demographic insights can guide policymakers, humanitarian organizations, and aid agencies in tailoring interventions that address the specific needs and challenges faced by different subgroups within the displaced population. Understanding the distribution of key demographics such as age, gender, education, and family composition is essential for designing effective and inclusive strategies to support the well-being and resilience of displaced individuals and families.

The table below (Table 1) presents, in tabular format, all the analyses conducted above

Table 1 - The socio-demographic profile of IDPs

Variable	Category	Frequency (occurrences)	Percent
Interview language	Amharic	225	75.00
	Tigrigna	75	25.00
Age (18+ at minimum)	18 – 22	58	19.33
	23 – 30	62	20.67
	30 – 35	61	20.33
	35 – 45	59	19.67
	45 and above	60	20.00
Gender	Female	154	51.33
	Male	146	48.67
Education level	I don't Study	162	54.00
	University Graduate	49	16.33
	University Student	42	14.00
	High School	21	7.00
	TVET (Technical and Vocational Education Training) Graduate	14	4.67
	TVET (Technical and Vocational Education Training) Student	7	2.33
	Middle School	3	1.00
	Catering	2	0.67
Where do you live now? (Location)	Addis Ababa	58	19.3
	Outside of Addis Ababa (Woldia, Debre Birhan, Bahir Dar, Dessie and Debre Sina)	242	80.7
How long do you stay in your current location?	1 – 5 months	70	23.33
	5 Months – 1 year	117	39.00
	> 1 year	113	37.67
Are you married	No, I'm not	155	51.67
	Yes	145	48.33
Do you have children?	I don't have	163	54.33
	Yes, I do	137	45.67

4.2 Job and income-related characteristics of the IDPs

In their hometown, they have private businesses 111 (37%), students 60 (20%), housewives 38 (12.67%), Teachers 21(7%), government office workers 19 (6.3%), farmers 22 (7.33%), and craft persons 24 (8%). Even though they had jobs in their hometown, the majority of them 252 (84%) now become jobless after displacement from their origin.

```
0s job_counts = df['previous job'].value_counts()
total_count = education_counts.sum()

job_table = pd.DataFrame({
    'Previous job': job_counts.index,
    'Frequency': job_counts.values,
    'Percentage': (job_counts.values / total_count) * 100
})
job_table['Percentage'] = job_table['Percentage'].round(2)
print("Previous Job (Frequency and Percentage):")
print(job_table)
```

Figure 31 - Generating a table of the participants' previous jobs using Google Colaboratory

Figure 32 shows the participants' previous jobs and gives insights into the range of professional backgrounds and work experiences among those surveyed.

```
Previous Job (Frequency and Percentage):
```

	Previous job	Frequency	Percentage
0	Private Business (Shop, Cafe..)	111	37.00
1	Just Student	60	20.00
2	House wife	38	12.67
3	Craft person	24	8.00
4	Farmer	22	7.33
5	Teacher (KG, Middle school, high school or uni...	21	7.00
6	Government office worker	19	6.33
7	Nurse	2	0.67
8	Cleaning	2	0.67
9	Doctor	1	0.33

Figure 32 - Respondents' previous jobs

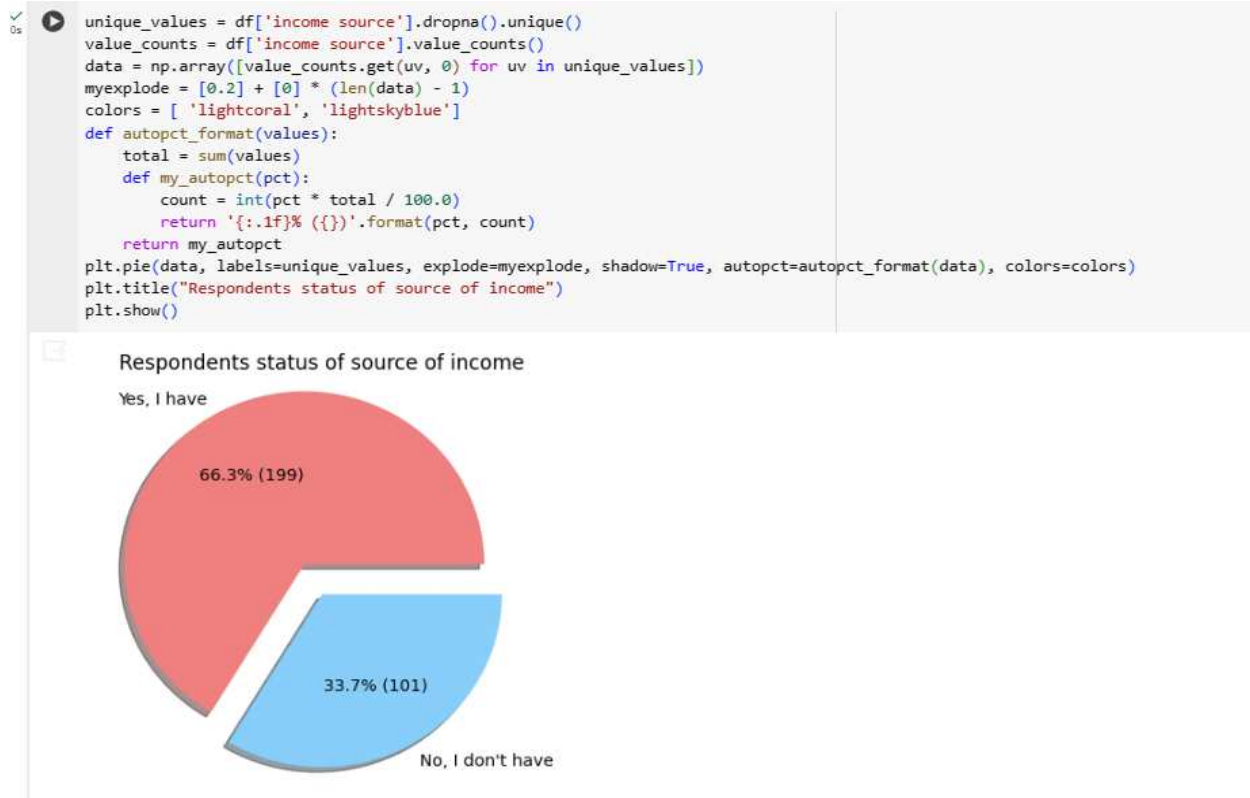


Figure 33 - Respondents' current source of income status

Among the participants, 199 (66.33%) of them had sources of income and 113 (56.78%) got their income from their family.

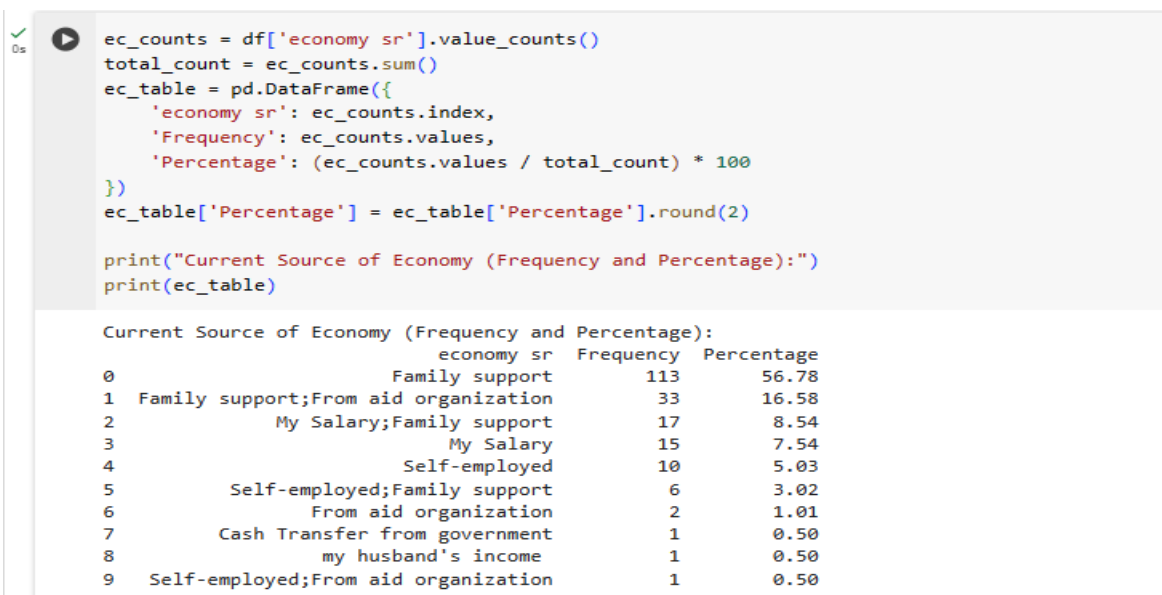


Figure 34 - Source of income

When sending money to someone, around 128 IDPs (45.71%) use both in-person banking and mobile banking for their transactions. In this study, 160 IDPs, or 57.14% of the sample, use mobile phones to send or receive money. This information is detailed in Table (2).

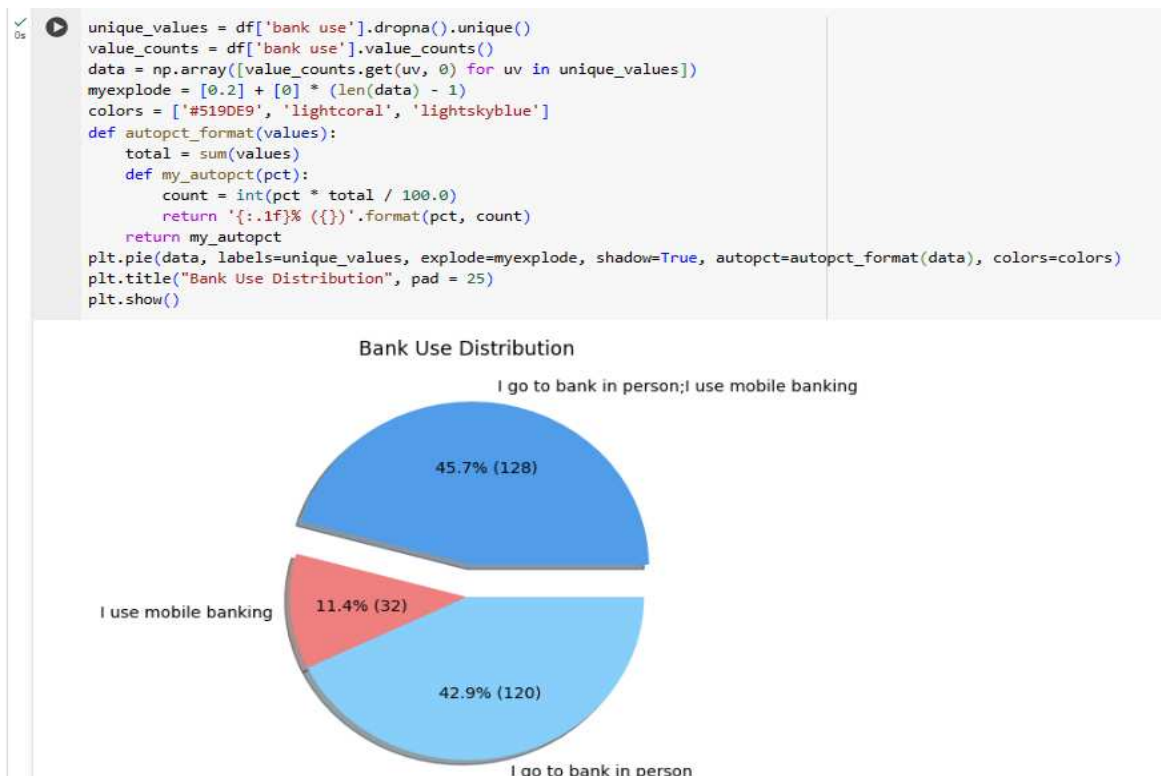


Figure 35 - Participants' bank system usage preferences

Approximately 117 IDPs, accounting for 39% of the total, receive money from abroad through legal money transfer services, and 38 (12.7%) use via black market.

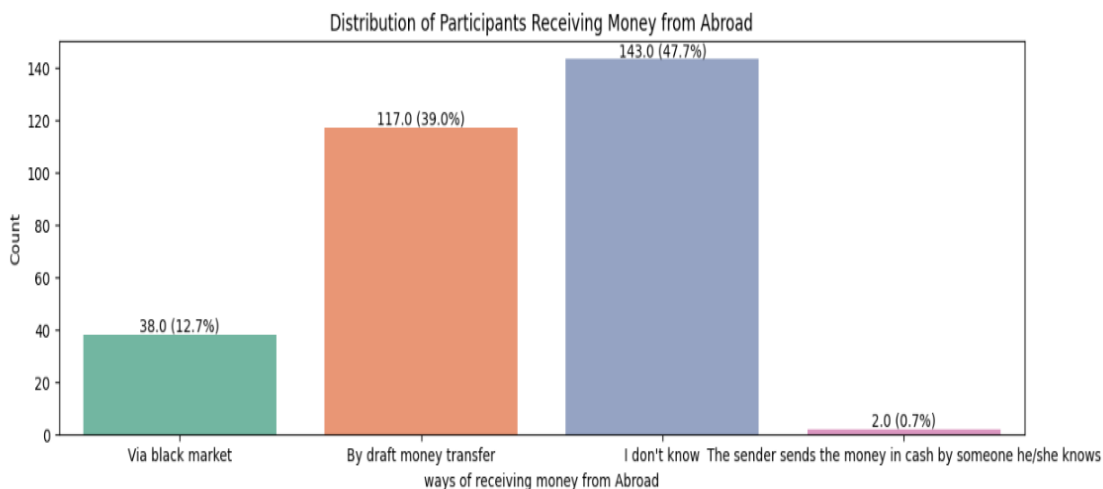


Figure 36 - How participants receive money from abroad

The table below (Table 2) presents, in tabular format, all the analyses conducted above

Table 2 - The Job and income-related characteristics of the IDPs

Variable	Category	Frequency	Percent
Job in your hometown?	Broker	1	0.33
	Cleaning	2	0.67
	Craft person	24	8.00
	Farmer	22	7.33
	Government office worker	19	6.33
	Health professionals	3	0.33
	Housewife	38	12.67
	Student	60	20.00
	Private Business	110	36.00
	Teacher	21	7.00
Do you currently have a source of income?	No, I don't have	101	33.67
	Yes, I have	199	66.33
What is your source of economy? (More answers are possible)	Cash Transfer from the government	1	0.50
	Family support	113	56.78
	Family support from the aid organization	33	16.58
	From aid organization	2	1.01
	My Salary	15	7.54
	My Salary and Family support	17	8.54
	Self-employed	10	5.03
	Self-employed and Family support	6	3.02
	Self-employed and from an aid organization	1	0.50
	My husband's income	1	0.50
How do you mostly receive money from abroad?	By legal money transfer	117	39.00
	I don't know	143	47.67
	The sender sends the money in cash via someone I know	2	0.67
	Via black market	38	12.67
When you send money to someone, which way do you use the most?	I haven't transferred	3	1.00
	I prefer to send cash via someone I know	13	4.33
	I use banks	271	90.33
	I use the Tele birr application	13	4.33
In which way do you use the bank the most?	I go to the bank in person	120	42.86
	I go to the bank in person, and I use mobile banking	128	45.71
	I use mobile banking	32	11.43

4.3 Smartphone technology usage by IDP

A higher number of participants preferred to use the TECNO 125 (41.7%) followed by Huawei brand of mobile 49 (15.33%) and Samsung 44 (14.67%).

```
mobile_brands = df['phone brand'].value_counts()
total_count = mobile_brands.sum()

mobile_table = pd.DataFrame({
    'Mobile Brand': mobile_brands.index,
    'Frequency': mobile_brands.values,
    'Percentage': (mobile_brands.values / total_count) * 100
})
education_table['Percentage'] = education_table['Percentage'].round(2)

print("Mobile brand (Frequency and Percentage):")
print(mobile_table)
```

```
Mobile brand (Frequency and Percentage):
Mobile Brand  Frequency  Percentage
0          TECNO         125    41.666667
1          Huawei         49    16.333333
2          Samsung         44    14.666667
3           SMADL         18     6.000000
4           Oppo         17     5.666667
5          Xiaomi         16     5.333333
6           HTC         16     5.333333
7        Infinix          6     2.000000
8           Iphone         4     1.333333
9           Nokia          3     1.000000
10         Realme          1     0.333333
11           itel          1     0.333333
```

Figure 37 - Mobile brands used by participants

Participants highly prefer to use both Amharic and English languages 113 (37.67%) and Amharic only 107 (35.57%) their mobile phone. The majority of the participants said the mobile usage trend has increased since their displacement to their current location 205 (68.33%).

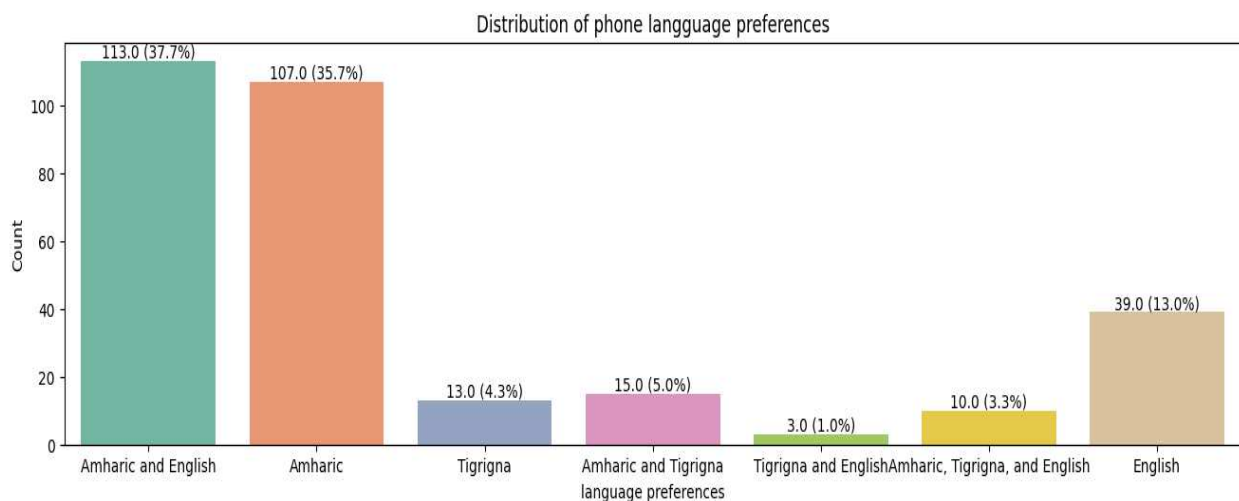


Figure 38 - Participants' phone language preference

They spend money on mobile ranging from 0.16-0.33 EUR (42.67%), 0.082-0.16 EUR (29.33%), 0.33-0.82 EUR (20.67%), >0.82 EUR (7.33%).

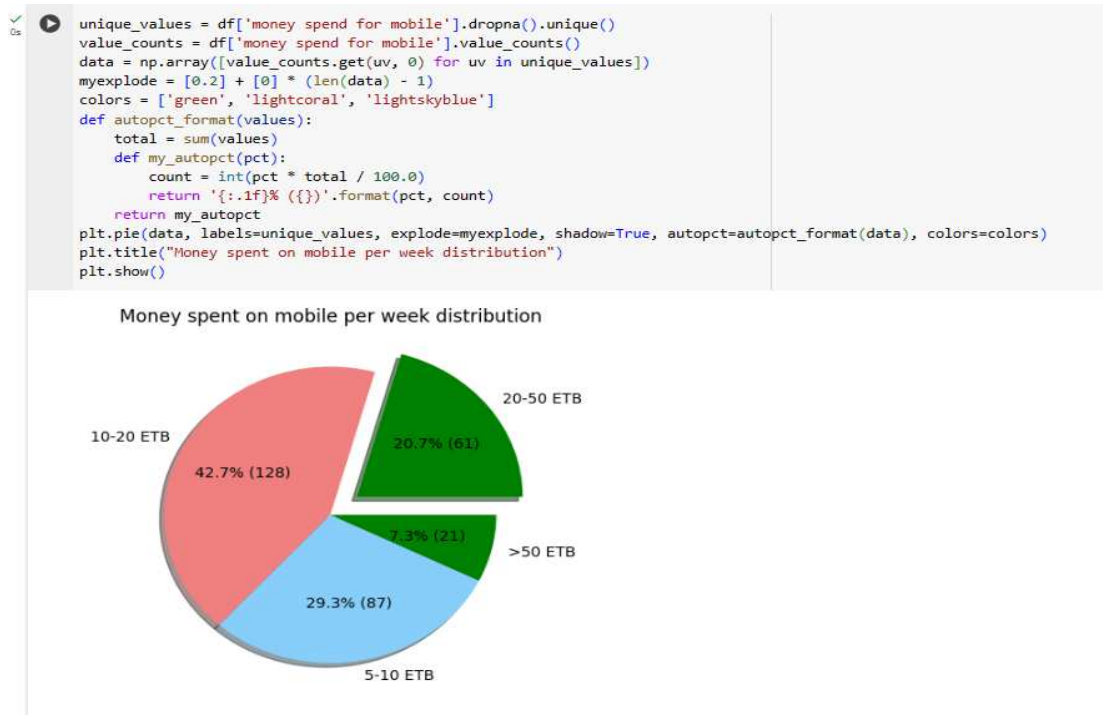


Figure 39 - Amount of money spent on mobile per week

Figure 40 shows the responses of participants regarding the helpfulness of smartphones. This visual representation helps understand the perceived value and utility of mobile phones among the surveyed group.

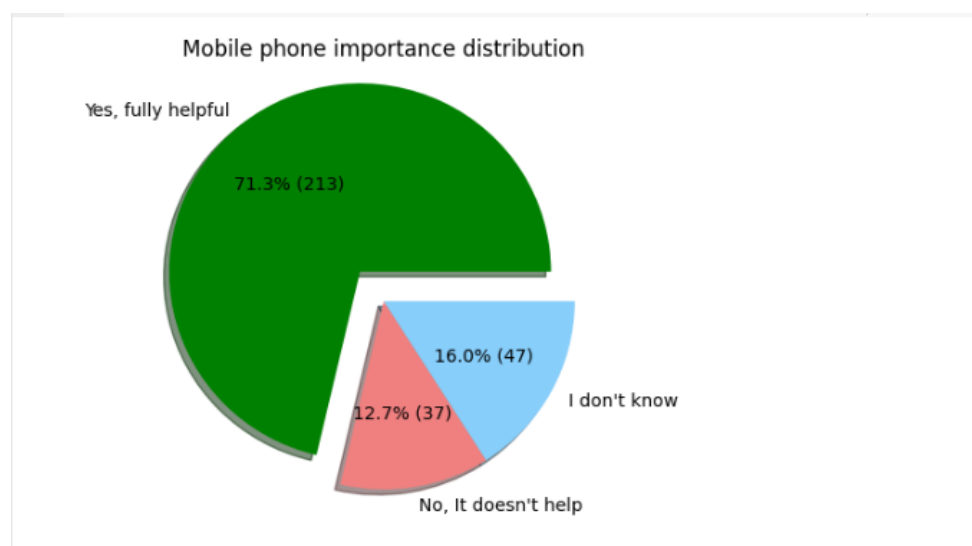


Figure 40 - Is a mobile phone helpful?

The reasons for using smartphones during migration among migrants were to access the internet, call and SMS, Games, take pictures, and videos, place navigation, and work activities.

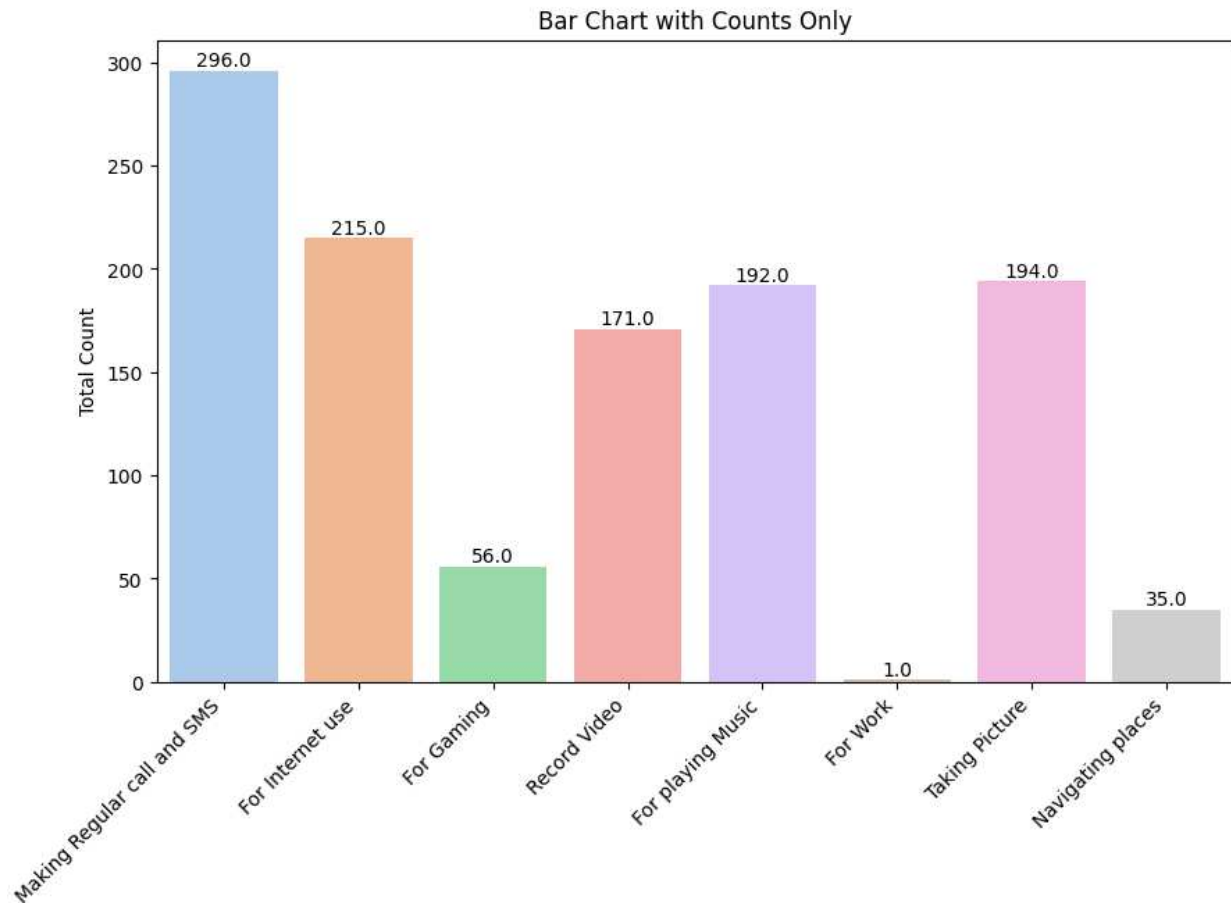


Figure 41 - Participant's reasons for using mobile phone

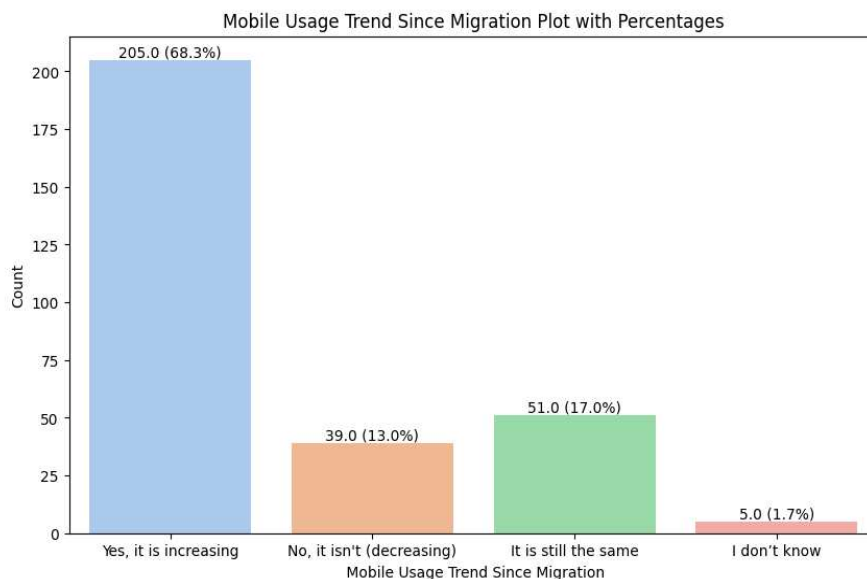


Figure 42 - Mobile usage trend of the participants

The table below (Table 3) presents, in tabular format, all the analyses conducted above

Table 3 - Smartphone technology usage by IDP

Variables	Category	Frequency	Percent
What brand of mobile device do you use?	TECNO	125	41.7
	Huawei	49	16.33
	Samsung	44	14.7
	Smadl	18	6
	Oppo	17	5.7
	xiaomi	16	5.3
	htc	16	5.3
	infinix	6	2
	Iphone	4	1.33
	Nokia	3	1
	realme	1	0.33
	Itel	1	0.33
In which language do you prefer to use your mobile device? (More answers are possible)	Amharic	107	35.67
	Amharic and English	112	37.67
	Amharic and Tigrigna	14	5
	Amharic, Tigrigna English	10	3.33
	English	39	13
	Tigrigna	13	4.33
	Tigrigna and English	3	1

Do you think your mobile usage has been increasing since you migrated	I don't know	5	1.67
	It is still the same	51	17
	No, it isn't (decreasing)	39	13
	Yes, it is increasing	205	68.33
How much money do you spend on mobile service in a week?	10-20 ETB (0.16-0.33 EUR)	128	42.67
	20-50 ETB (0.33-0.82 EUR)	62	20.67
	5-10 ETB (0.082-0.16 EUR)	88	29.33
	>50 ETB (>0.82 EUR)	22	7.33
The reasons for using smartphones during migration among migrants	Internet	215	71.67
	Call	296	98.67
	Game	56	18.67
	Picture	194	64.67
	Video	171	57
	Music	192	64
	Navigate place	35	11.67
	Working activity	1	0.33
Do you think using a mobile device is helpful for your current situation?	I don't know	48	16
	No, it doesn't help	38	12.67
	Yes, helpful (214)		

4.4 The increased usage of mobile phones after migration and migrants' characteristics

As mentioned above the majority of the participants said the mobile usage trend has increased since their displacement to current location 205 (68.33%). However, regarding the trend of smartphone usage since migration, the general characteristics of the participants are illustrated through correlation. Moreover, the age groups between 23 and 30 and between 30 and 35, as well as those residing in Addis Ababa and having a source of income, are significantly associated with an increase in smartphone usage (see Figure 43). These factors suggest that younger respondents who migrated to Addis Ababa (Capital city) with a stable income are more likely to experience a growth in their use of smartphones.

Figure 43 demonstrates that staying for more than one year, living outside of Addis Ababa, and being respondents who haven't pursued higher education all contribute to a significant decrease in the probability of an increase in smartphone usage.

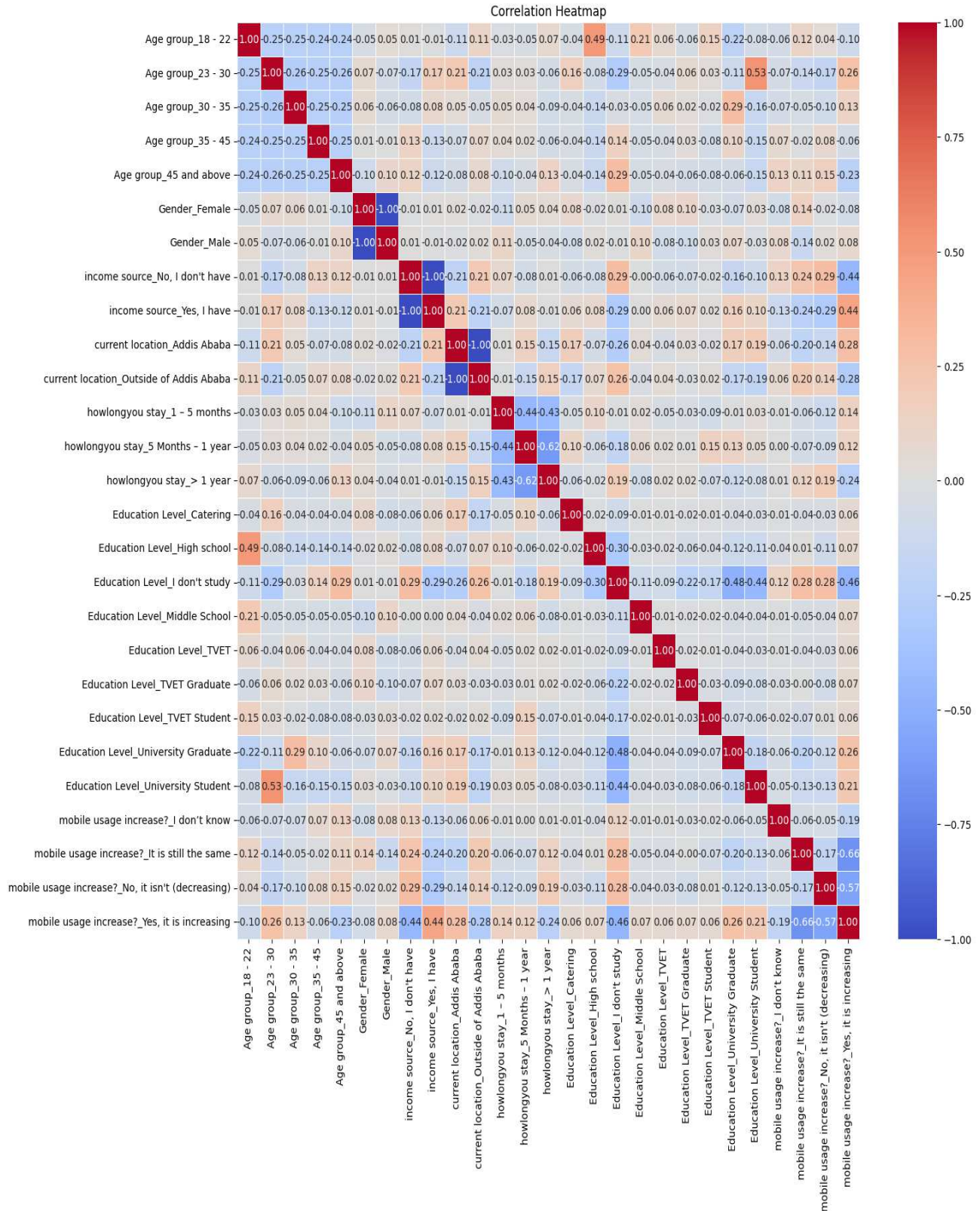


Figure 43 - Correlation between the trend of smartphone usage after migration and characteristics

4.5 The social media usage of the IDPs

Here, the IDP's general social media use is examined (i.e., the migrants' use of any social media service, including Facebook, YouTube, Instagram, Facebook, Imo, TikTok, Twitter, and WhatsApp) and separate use of each of these services. Above two-thirds, 216 (72%) of the participants have social media accounts (used at least one of the seven studied social media services).

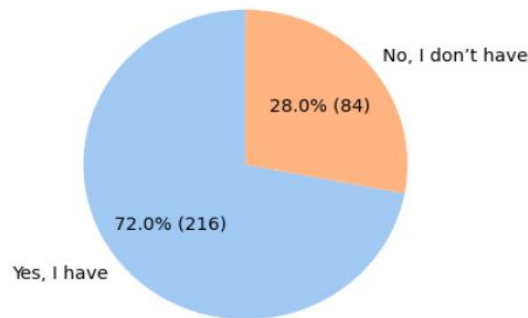


Figure 44 - having social media

The most commonly used social media applications in the study are Facebook 193, Telegram 166, WhatsApp 135, YouTube 113, Tiktok 111, and Instagram 54,

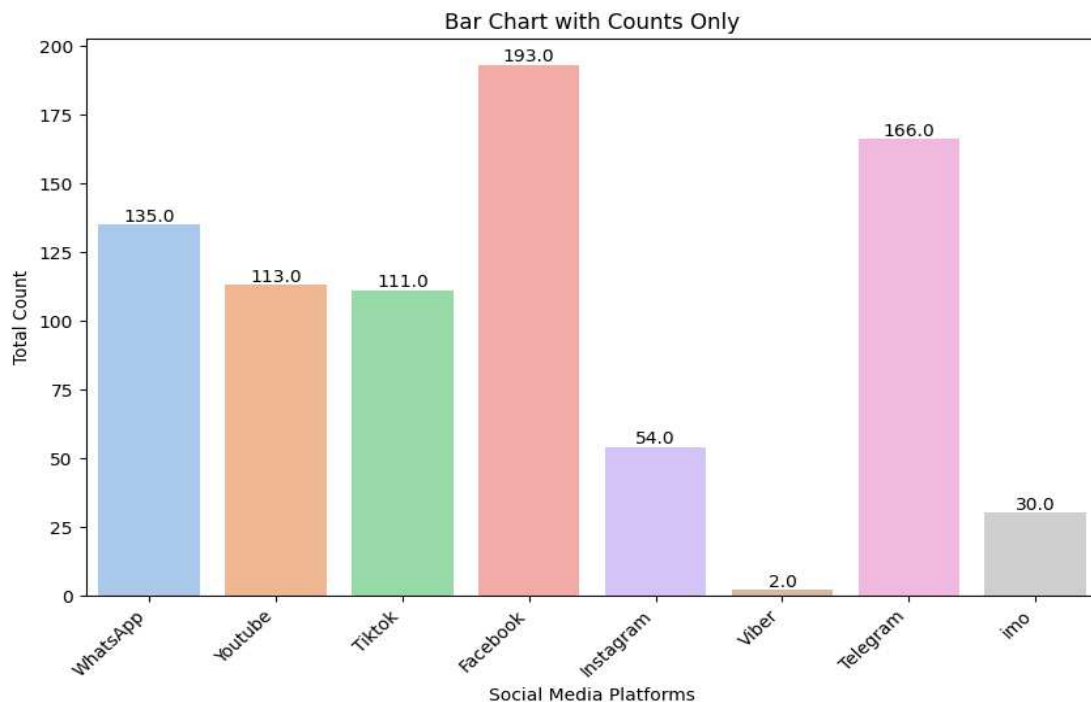


Figure 45 - Participants' social media platform preferences

In addition, the correlation in the figure down below (Figure 46) shows statistically significant differences between the ages of the IDPs and Genders with respect to the use of different social media platforms. The results show all age groups of the respondents have diverse preferences across platforms. However, their first two preferred social media platforms are Facebook and Telegram.

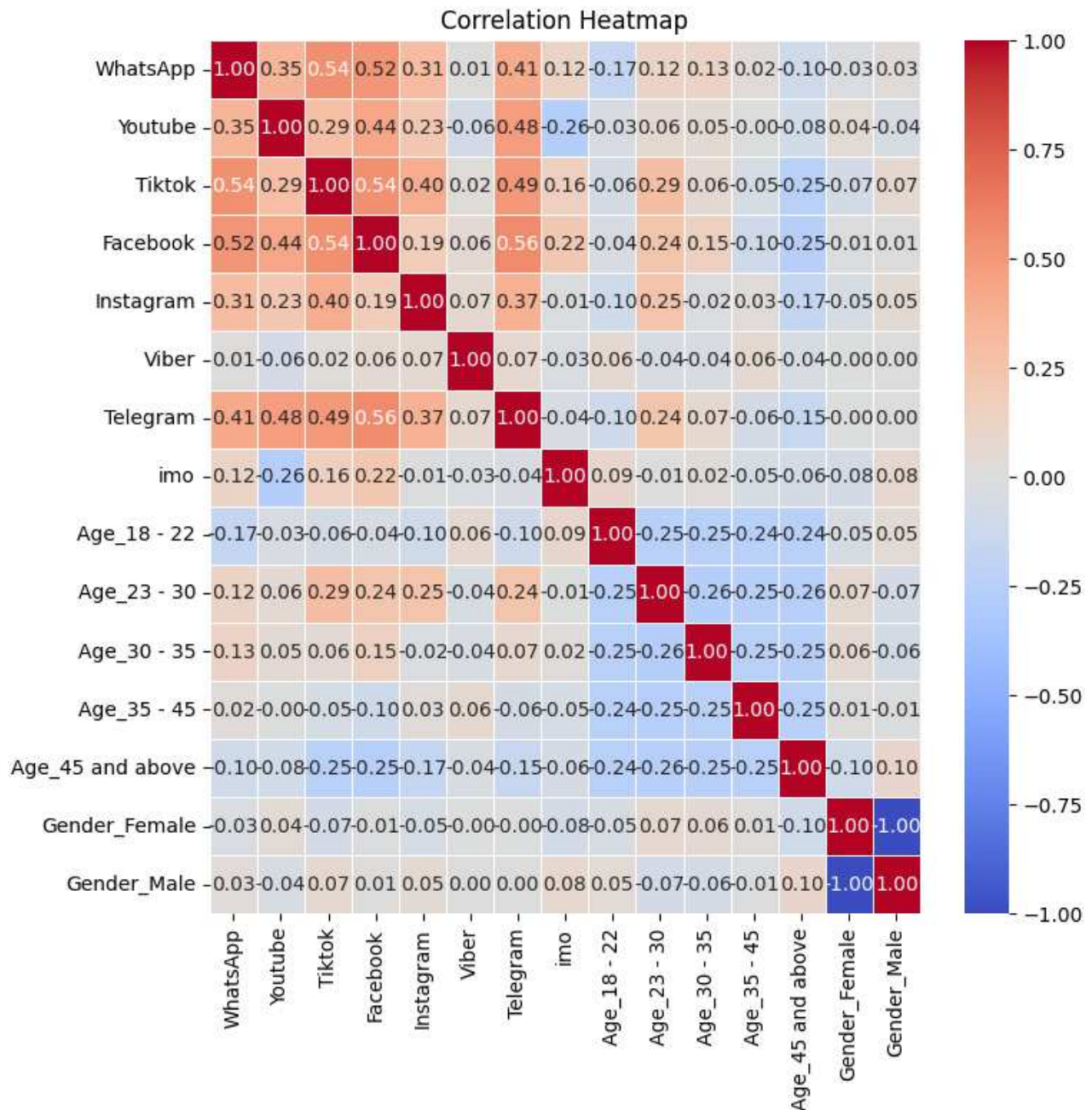


Figure 46 - Correlation between social media platforms and participants' Age and Gender

Moreover, the number of females and males using Facebook was nearly identical, indicating that there's no significant gender-based disparity in Facebook usage among the participants. This observation suggests that both genders have embraced social media platforms Facebook at similar rates, reflecting a balanced representation in online activity as shown in the Figure 47.

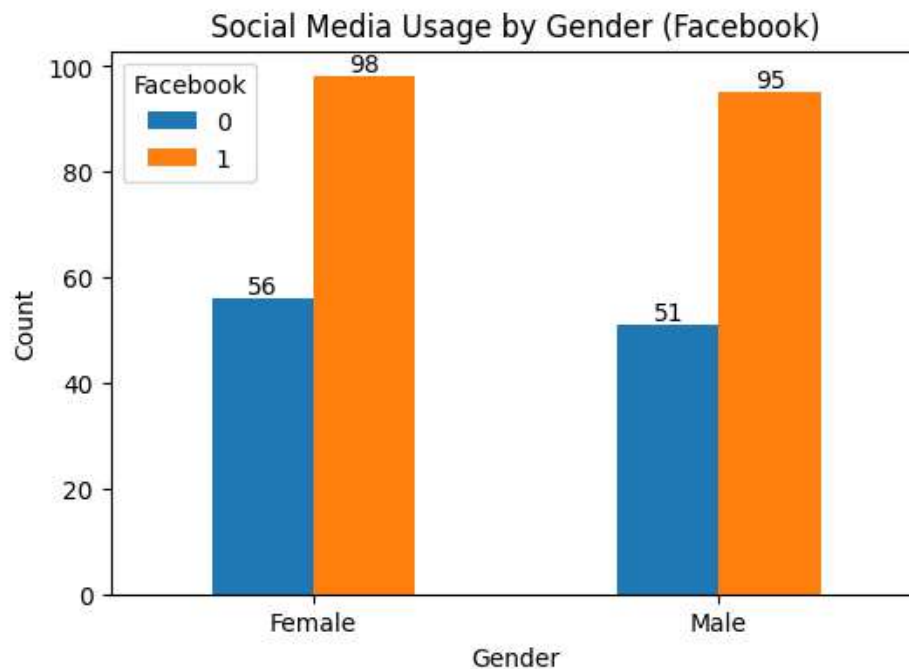


Figure 47 - Facebook usage among participants.

The correlation shows staying more than one year decreases the probability of using social media. Furthermore, the age group 23-20, having an income, university students, and university graduates are highly connected with social media use (Figure 48).

Those having a duration of stay greater than one year are less likely (less than 5 months of stay) to use at least one social media. Of those who don't have social media accounts 84 (28%), majority 24 (28.57%) reason that social media is not crucial for them (see Figure 49).

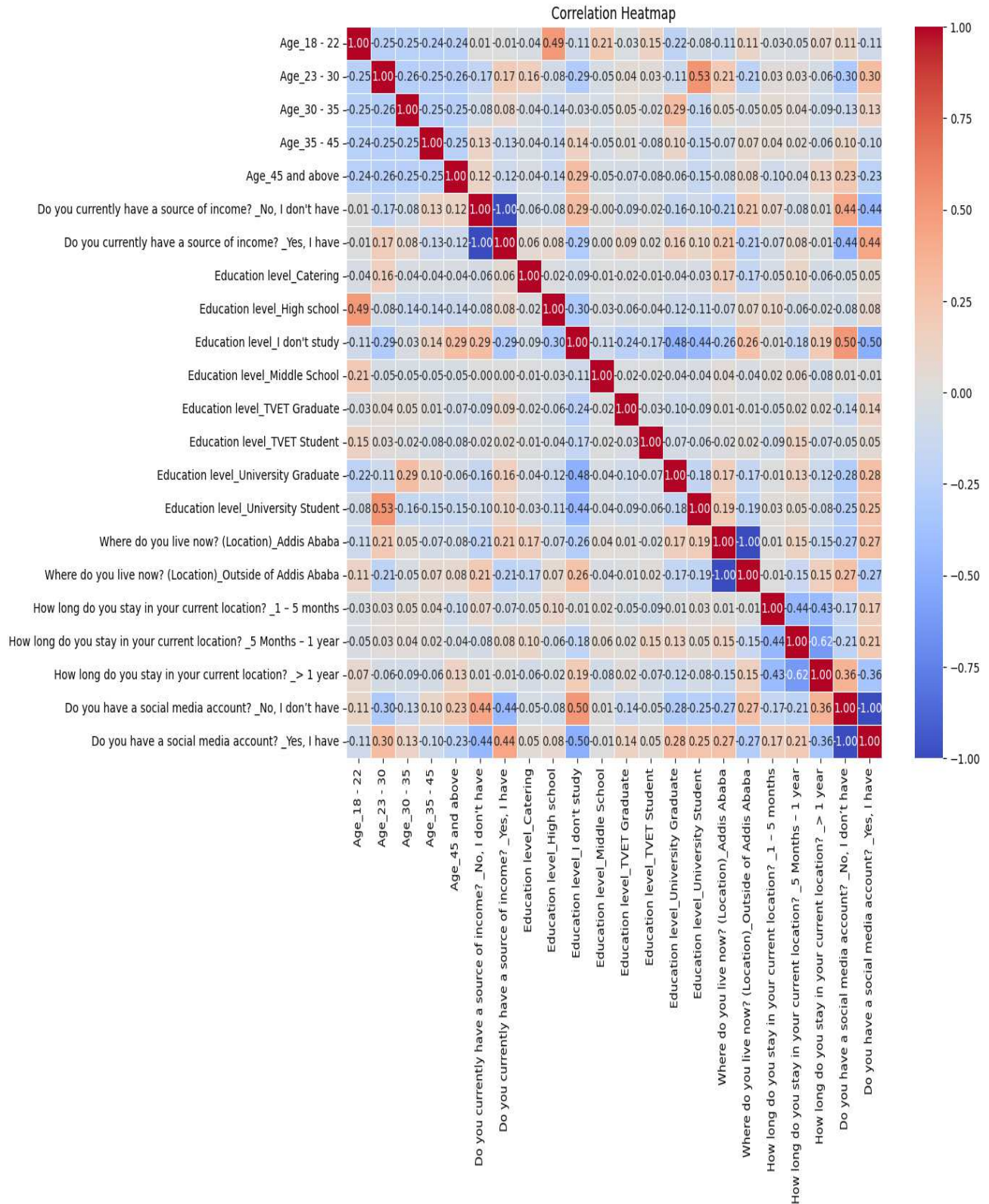


Figure 48 - Correlation between having social media and migrants' characteristics

```

reasons_counts = df['Reason for Not Having Social Media?'].value_counts()
total_count = reasons_counts.sum()
reasons_table = pd.DataFrame({
    'Reasons': reasons_counts.index,
    'Frequency': reasons_counts.values,
    'Percentage': (reasons_counts.values / total_count) * 100
})
education_table['Percentage'] = education_table['Percentage'].round(2)
print("Reasons Level Distribution (Frequency and Percentage):")
print(reasons_table)

```

Reasons Level Distribution (Frequency and Percentage):

	Reasons	Frequency	Percentage
0	I don't think it is crucial for me	24	28.571429
1	I can't afford the internet price;I don't thin...	22	26.190476
2	I don't think it is crucial for me;I don't kno...	16	19.047619
3	I don't know how to use it	11	13.095238
4	I can't afford the internet price;I don't know...	5	5.952381
5	I can't afford the internet price;I don't thin...	3	3.571429
6	I can't afford the internet price	3	3.571429

Figure 49 - Reasons given by participants for not having social media accounts

As shown in Figure 50, the main reasons for the participants to use social media are to meet people (208 participants), to get information (204 participants), and for entertainment (107 participants). These responses highlight the diverse motivations behind social media usage within the group surveyed.

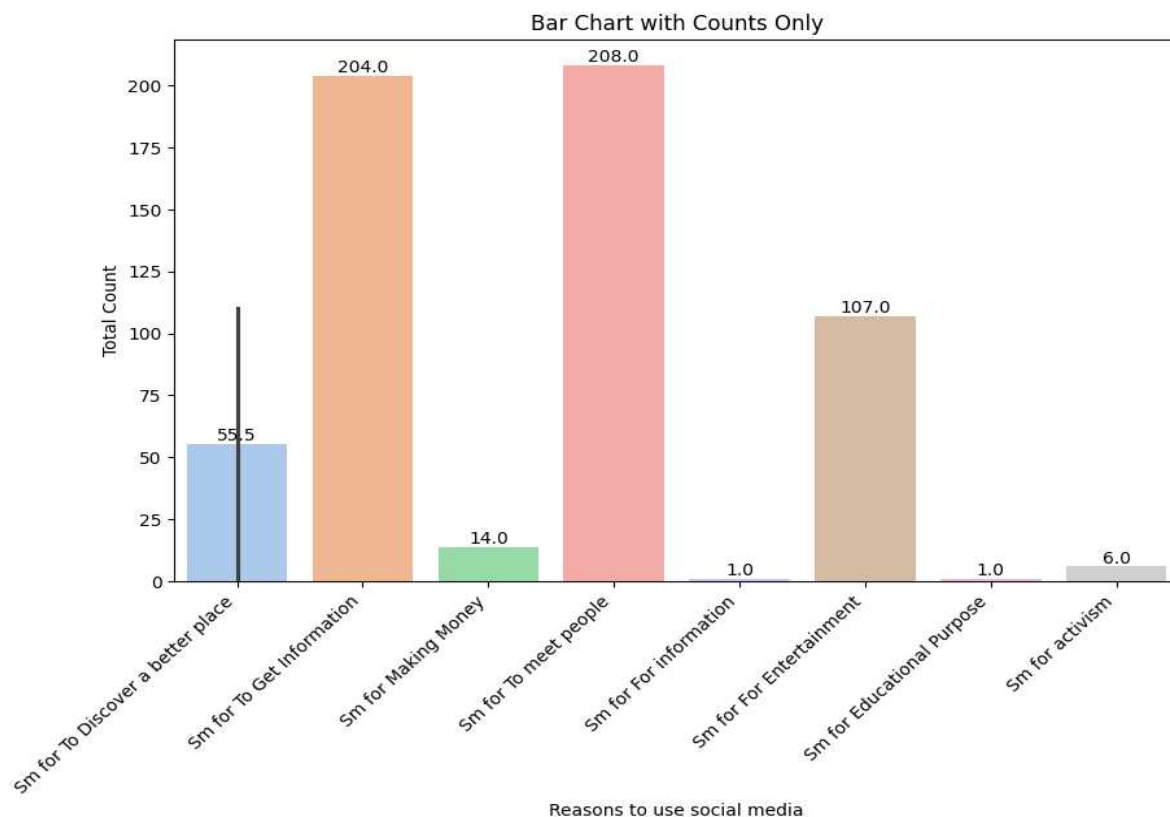


Figure 50 - Reasons participants use social media

The table below (Table 4) presents, in tabular format, all the analyses conducted above

Table 4 - Social media usage among IDPs

Do you have a social media account?	No, I don't have	84	28.00
	Yes, I have	216	72.00
If you don't have one, which one would be the reason? (more answers are possible)	I don't know how to use it	11	13.10
	I don't think it is crucial for me	24	28.57
	I can't afford the internet price	3	3.57
	I can't afford the internet price and I don't think it is crucial for me	3	2.38
	I can't afford the internet price and I don't know how to use it	5	3.57
	I don't think it is crucial for me and I don't know how to use it	16	19.05
	I can't afford the internet price and I don't think it is crucial for me and I don't know how to use it	22	26.19
The reasons for using social media	To meet friends, family	208	
	To entertainment	107	
	To get information	204	
	To discover places	55	
	To business purposes	14	
	To activism	36	
	To education purpose	1	
The most commonly used mobile applications.	Telegram	166	
	Facebook	193	
	YouTube	113	
	WhatsApp	135	
	Imo	30	
	Instagram	54	
	TikTok	111	

4.6 The attitude toward the usage of technology among migrants

The challenges of technology usage among migrants are ranked as follows: Internet connection problems, with 173 (57.67%) people agreeing and 68 (22.67%) strongly agreeing. This is followed by a shortage of electric power to charge phones, with 163 (54.3%) people agreeing and 71 (23.67%) strongly agreeing. Finally, the cost of devices, with 133 (44.3%) people agreeing and 60 (20%) strongly agreeing, is identified as another significant challenge.

Table 5 - Challenges of technology usage among migrants

Variables	Category	Freq.	Percent
The cost of the Device is not affordable	Agree	133	44.33
	Disagree	10	3.33
	Neutral	97	32.33
	Strongly Agree	60	20.00
Shortage of Electric power to charge the phone	Agree	163	54.33
	Disagree	4	1.33
	Neutral	62	20.67
	Strongly Agree	71	23.67
Internet connection problem	Agree	68	22.67
	Disagree	25	8.33
	Neutral	31	10.33
	Strongly Agree	173	57.67
	Strongly Disagree	3	1.00

4.7 Building the models

Training and building the models aim to predict the migrants' reasons for smartphone usage. Since the applications migrants use fall under one of the reasons for smartphone usage, which is 'For Internet Access,' the study will predict the most commonly used applications by migrants.

Building a model is necessary to provide important insights for well-informed decision-making. The selected models; Decision tree and Random forest, are robust, interpretable, and appropriate for use with categorical data and small datasets. They can also produce precise predictions.

Procedures for building models for predicting reasons for smartphone usage

The following procedures are followed to train the model to predict the reasons for smartphone usage.

Feature selection

The process of choosing important features (variables) from the original dataset for use in model training and prediction is known as feature selection. Figure 51 shows the the splitting process of dependent and independent variables into X and Y respectively.

```
✓ [13] X = df.loc[:, df.columns != 'mobile usage reason']  
0s y = df.loc[:, df.columns == 'mobile usage reason']
```

Figure 51 - Splitting the data set into X and Y

The dataset consists entirely of categorical data. Since the chi-squared test (chi2) is commonly used for feature selection with categorical data, It is implemented for selecting the important columns. the higher the value, the more significant the association between that column and the target variable.

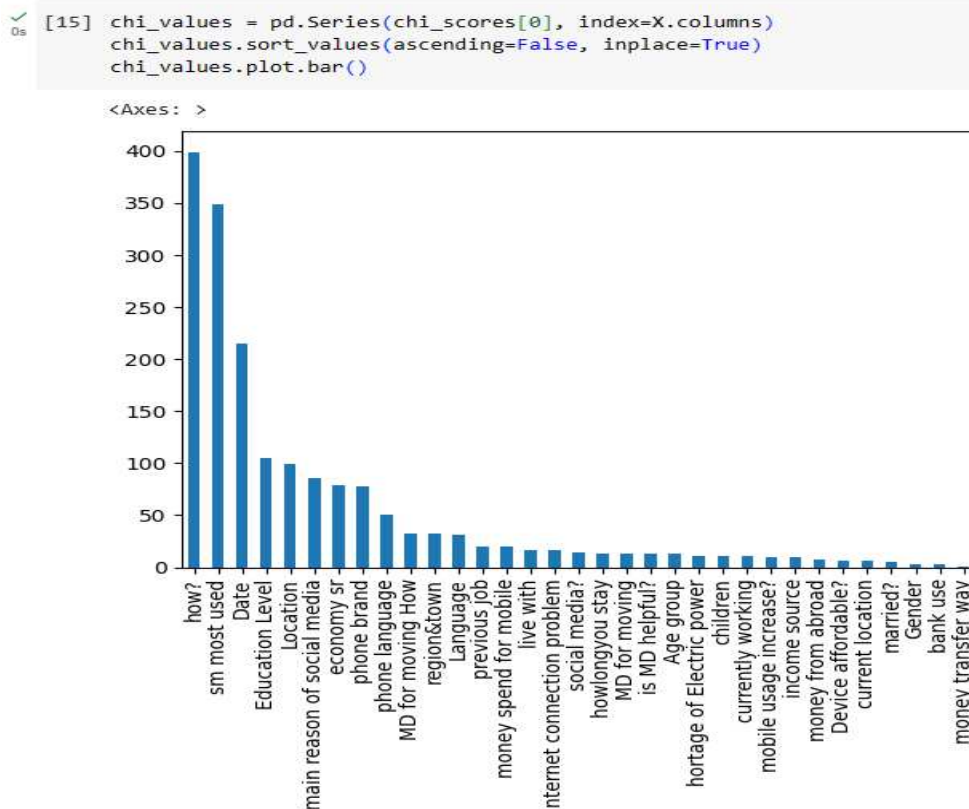


Figure 52 - Applying chi2 for feature selection

```

✓ [16] k = 10
0s

chi2_selector = SelectKBest(chi2, k=k)
X_new = chi2_selector.fit_transform(X, y)

selected_features = df.columns[:-1][chi2_selector.get_support()]
print("\nSelected Features:")
print(selected_features)

Selected Features:
Index(['Date', 'Location', 'Education Level', 'economy sr', 'phone brand',
      'phone language', 'sm most used', 'main reason of social media', 'how?',
      'MD for moving How'],
      dtype='object')

```

Figure 53 - Selected top 10 features using chi-square

Addressing the class imbalance

Before training the model, the other step that is followed is dealing with the class imbalance. In the first case, the target class has 8 classes are the the values of reasons why migrants use smartphones. From “Class 0” - to “Class 7” (For making regular calls and SMS, Internet use, Playing Music, Taking pictures, recording videos, gaming, navigating places, and for work).

Addressing class imbalance is a crucial process. because if it’s not fixed, the model will tend to predict the majority class, leading to misleading evaluation metrics. Figure 54 shows the Class imbalance in the target variable and after it is addressed through the oversampling technique. The distribution of classes is significantly skewed, one class has far more samples than the others.

By using over-sampling techniques, the representation of minority classes is increased to balance the distribution by duplicating the samples from the minority class.

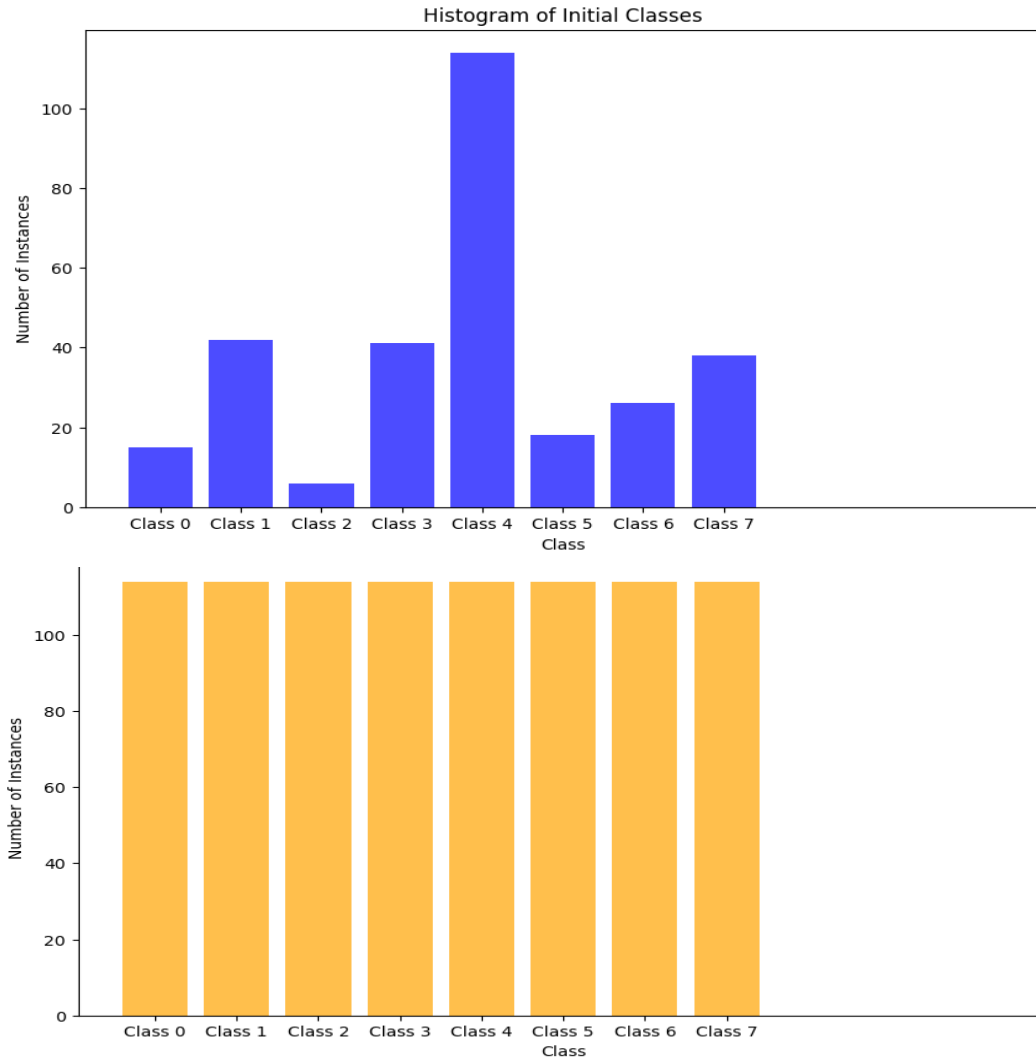


Figure 54 - Initial imbalanced classes and classes after applying the oversampling technique

Splitting into train and test sets

Figure 55 shows the splitting of the dataset into training and testing sets.

```
[20] from sklearn.model_selection import train_test_split
      X_train, X_test, y_train, y_test = train_test_split(X_resampled, y_resampled, test_size=0.33, random_state=42)
```

Figure 55 - Splitting into train and test sets

- **Decision tree**

After the decision tree model implementation on the prediction of the reason for social media usage, the model gives the following result.

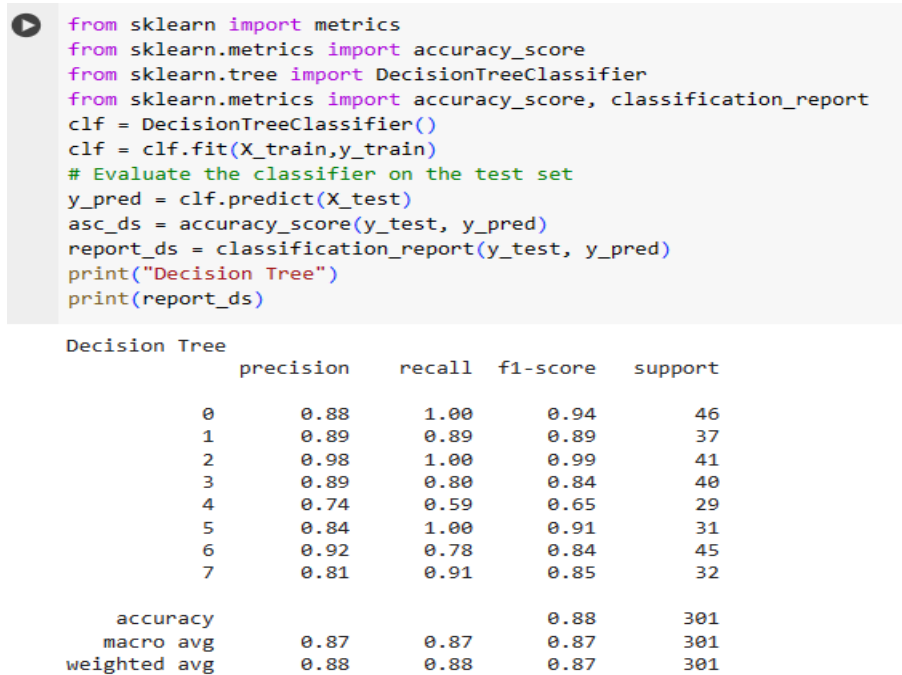


Figure 56 - Decision tree

After getting the score (88% accuracy score) using the decision tree model, to improve the result, GridSearchCV hyperparameter tuning is implemented.

The code snippet outlines the process of hyperparameter tuning for a decision tree classifier using GridSearchCV. However, the improvement step couldn't improve the result. It gives an 87% accuracy score as shown in Figure 57.

```

from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score

param_grid = {
    'clf__criterion': ['gini', 'entropy'],
    'clf__max_depth': [None, 10, 20],
    'clf__min_samples_split': [2, 5, 10],
    'clf__min_samples_leaf': [1, 2, 4],
    'clf__max_features': [None, 'sqrt', 'log2']
}

pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('clf', DecisionTreeClassifier(random_state=42))
])

grid_search = GridSearchCV(pipeline, param_grid, cv=5, scoring='accuracy', n_jobs=-1)
grid_search.fit(X_train, y_train)
best_params = grid_search.best_params_
print("Best Hyperparameters:", best_params)
best_clf = grid_search.best_estimator_
y_pred = best_clf.predict(X_test)
test_accuracy = accuracy_score(y_test, y_pred)
print("Test Accuracy:", test_accuracy)

Best Hyperparameters: {'clf__criterion': 'gini', 'clf__max_depth': None, 'clf__max_features': 'sqrt', 'clf__min_samples_leaf': 1, 'clf__min_samples_split': 2}
Test Accuracy: 0.8676470588235294

```

Figure 57 - Hyperparameter tuning on decision tree

▪ Random forest

To get an improved result, the Random forest model is implemented as shown in Figure 58.

```

[80] from sklearn.metrics import classification_report
rf_classifier = RandomForestClassifier(max_depth=68, random_state=83)
rf_classifier.fit(X_train, y_train)
y_train_pred = rf_classifier.predict(X_train)
y_test_pred = rf_classifier.predict(X_test)
train_accuracy = accuracy_score(y_train, y_train_pred)
test_accuracy_rand = accuracy_score(y_test, y_test_pred)
print("Accuracy on Train Set:", train_accuracy)
print("Accuracy on Test Set:", test_accuracy_rand)
print("\nClassification Report:")
print(classification_report(y_test, y_test_pred))

```

Accuracy on Train Set: 1.0
Accuracy on Test Set: 0.8823529411764706

Classification Report:				
	precision	recall	f1-score	support
0	0.82	0.67	0.74	21
1	0.89	1.00	0.94	17
2	0.93	0.76	0.84	17
3	0.91	1.00	0.95	10
4	0.94	1.00	0.97	16
5	0.76	0.76	0.76	17
6	0.95	0.95	0.95	22
7	0.84	1.00	0.91	16
accuracy			0.88	136
macro avg	0.88	0.89	0.88	136
weighted avg	0.88	0.88	0.88	136

Figure 58 - Random forest prediction

After implementing Random forest model, the model gives 88 % of accuracy score. To improve the accuracy score, hyperparameter tuning is done. The following steps and Figure 59 shows the steps followed for this process.

```
[78] from sklearn.model_selection import GridSearchCV
param_grid = {
    'n_estimators': [100, 200, 300],
    'max_depth': [10, 50, 100, None],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4],
    'max_features': ['auto', 'sqrt', 'log2']
}
rf_classifier = RandomForestClassifier(random_state=83)

grid_search = GridSearchCV(estimator=rf_classifier,
                           param_grid=param_grid,
                           cv=5,
                           scoring='accuracy',
                           n_jobs=-1,
                           verbose=2)

grid_search.fit(X_train, y_train)
best_params = grid_search.best_params_
print("Best parameters found:", best_params)
best_rf_classifier = grid_search.best_estimator_
y_test_pred = best_rf_classifier.predict(X_test)
test_accuracy1 = accuracy_score(y_test, y_test_pred)
print("Test accuracy with tuned Random Forest:", test_accuracy1)
```

Fitting 5 folds for each of 324 candidates, totalling 1620 fits
Best parameters found: {'max_depth': 50, 'max_features': 'auto', 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 100}
Test accuracy with tuned Random Forest: 0.8823529411764706

Figure 59 - Hyperparameter tuning

As Figure 59 shows, after applying hyperparameter tuning, the accuracy of the the model is still the same.

The comparison between the implemented models. As shown in Figure 60, both models give the same accuracy scores.

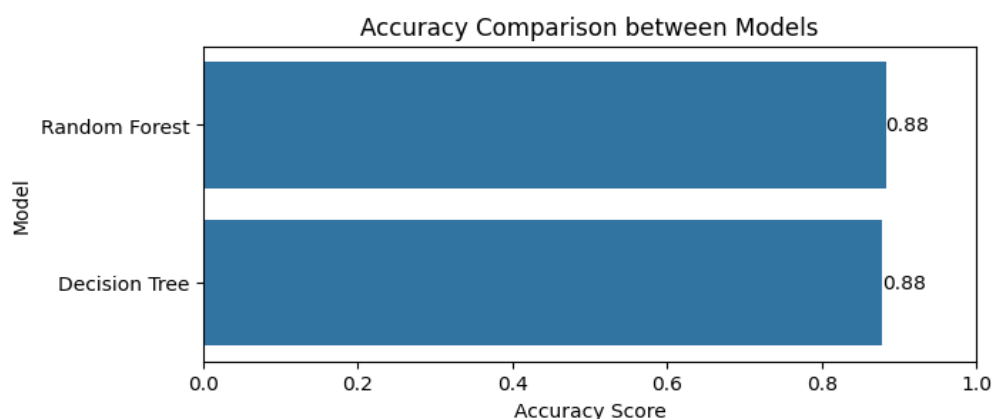


Figure 60 - Comparison between the models

Procedures for building a model and predicting for mostly used applications

The following processes steps for building a model to predict mostly used applications among migrants. The steps followed in this process are the same as the above steps that are followed for the reason of smartphone usage prediction.

Feature Selection

Figure 61 shows the feature selection process.

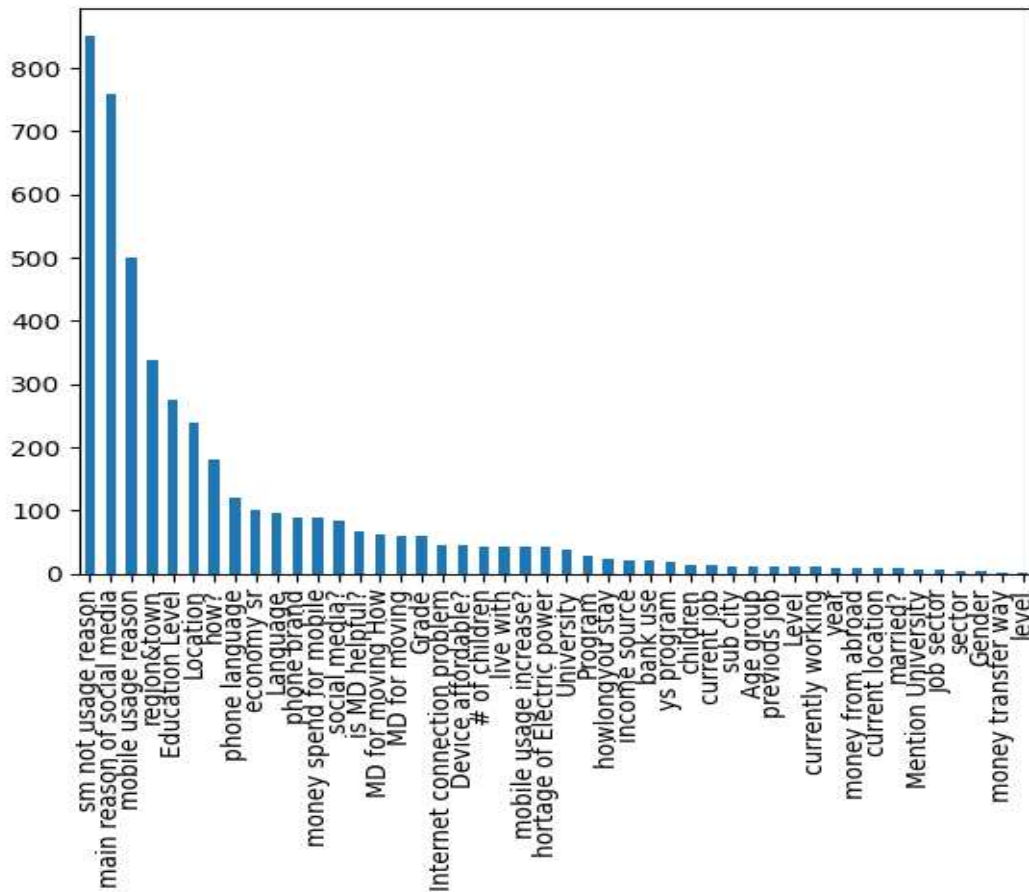


Figure 61 - Feature selection for mostly used application

Figure 62 shows the selection process of the top 10 relevant features by using chi-square.

```
43] k = 15

chi2_selector1 = SelectKBest(chi2, k=k)
X_neww = chi2_selector1.fit_transform(X, y)

selected_features1 = df.columns[:-1][chi2_selector1.get_support()]
print("\nSelected Features:")
print(selected_features1)

Selected Features:
Index(['Language', 'Location', 'Education Level', 'region&town', 'economy sr',
      'phone brand', 'mobile usage reason', 'phone language', 'social media?',
      'sm not usage reason', 'main reason of social media',
      'money spend for mobile', 'is MD helpful?', 'how?',
      'MD for moving How'],
      dtype='object')
```

Figure 62 - select the top 10 features

Handling class imbalance

Figure 63 shows an imbalanced class and after the class imbalance is handled. The classes here are the applications used by migrants.

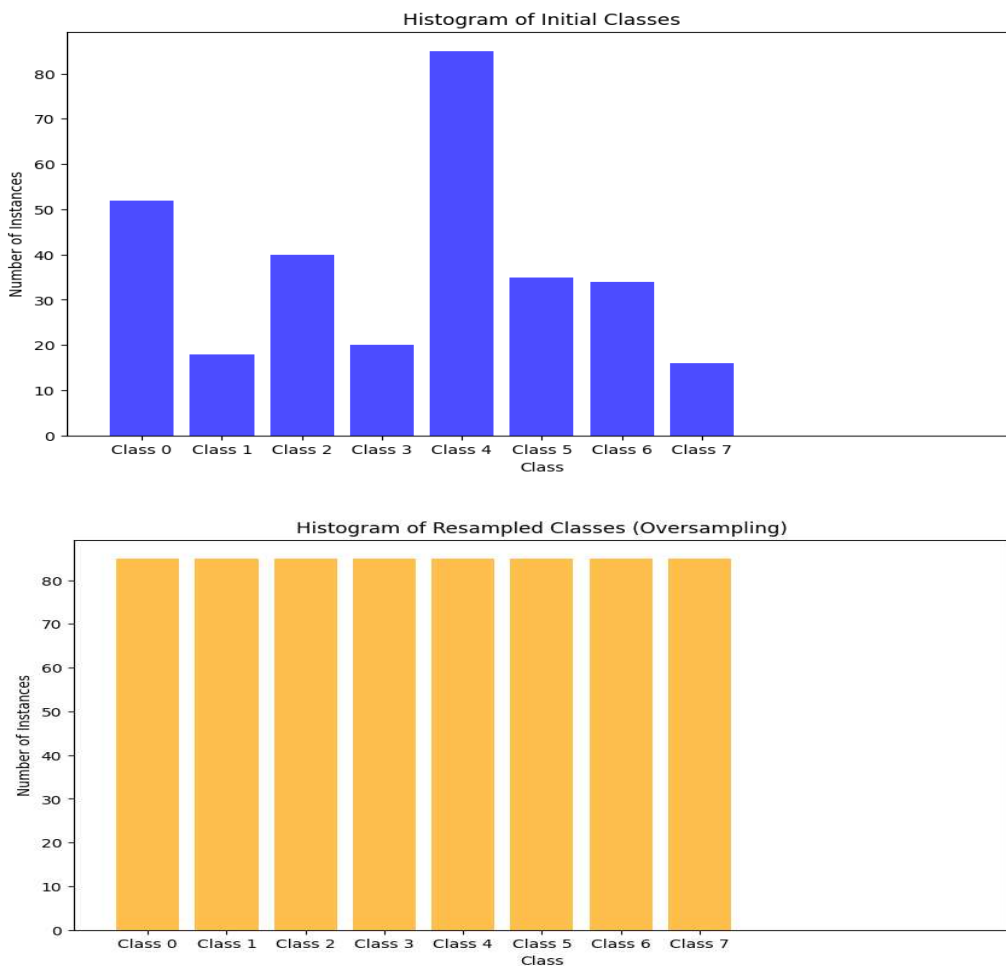


Figure 63 – Oversampling technique

- **Random forest**

The code snippet in Figure 64 shows the process of training and testing the accuracy of the model.

```
[50] rf_classifier = RandomForestClassifier(max_depth=68, random_state=83)
      rf_classifier.fit(X_train, y_train)
      y_train_pred = rf_classifier.predict(X_train)
      y_test_pred = rf_classifier.predict(X_test)
```

```
▶ train_accuracy = accuracy_score(y_train, y_train_pred)
  print("Accuracy on Train Set:", train_accuracy)
  test_accuracy = accuracy_score(y_test, y_test_pred)
  print("Accuracy on Test Set:", test_accuracy)
```

```
Accuracy on Train Set: 1.0
Accuracy on Test Set: 0.8823529411764706
```

Figure 64 - Random forest model

After implementing the random forest model, it gives an 88% accuracy score. To improve the result hyperparameter tuning is implemented as shown in Figure 65. However, it gives the same result.

```
✓ [53] from sklearn.model_selection import GridSearchCV
11m
      param_grid = {
          'n_estimators': [100, 200, 300],
          'max_depth': [10, 50, 100, None],
          'min_samples_split': [2, 5, 10],
          'min_samples_leaf': [1, 2, 4],
          'max_features': ['auto', 'sqrt', 'log2']
      }
      rf_classifier = RandomForestClassifier(random_state=83)

      grid_search = GridSearchCV(estimator=rf_classifier,
                                param_grid=param_grid,
                                cv=5,
                                scoring='accuracy',
                                n_jobs=-1,
                                verbose=2)

      grid_search.fit(X_train, y_train)
      best_params = grid_search.best_params_
      print("Best parameters found:", best_params)
      best_rf_classifier = grid_search.best_estimator_
      y_test_pred = best_rf_classifier.predict(X_test)
      test_accuracy = accuracy_score(y_test, y_test_pred)
      print("Test accuracy with tuned Random Forest:", test_accuracy)

      Fitting 5 folds for each of 324 candidates, totalling 1620 fits
      Best parameters found: {'max_depth': 50, 'max_features': 'auto', 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_es
      Test accuracy with tuned Random Forest: 0.8823529411764706
```

Figure 65 - Hyperparameter tuning on random forest model

- **Decision tree**

Figure 66 shows the decision tree model implementation process and the accuracy score. The model gives an 89 % accuracy score.

```
from sklearn.tree import DecisionTreeClassifier
clf = DecisionTreeClassifier()
clf = clf.fit(X_train,y_train)
y_pred = clf.predict(X_test)

[61] from sklearn import metrics
from sklearn.metrics import accuracy_score
print("Accuracy:", metrics.accuracy_score(y_test, y_pred))

Accuracy: 0.8897058823529411
```

Figure 66 - Decision tree prediction

As shown in Figure 67 hyperparameter tuning is implemented in this model to improve the accuracy of the model. But the model gives less value.

```
[ ] from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score

param_grid = {
    'clf__criterion': ['gini', 'entropy'],
    'clf__max_depth': [None, 10, 20],
    'clf__min_samples_split': [2, 5, 10],
    'clf__min_samples_leaf': [1, 2, 4],
    'clf__max_features': [None, 'sqrt', 'log2']
}

pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('clf', DecisionTreeClassifier(random_state=42))
])

grid_search = GridSearchCV(pipeline, param_grid, cv=5, scoring='accuracy', n_jobs=-1)
grid_search.fit(X_train, y_train)
best_params = grid_search.best_params_
print("Best Hyperparameters:", best_params)
best_clf = grid_search.best_estimator_
y_pred = best_clf.predict(X_test)
test_accuracy = accuracy_score(y_test, y_pred)
print("Test Accuracy:", test_accuracy)

Best Hyperparameters: {'clf__criterion': 'gini', 'clf__max_depth': None, 'clf__max_features': 'sqrt', 'clf__min_samples_leaf': 1, 'clf__min_samples_split': 2}
Test Accuracy: 0.8676470588235294
```

Figure 67 - Hyperparameter tuning on decision tree

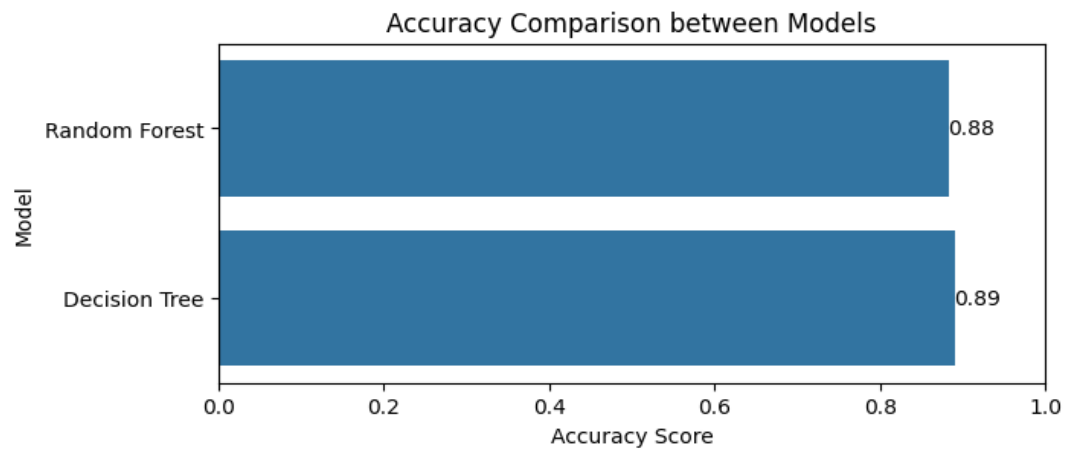


Figure 68 - Implemented models comparison (2)

5 Discussion

The age seems equally distributed among groups of interviewees which is somewhat high between 23 to 30 (20.67%). This age range often encapsulates individuals in their prime working years, highlighting the impact of displacement on a significant segment of the population. Most of the participants were females 154 (51.33%). The predominance of female participants in the study sheds light on the gender dynamics of displacement. This finding underscores the importance of considering gender-specific needs and vulnerabilities in the context of internal displacement. The educational levels of the study were majorly shared by never educated 162 (54%).

The high prevalence of individuals with no formal education (54%) among the participants indicates a potential lack of access to education and highlights the challenges faced by the displaced population in terms of educational attainment and skills development. The majority of interviews (80.7%) conducted outside of Addis Ababa and the remaining (19.3%) interviews conducted in Addis Ababa point to the widespread impact of the conflict, extending beyond the conflict area. The conflict and war in the northern part of Ethiopia have unleashed a humanitarian crisis, leading to a significant displacement of people. As UNHCR (2021) report shows Internally Displaced Persons (IDPs) are forced to flee their homes, seeking safety and shelter in other regions of the country, including the bustling capital of Addis Ababa.

This distribution underscores the need to address displacement-related issues in both rural and urban settings. The findings on marital status (48.33% married) and parenthood (45.17% with children) provide insights into the family structures of the displaced population. Understanding the family dynamics and responsibilities can inform targeted support and assistance programs. Demographic insights can guide policymakers, humanitarian organizations, and aid agencies in tailoring interventions that address the specific needs and challenges faced by different subgroups within the displaced population. Understanding the distribution of key demographics such as age, gender, education, and family composition is essential for designing effective and inclusive strategies to support the well-being and resilience of displaced individuals and families.

5.1 Reasons for migrants to use smartphone

Mobile phones have become essential tools for networking, communication, and entertainment in today's digital era. This study implements Python to analyze the specific uses of mobile phones among migrants in northern Ethiopia. The results show that the main purposes for which migrants use their phones are making calls and sending SMS (296 instances), accessing the internet (215 instances), taking pictures (194 instances), and engaging in entertainment activities like listening to music (192 instances). These purposes demonstrate the significance of mobile technology in assisting migrants with information access, and communication during their journeys.

This finding is supported by a study done by Frouws et al. (2016) offered a typology of how ICT is used along mixed migration routes, these uses included obtaining updated information about their journey, reaching out to smugglers, getting data on the target destination places, using them

to facilitate safety and rescue on route as well as providing a platform for migrant and refugee voices.

Even in a world where internet-based messaging apps are becoming more and more prevalent, the overwhelming preference for calls and SMS shows how important conventional modes of communication are. This could be because people are more or less comfortable with traditional techniques. After all, real-time communication is necessary, because different people have different amounts of internet accessibility.

With 215 people using mobile phones to access the internet, it's easy to see that these smartphones serve as entry points to a wider range of social media, online services, and information. Internet connectivity is essential for migrants to entertain, maintain connections with support systems, and keep informed.

The article by Gerlich et al. (2015) agrees with the result. It mentions how smartphones are essential and also make it easier to socialize both formally and informally, enjoy games and other forms of entertainment, communicate via messengers and social media, locate using maps, and gather information from social media and news media applications.

Another supporting research by Fedorova & Potemkin (2020) shows how migrants integrate and adjust to their new environment while maintaining connections to their families with technologies like the Internet and video cameras. The study conducted on Syrian migrants indicates that using smartphones is crucial for refugees to stay in touch with friends and family back home, get advice from relatives who were already residing in Sweden, and interact with people they met while traveling across borders to reach Europe (Dekker et al., 2018). As Dekker's (2018) study mentions, smartphones became widely used and important during the migration process, as they provide migrants with tools such as video cameras, and internet connections, enabling them to make smart and careful decisions.

Out of 300 participants, 214 (71.33%) think using a mobile device is helpful for their current situation and the majority 186 (62.00%) were helped by their mobile phones to move to their current place. This finding agrees with different studies that have shown how online media provide an infrastructure that extends the availability of network ties and information for prospective migrants (Burrell & Anderson, 2008; Dekker & Engbersen, 2014; Hiller & Franz, 2004). Syrian migrants who are living in the Netherlands, as studied by Alencar et al. (2019), explained that they only felt somewhat safe at least when they could contact family and other help during stressful and dangerous situations. In certain cases, the smartphone was helpful and a lifesaver in potentially dangerous situations.

The findings underscore the critical role of mobile phones as essential tools for migrants, providing access to information, communication, navigation, and assistance in various aspects of the migration journey. This study's finding is supported by a study done by many researchers on how

ICT is used along mixed migration routes, these uses included obtaining updated route information, making contact with traffickers, getting information about the target countries, utilizing them to assist in safety and rescue while traveling, and providing a platform for the voices of migrants and refugees and their use as information sources (Ennaji, 2021; Iqbal et al., 2019; Stremlau & Tsalapatanis, 2022).

The increased mobile phone usage among migrants indicates the reliance on mobile technology for communication, information access, and possibly even for social integration in their home destiny.

In this study's findings, challenges of technology usage among migrants are ranked as follows: Internet connection problems, with 173 (57.67%) people agreeing and 68 (22.67%) strongly agreeing. This is followed by a shortage of electric power to charge the phone, with 163 (54.3%) people agreeing and 71 (23.67%) strongly agreeing. Finally, the cost of devices, with 133 (44.3%) people agreeing and 60 (20%) strongly agreeing, is identified as another significant challenge. As the World Economic Forum (2024) mentions, that in Ethiopia, small towns have better internet connections than big cities. The finding shows that migrants face challenges in different ways. Supportive study by Turner & Gülerce (2021) shows Syrians seeking asylum in Turkey mentioning that they are struggling due to their economic status to afford media tools like mobile phones and stable internet access.

5.2 Most commonly used mobile applications

The most commonly used mobile applications in this study are Facebook 193 (64.33%), Telegram 166 (55.33%), and WhatsApp 135 (37.3%), indicating the popularity of these platforms for communication and information sharing during migration. Additionally, 160 of the respondents use mobile banking applications. The results indicate that the respondents primarily utilize social media and mobile banking applications. Specifically, they use social media platforms primarily for connecting with friends and family (208), obtaining information (204), exploring new places (110), entertainment (107), and engaging in activism purposes (36).

The study conducted by Sanchez et al. (2018) comprised a social media and social network analysis of 21 people: 11 from Bangladesh and 10 from Nigeria, where Nigerian immigrants live in the Italian region of Apulia, and Bangladeshis reside in the cities of Como and Ventimiglia. Every person in the sample uses Facebook frequently. Many migrants prefer Facebook; therefore, it's important to pay closer attention to how pages and profiles are being used (Stremlau & Tsalapatanis, 2022).

However, the finding is opposed by a report done by the European Commission (2017), which shows Facebook continues to be the most significant social media source for educating migrants on their journey and destination, but the report's analysis of the frequency of usage of these platforms revealed that it is still infrequently utilized by migrants.

Facebook data help to address the lack of statistics regarding the global mobility of migrant groups by providing opportunities to gather fresh perspectives on the sociodemographic and behavioral attributes of the immigrant community segmented by age, gender, country of origin, and location, regardless of its limitations (Spyratos et al., 2020).

Another piece of evidence from Syrian Youth migrants who are living in Canada mentioned the importance of social media, including Facebook and WhatsApp, in enabling them to maintain relationships with friends and relatives who have left Syria or who have been displaced and relocated abroad. Some explain that they utilize it to learn about Canadian culture and to improve their English (Veronis et al., 2018).

In this study 165 use mobile phones to receive or send money to others. This indicates the importance of mobile financial services for migrant communities, offering convenience and accessibility for sending money. As shown by Baldé (2009), mobile money transfer services are becoming increasingly popular as alternatives to traditional banking systems for both domestic and international transactions.

Mobile-based financial solutions are becoming more and more accessible and practical. Mobile banking has changed the banking industry in Ethiopia by providing vital financial services to those who don't have access, despite the country's weak traditional banking system (Fernandes Malaquias & Frances Silva, 2020).

An important step towards Ethiopia's financial inclusion and the development of digital financial services was achieved when Safaricom introduced its mobile money service in Ethiopia. They have joined the ranks of the leading telco-led mobile money provider, alongside the state-owned Telebirr, a division of Ethio Telecom (Daniele Tricarico, 2023). The practical implications of this finding are mobile phone money transfer services play a crucial role in enabling financial inclusion for migrants, allowing them to send money securely and conveniently to their families.

The practical implications of this finding are mobile phone money transfer services play a crucial role in enabling financial inclusion for migrants, allowing them to securely and conveniently send money to their families. The increased mobile phone usage among migrants indicates the reliance on mobile technology for communication, information access, and possibly even for social integration in their home destiny. Theoretical implications are the findings might contribute to the existing literature on the impact of mobile technology on migrant communities, shedding light on how mobile phones serve as tools for financial empowerment and connectivity.

The widespread use of social media facilitated the rapid sharing of information, enabling individuals to organize protests, mobilize supporters, and raise awareness about social and political issues. The study by Kassem Chebib & Minatullah Sohail (2011) illustrates that people extensively utilize social media platforms for sharing pictures and videos, a phenomenon that has significantly contributed to the Arab Spring.

Above two-thirds, 216 (72%) of the participants have social media accounts (used at least one of the seven studied social media services) demonstrating a significant presence on various platforms. According to recent surveys, online media is becoming more and more popular among non-migrants in the countries of origin and migrants in Western Europe for cross-border contact (Dekker & Engbersen, 2014). Migrants search for support or information on migration from social media while still residing in their homeland, during the migration process, and after they have moved.

Furthermore, those having a duration of stay greater than one year 92 % are less likely (less than 5 months of stay) to use at least one social media. According to Dekker & Engbersen (2014) study, social media usage varied depending on their age, location of residence, and other reasons.

Having an income is significantly connected with social media use and to use social media compared to not having an income source. Socioeconomic factors play a crucial role in shaping technology adoption and usage patterns within migrant communities. This finding is supported by a study conducted by Obi et al. (2020), which demonstrates that the usage of smartphones is highly influenced by the income levels of Nigerian migrants.

Of those who don't have social media accounts 84 (28%) use them, the 24 (28.57%) reason that social media is not crucial for them. The duration of stay, age group, and income source significantly impact social media usage among IDPs. Longer stays and certain age brackets influence the probability of engaging with social media platforms while having a source of income increases the likelihood of social media use.

The high adoption of social media among IDPs underscores the role of digital communication in facilitating social connections, access to information, and support networks during the migration journey. Understanding age, gender, and duration of stay influences on social media use can inform targeted strategies for enhancing digital literacy, communication support, and tailored information dissemination among migrant populations.

5.3 Patterns of smartphone usage

The patterns of smartphone usage among different characteristics of migrants show the majority of the participants' trend of using mobile has increased since their displacement to their current location. However, the analysis conducted by Merisalo & Jauhiainen (2020), contradicts the study's findings, African responders utilized the internet less when traveling than they did back home. This discrepancy might be due to the among migrants who cross the boundary fear about tracking and, smugglers but in this study's case, internally displaced rather than crossed the boundary. The study by Steel et al. (2017) shows Middle Eastern migrants (68%) were more likely than African migrants (55%) to own a smartphone.

The study by Jauhiainen & Tedeschi (2021) shows after migrants depart their country, there is a rise in smartphone ownership and use. The research mentions that displaced Syrians in Turkey also had higher smartphone ownership and better internet access after leaving their home country.

However, concerning Smartphone usage increase, the analysis shows staying more than one year are less likely to have a smartphone usage increment than those who stay less than 5 months of stay. Having an income source is more likely to increase Smartphone usage compared to not having an income source.

All age groups of the respondents have diverse preferences across platforms. However, their first two preferred social media platforms are Facebook and Telegram. This finding is supported by a study done by (Merisalo & Jauhiainen, 2020). The use of social media accounts showed statistically significant differences between the ages of the IDPs to the use of all studied social media platforms. These results support the hypothesis that the majority of migrants use social media, but age, location, and sex-specific differences can influence preferences for social media platforms.

The study conducted by Andone et al. (2016) confirms that younger people are utilizing social and communication applications more frequently due to increased communication. Additionally, compared to the general population, they play more games while Elderly individuals frequently use their smartphones only as a means of accessing information because they have different demands.

Almost a close number of females (50.78%) and males (49.22%) are using Facebook and there is no significant difference between them. However, in comparison to men, a larger share of women use the studied social media platforms except Instagram and WhatsApp. This finding could indicate that women IDPs have particular needs, interests, or social networks that are facilitated or expressed through social media platforms.

The study by Andone et al. (2016), which involves 16,147 male and 14,523 female participants, supports this study's findings. The study shows that social media usage is higher among females than males. Whereas Jauhiainen et al. (2022) study on Syrian migrants in Turkey contradicts the findings. It demonstrates that women use the internet less frequently than males. Furthermore, this study's statistics show that 72% of both men and women use at least one of the studied social media services.

When they receive money from abroad, the majority of this study's respondents 143 (47.67%) don't have experiences, but the rest have different preferences. Specifically, 117 (39%) prefer to use a legal money transfer system, while 38 (12.67%) prefer the black market or illegal way. This data indicates that a significant number of survey participants had no experience with receiving money from abroad. However, there are a variety of preferences among people who have experience. Many prefer to use legal money transfer systems, show trust in formal channels, and

adhere to legal processes. However, a small but significant portion favors the illegal or black market channels.

The study by Geda (2011) shows in Ethiopia the competition from back-market remittance service providers (RSP) is an ongoing challenge for many remittance service providers in the formal sector. This competition arises because the black market provides foreign exchange rates at a better rate and transfer services at a cheaper cost.

5.4 Prediction results of the implemented models

Random forest and decision tree are implemented to predict reasons for smartphone usage and mostly used applications. The quantity and quality of the data affect the model's performance. and models for small datasets such as decision trees and random forests are suitable (Xu et al., 2023).

For predicting reasons for smartphone usage, both models yield the same result. Despite implementing GridsearchCV hyperparameter tuning, the outcome couldn't be improved. The decision tree model performed better than the random forest model, as a prediction on the preferred applications, and migrants' is improved.

When we talk about migration modeling and forecasting research, we're talking about trying to predict large-scale patterns with respect to migration and doing it from the perspective of humanitarian aid and controlling migration. Pham & Luengo-Oroz (2022) studies at the ways that machine learning can be applied to forecast the flow of refugees. Prediction using machine learning models in smart migration and digital technology can provide insightful information.

5.5 The interaction between migrants and smartphones using UML

Unified Modelling Language (UML) has become the standard modeling technique and a necessary tool for any IS professional's toolkit (Shen et al., 2018). The use case and class diagrams are still the most popular among academics and practitioners, even though the fact that the most recent version of UML takes into account other distinct diagrams that show the many abstractions of a system (Alturas, 2023).

One kind of behavioral diagram that shows how a system and its actors interact is a use case diagram. As the study by Alturas (2023) shows, every use case depicts how users interact with the system. Included in the use case diagram contains actors, use cases, and relationships between them. An actor presents a character that is human (or non-human) in relation to a framework. Actors implement use cases (Richard Vidgen, 2023).

The figure illustration is important to easily understand the general idea that is analyzed in the analysis section and discussed in the discussion section. The use case diagram helps to easily depict the interaction between Actors (migrants and other stakeholders) and smartphones.

Table 6 shows the lists of actors that are involved in the migration process and smartphone usage and their descriptions.

Table 6 - Lists of actors

Actors	Description
Migrants	Internally displaced persons who have migrated due to the war in northern Ethiopia
Banks	Financial institution that offers a mobile banking service.
Service providers	A telecommunication company that provides both Internet and traditional telecommunication services (voice calls, SMS).

The following table shows the use cases and the description of them.

Table 7 - Lists of use cases

Use cases	Description
Internet Access	Using the internet to access social media platforms, bank applications
Navigations	Determining a route and guiding the movement
For work	Using a smartphone for work purposes
Entertainment	Using a smartphone for leisure activities such as listening to music, taking pictures, recording videos, and playing games.

Figure 69 illustrates the relationship or interaction between migrants and other stakeholders (Banks and service providers) using a use case diagram.

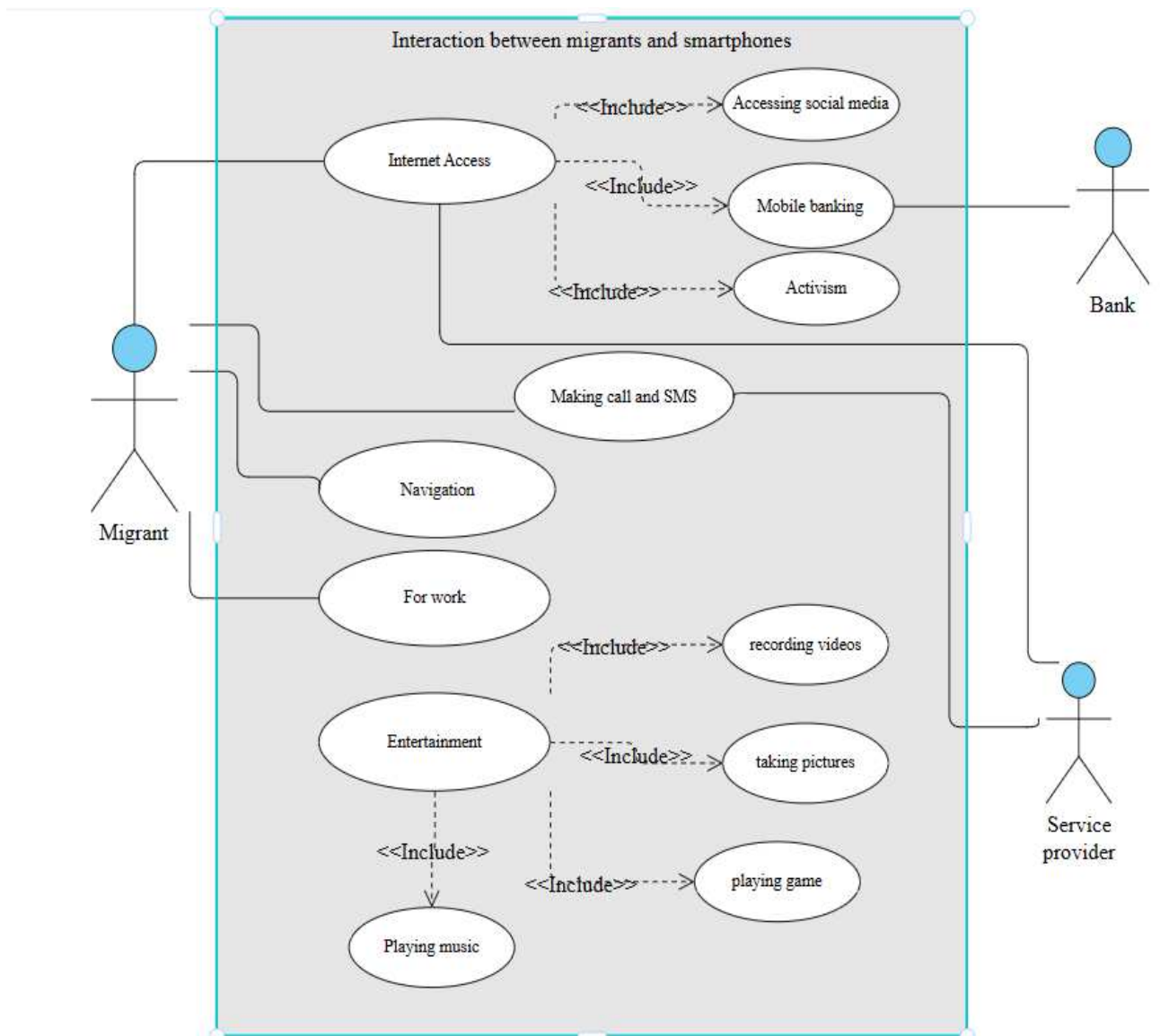


Figure 69 - interaction between migrants, smartphones, and other stakeholders

6 Conclusion and Recommendation

In conclusion, the study sheds light on the main reasons for using mobile phones, the most commonly used mobile applications, and their patterns among internally displaced persons (IDPs) affected by the northern Ethiopian war. By recognizing the diverse platform preferences and demographic variances influencing social media use, the research highlights the need for tailored communication strategies that cater to the unique needs and preferences of IDPs across age, gender, and duration of displacement.

Respondents of this study mostly use mobile phones for different purposes like making calls and SMS, internet access, taking pictures, and enjoying entertainment such as music. These results show how important mobile phones are for communication, getting information, and having fun for people in this area.

The most common applications being used by IDPs are Facebook, Telegram, mobile banking applications, and WhatsApp. 216 (72%) participants have social media accounts which underscores the significance of digital channels in enabling information exchange, social interaction, and support networks within the displaced community. They use social media platforms primarily for connecting with friends and family (208), obtaining information (204), exploring new places (110), entertainment (107), and engaging in activism purposes (36).

By recognizing the diverse platform preferences and demographic variances influencing social media use, the research highlights the need for tailored communication strategies that cater to the unique needs and preferences of IDPs across age, gender, and duration of displacement. A significant number of IDPs have increased their use of mobile since they migrated from their current location.

Even though most migrants don't have jobs in their current location, they still receive family support 113 (56.78%) and have smartphones. About 128 (42.7%) migrants spend 0.16 up to 0.33 EUR (10 - 20 ETB) per week on mobile use. The analysis results also show that all age groups of respondents have diverse preferences across platforms. However, their first two preferred social media platforms are Facebook and Telegram. Almost an equal number of females and males use Facebook.

Concerning smartphone usage increase, the data reveals that individuals staying for more than one year are less likely to experience an increase in smartphone usage compared to those staying for less than five months. Additionally, individuals with a stable income source are approximately five times more likely to observe an increase in smartphone usage compared to those without a reliable income source. These findings underscore the significant impact of both duration of stay and financial stability on smartphone usage patterns among the surveyed population.

Moreover, the result shows the importance of addressing barriers to access and leveraging digital technologies as tools for enhancing communication and social engagement in displacement

settings. Factors such as income source and length of stay emerge as key determinants shaping social media engagement among IDPs, emphasizing the necessity of promoting digital literacy and inclusivity to ensure equitable access to the benefits of digital connectivity.

From the implemented machine learning models, to predict *reasons for smartphone usage*, both models performed the same. For *most used applications*, Decision tree performance is the highest.

This diploma thesis will serve as a stepping-stone for other researchers to conduct further studies in this area with similar topics. Additionally, to establish evidence-based policy and programming interventions, it is essential to understand the potential challenges and opportunities of digital technology and how these digital technologies can potentially be utilized to address the situation of the people who have been displaced in northern Ethiopia during times of conflict.

1. It is recommended to conduct detailed research on a larger group of respondents among conflict-affected migrants in northern Ethiopia.

It is advised that future studies collect more data with more respondents and implement different algorithms and accuracy score improvement techniques to increase the machine learning predictive model's accuracy. A bigger data set would offer more diversity, lessen the effect of outliers, and maybe improve the model's generalizability in a variety of contexts.

A larger dataset would probably enable the algorithm to identify patterns and trends in the use of smartphones by migrants. By better understanding the needs and behaviors of migratory groups, decision-makers may be able to take more practical action based on increased accuracy. Therefore, collecting deeper data and improving the predictive model could greatly increase the effectiveness of future studies in this field.

This research presents a view of the frequency of smart device usage among them, preferred applications, behaviors, and challenges related to smart device use. However, it is important to include a larger number of study participants who are interviewed in-depth and to carry out comprehensive interviews with larger groups of migrants to get an overall picture.

To gain insight into the experiences, perspectives, and motivations of the participants, it is critical to conduct in-depth interviews with a greater number of participants. Interviews yield qualitative data that can bring out nuanced aspects of migrants' characteristics, behaviors, and experiences. This level of understanding is necessary for complete knowledge of the factors that influence migration trends and outcomes.

Moreover, further studies should be done to investigate how digital technology impacts the lives of internally displaced people (IDPs) in conflict-affected areas and discover new ways through which mobile applications can be used to promote their well-being and resilience. We can create more inclusive and supportive environments for those affected by conflict if we know what problems and possibilities come with online connection in times of displacement.

2. Expanding the research on the entire Ethiopia

There are many different ethnic groups in Ethiopia, and they all have unique linguistic traditions, cultural customs, and migration histories. Doing research throughout the country facilitates the capture of the subtleties of migration experiences among various communities, encompassing both urban and rural migrants, internal migrants, and international refugees.

Even though migration from northern Ethiopia may have occurred as a result of particular conflicts, it is important to note that different places have different abilities to get onto social media and use smartphones. To have even more detailed knowledge of how much people know about and use technology requires doing research all over the country.

Studying different parts of the country can give important insights into how we can develop flexible, scalable interventions to address common problems, taking into account specific regional features. By conducting studies in several areas, researchers can explore the impact of social networks, beliefs, and cultural norms on the digital behaviors of mobile populations. This kind of understanding should be used to develop interventions that are culturally sensitive and deliver positive outcomes when working with migrant groups.

3. Creating a digital technology policy is essential for addressing several issues and getting chances to help displaced people.

In conclusion, based on this research, it's clear that technology usage among migrants is influenced by different challenges like internet problems, unavailability of electricity, and expensive phone prices. Despite all the mentioned challenges, migrants communicate on social platforms Facebook, Telegram, and WhatsApp for communication, entertainment, and for sharing informative materials as well as entertainment and political activism activities.

The above-mentioned sites are essential as they help the immigrants feel close to their relatives and buddies, additionally, they are a source of information and a guide to the new environment. Policymakers and organizations need to consider the unique needs and preferences of migrant populations when designing technology-based interventions to support their integration and well-being.

According to this study's results, aid groups and lawmakers should start digital literacy programs and efforts meant to close the technology gap among people who have been forced from their homes in north Ethiopia. They would have to center these around getting cheap smartphones into as many hands as possible, providing internet access, and teaching how to use apps correctly when communicating, sharing information, or being sociable online.

References

- Aaron Matteo Terrazas. (2007). Article: Beyond Regional Circularity: The Emergenc.. | migrationpolicy.org. <https://www.migrationpolicy.org/article/beyond-regional-circularity-emergence-ethiopian-diaspora>
- Abdul-Jabbar, S. S., & K. Farhan, A. (2022). Data Analytics and Techniques. ARO-THE SCIENTIFIC JOURNAL OF KOYA UNIVERSITY, 10(2), 45–55. <https://doi.org/10.14500/aro.10975>
- Adugna Girmachew. (2021). Once Primarily an Origin for Refugees, Ethiopia Experiences Evolving Migration Patterns - Ethiopia | ReliefWeb. <https://www.migrationpolicy.org/article/ethiopia-origin-refugees-evolving-migration>
- Alencar, A. (2017). Refugee integration and social media: a local and experiential perspective. <https://doi.org/10.1080/1369118X.2017.1340500>
- Alencar, A. (2020). Mobile communication and refugees: An analytical review of academic literature. *Sociology Compass*, 14(8). <https://doi.org/10.1111/soc4.12802>
- Alencar A. (2023). Technology Can Be Transformative for Refugees, but It Can Also Hold Them Back. <https://www.migrationpolicy.org/article/digital-technology-refugees>
- Alencar, A., Kondova, K., & Ribbens, W. (2019). The smartphone as a lifeline: an exploration of refugees' use of mobile communication technologies during their flight. *Media, Culture and Society*, 41(6), 828–844. <https://doi.org/10.1177/0163443718813486>
- Alpes, M. J. (2017). Why aspiring migrants trust migration brokers: The moral economy of departure in Anglophone Cameroon. *Africa*, 87(2), 304–321. <https://doi.org/10.1017/S0001972016000978>
- Alturas, B. (2023). Connection between UML use case diagrams and UML class diagrams: a matrix proposal. *International Journal of Computer Applications in Technology*, 72(3), 161–168. <https://doi.org/10.1504/IJCAT.2023.133294>
- Alula Pankhurst, & François Piguet. (2004). People, space, and the state : migration, resettlement, and displacement in Ethiopia : proceedings of the workshop held by the Ethiopian Society of Sociologists, Social Workers, and Anthropologists and the United Nations Emergencies Unit for Ethiopia, 28-30 January 2003 : including reviews of the 2003 resettlement experience /. Ethiopian Society of Sociologists, Social Workers and Anthropologists,.
- Andone, I., Blaszkiewicz, K., Eibes, M., Trendafilov, B., Markowetz, A., & Montag, C. (2016). How age and gender affect smartphone usage. *UbiComp 2016 Adjunct - Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, 9–12. <https://doi.org/10.1145/2968219.2971451>

- Awad, I., & Tossell, J. (2021). Is the smartphone always a smart choice? Against the utilitarian view of the 'connected migrant'. *Information Communication and Society*, 24(4), 611–626. <https://doi.org/10.1080/1369118X.2019.1668456>
- Azhar, F., Vannini, S., Newell, B. C., & Gomez, R. (n.d.). Navigating borders/navigating networks: migration, technology and social capital. 92–106. <https://doi.org/10.4337/9781839100611.00016>
- Baldé, Y. (2009). Migrants' Remittances and Economic Growth in Sub-Saharan Africa.
- Barrett, P. E., Shopbell, P. L., Britton, M. C., Ebert, R., Barrett, P., Hunter, J., Miller, J. T., Hsu, J.-C., & Greenfield, P. (2005). *matplotlib-A Portable Python Plotting Package* (Vol. 347). <https://www.researchgate.net/publication/234238535>
- Bernal, V. (2020). African digital diasporas: Technologies, tactics, and trends. *African Diaspora*, 12(1–2), 1–10. <https://doi.org/10.1163/18725465-bja10007>
- Blackwell, B., Bernheim, D., Bloom, D. E., Galor, O., Katz, E., Lauby, J., Levhari, D., B Lucas, R. E., Rosenzweig, M. R., Shukla, V., Edward Taylor, J., & Yitzhaki, S. (1991). *The Migration of Labor Oded Stark*.
- Borgatti, S. P., & Halgin, D. S. (2011). On network theory. *Organization Science*, 22(5), 1168–1181. <https://doi.org/10.1287/orsc.1100.0641>
- Breiman, L. (2001). *Random Forests* (Vol. 45).
- Bundervoet, T. (2018). *Internal Migration in Ethiopia Evidence from a Quantitative and Qualitative Research Study*. www.worldbank.org
- Burrell, J., & Anderson, K. (2008). 'I have great desires to look beyond my world': Trajectories of information and communication technology use among Ghanaians living abroad. *New Media and Society*, 10(2), 203–224. <https://doi.org/10.1177/1461444807086472>
- Camana Acosta, M. R., Ahmed, S., Garcia, C. E., & Koo, I. (2020). Extremely randomized trees-based scheme for stealthy cyber-attack detection in smart grid networks. *IEEE Access*, 8, 19921–19933. <https://doi.org/10.1109/ACCESS.2020.2968934>
- Carter, B., Rohwerder, B., & Kuschminder, K. (2016). Rapid fragility and migration assessment for Ethiopia. http://www.gsdr.org/gsdrc_pub_type/literature-reviews/
- Carvalho, T. P., Soares, F. A. A. M. N., Vita, R., Francisco, R. da P., Basto, J. P., & Alcalá, S. G. S. (2019). A systematic literature review of machine learning methods applied to predictive maintenance. *Computers and Industrial Engineering*, 137. <https://doi.org/10.1016/j.cie.2019.106024>

- Cernea, M. M. (2009). Introduction: Resettlement An Enduring Issue in Development. *Asia Pacific Journal of Anthropology*, 10(4), 263–265. <https://doi.org/10.1080/14442210903079756>
- Daniele Tricarico. (2023). GSMA | Mobile Money in Ethiopia: What we learnt from our expert roundtable | Mobile for Development. <https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-for-development/blog/mobile-money-in-ethiopia-what-we-learnt-from-our-expert-roundtable/>
- Dekker, R., & Engbersen, G. (2014). How social media transform migrant networks and facilitate migration. *Global Networks*, 14(4), 401–418. <https://doi.org/10.1111/GLOB.12040>
- Dekker, R., Engbersen, G., Klaver, J., & Vonk, H. (2018). Smart Refugees: How Syrian Asylum Migrants Use Social Media Information in Migration Decision-Making. *Social Media and Society*, 4(1). <https://doi.org/10.1177/2056305118764439>
- Dekker, R., & Engbersen Themis, G. (2012). How social media transform migrant networks and facilitate migration Theorizing the Evolution of European Migration Systems The IMI Working Papers Series. www.imi.ox.ac.uk
- Dessalegn, M., Debevec, L., Nicol, A., & Ludi, E. (2023). A Critical Examination of Rural Out-Migration Studies in Ethiopia: Considering Impacts on Agriculture in the Sending Communities. In *Land* (Vol. 12, Issue 1). MDPI. <https://doi.org/10.3390/land12010176>
- Diminescu, D. (2008). The connected migrant: An epistemological manifesto. *Social Science Information*, 47(4), 565–579. <https://doi.org/10.1177/0539018408096447>
- Elger, T. (2015). Dual Labor Market Theories. In *The Wiley Blackwell Encyclopedia of Race, Ethnicity, and Nationalism* (pp. 1–3). Wiley. <https://doi.org/10.1002/9781118663202.wberen097>
- EMM. (2024). Types of movements | EMM2. <https://emm.iom.int/handbooks/global-context-international-migration/types-movements-0>
- Ennaji, M. (2021). Mobile phones: An indispensable tool for migrants. <https://en.unesco.org/courier/subscribe>
- European Commission. (2017). How West African migrants engage with migration information en-route to Europe. <https://doi.org/10.2837/97937>
- Fan, C., Chen, M., Wang, X., Wang, J., & Huang, B. (2021). A Review on Data Preprocessing Techniques Toward Efficient and Reliable Knowledge Discovery From Building Operational Data. In *Frontiers in Energy Research* (Vol. 9). Frontiers Media S.A. <https://doi.org/10.3389/fenrg.2021.652801>

- Fedorova, I., & Potemkin, V. (2020). Impact of digital technologies on the integration of labor migrants. *E3S Web of Conferences*, 217. <https://doi.org/10.1051/e3sconf/202021706001>
- Fernandes Malaquias, R., & Frances Silva, A. (2020). Understanding the use of mobile banking in rural areas of Brazil. <https://doi.org/10.1016/j.techsoc.2020.101260>
- Fokkema, T., & de Haas, H. (2015). Pre- and Post-Migration Determinants of Socio-Cultural Integration of African Immigrants in Italy and Spain. *International Migration*, 53(6), 3–26. <https://doi.org/10.1111/J.1468-2435.2011.00687.X>
- Fred Harter. (2023). The New Humanitarian | Unresolved status of western Tigray threatens Ethiopia's peace deal. <https://www.thenewhumanitarian.org/news-feature/2023/09/26/unresolved-status-western-tigray-ethiopia-peace-deal>
- Frouws, B., Phillips, M., Hassan, A., & Twigt, M. (2016). Getting to Europe the 'WhatsApp' way The use of ICT in contemporary mixed migration flows to Europe. <https://ssrn.com/abstract=2862592>
- Geda, A. (2011). Remittance and Remittance Service Providers in Ethiopia. <https://www.researchgate.net/publication/264004495>
- Gillespie, M., Ampofo, L., & Blanquerna, F. (2016). Mapping Refugee Media Journeys: Smartphones and Social Media Networks. <https://doi.org/10.13140/RG.2.2.15633.22888>
- Gillespie, M., Osseiran, S., & Cheesman, M. (2018). Syrian Refugees and the Digital Passage to Europe: Smartphone Infrastructures and Affordances. *Social Media and Society*, 4(1). <https://doi.org/10.1177/2056305118764440>
- Glückler, J., & Panitz, R. (2024). Knowledge and Digital Technology. In Klaus Tschira Symposia Knowledge and Space (Vol. 19).
- GSMA. (2017). The Importance of Mobile for Refugees: A Landscape of New Services and Approaches. www.gsma.com/disasterresponse
- Hagen-Zanker, J. (2017). Migration and the 2030 Agenda for Sustainable Development. <https://www.researchgate.net/publication/321268472>
- Hambali, I., Bilgiler Eğitimi Araştırmaları Dergisi, S., Hidayah, N., Hambali, I., & Ligya Radjah, C. (2019). Human Agency as a Self-Cognition of Human Autonomous Learning: A Synthesized Practical of Agentic Approach *Journal of Social Studies Education Research* Human Agency as a Self-Cognition of Human Autonomous Learning: A Synthesized Practical of Agentic Approach. In Article in *Journal of Social Studies Education Research* (Issue 10). www.jsser.org

- Haythornthwaite, C. (2016). Social networks and Internet connectivity effects. <https://doi.org/10.1080/13691180500146185>
- Hiller, H. H., & Franz, T. M. (2004). New ties, old ties and lost ties: The use of the internet in diaspora. In *New Media and Society* (Vol. 6, Issue 6, pp. 731–752). <https://doi.org/10.1177/146144804044327>
- Human Rights Watch. (2024). Tigray Conflict | Human Rights Watch. <https://www.hrw.org/tag/tigray-conflict>
- ICRC. (2010). Refugees and displaced persons protected under international humanitarian law | International Committee of the Red Cross. <https://www.icrc.org/en/document/protected-persons/refugees-displaced-persons>
- IDMC. (2022a). GRID 2022 Children and youth in internal displacement.
- IDMC. (2022b). Internal displacement | IDMC - Internal Displacement Monitoring Centre. Internal Displacement | IDMC - Internal Displacement Monitoring Centre. <https://www.internal-displacement.org/internal-displacement/>
- IDMC. (2023). Global Report on Internal Displacement 2023 IDMC Global Report on Internal Displacement 2023 IDMC II III.
- ILO. (2018). Public Employment Services Provision and Labour Market Information Collection and Utilization ETHIOPIA.
- ILO. (2020). Addressing the Root Causes of Migration in Ethiopia.
- IOM. (2021). Over 1.7 Million People Displaced Due to Conflict Need Urgent Assistance in Northern Ethiopia: IOM | International Organization for Migration. <https://www.iom.int/news/over-17-million-people-displaced-due-conflict-need-urgent-assistance-northern-ethiopia-iom>
- Iqbal, K., Hassan, S. T., Peng, H., & Khurshaid. (2019). Analyzing the role of information and telecommunication technology in human development: panel data analysis. *Environmental Science and Pollution Research*, 26(15), 15153–15161. <https://doi.org/10.1007/S11356-019-04918-4>
- Islam, A., Belhaouari, S. B., Rehman, A. U., & Bensmail, H. (2022). KNNOR: An oversampling technique for imbalanced datasets[Formula presented]. *Applied Soft Computing*, 115. <https://doi.org/10.1016/j.asoc.2021.108288>
- Italian Agency for Development Cooperation. (2017). ABOUT MIGRATION 7 Researches of 5 Ethiopian Universities on the roots causes Emergency Initiative in support of vulnerable people, refugees, IDPs and migrants addressing the root causes of irregular migration.

- Jauhiainen, J. S., Özçürümez, S., & Tursun, Ö. (2022). Internet and social media uses, digital divides, and digitally mediated transnationalism in forced migration: Syrians in Turkey. *Global Networks*, 22(2), 197–210. <https://doi.org/10.1111/glob.12339>
- Jauhiainen, J. S., & Tedeschi, M. (2021). IMISCOE Research Series Undocumented Migrants and their Everyday Lives. <http://www.springer.com/series/13502>
- JEFF SALTZ. (2024). What is a Machine Learning Life Cycle? - Data Science Process Alliance. <https://www.datascience-pm.com/machine-learning-life-cycle/>
- Jima, A. O. (2023). Ethiopian political crisis after reform: Causes of Tigray conflict. *Cogent Social Sciences*, 9(1). <https://doi.org/10.1080/23311886.2023.2209991>
- Joshi, A., Kale, S., Chandel, S., & Pal, D. (2015). Likert Scale: Explored and Explained. *British Journal of Applied Science & Technology*, 7(4), 396–403. <https://doi.org/10.9734/bjast/2015/14975>
- Kassem Chebib, N., & Minatullah Sohail, R. (2011). The Reasons Social Media Contributed To The 2011 Egyptian Revolution. *Rabia Minatullah Sohail International Journal of Business Research and Management (IJBRM)*, 2, 139.
- ken Stewart. (2023). Chi-squared test | Definition, Formula, Types, & Facts | Britannica. <https://www.britannica.com/topic/chi-squared-test>
- Klimczuk, A., & Klimczuk-Kochańska, M. (2016). Dual Labor Market. In *The Wiley Blackwell Encyclopedia of Gender and Sexuality Studies* (pp. 1–3). Wiley. <https://doi.org/10.1002/9781118663219.wbegss529>
- Lara, J. (2015). International migration and human capital in Mexico: Networks or parental absence? <https://doi.org/10.1016/j.ijedudev.2015.02.006>
- Lischer, S. K. (2007). Causes and Consequences of Conflict-Induced Displacement. *Civil Wars*, 9(2), 142–155. <https://doi.org/10.1080/13698240701207302>
- Madakam, S., Uchiya, T., Mark, S., & Lurie, Y. (2022). Artificial Intelligence, Machine Learning and Deep Learning (Literature: Review and Metrics). *Asia-Pacific Journal of Management Research and Innovation*, 18(1–2), 7–23. <https://doi.org/10.1177/2319510x221136682>
- Mahadevan Malamahadevan. (2024). Step-by-Step Exploratory Data Analysis (EDA) using Python -. <https://www.analyticsvidhya.com/blog/2022/07/step-by-step-exploratory-data-analysis-eda-using-python/>
- Manne, R. (2021). Machine Learning Techniques in Drug Discovery and Development. *International Journal of Applied Research*, 7(4), 21–28. <https://doi.org/10.22271/allresearch.2021.v7.i4a.8455>

- Maria Gerth-Niculescu. (2021). The New Humanitarian | Anger, fear run deep after months of ethnic violence in western Ethiopia. <https://www.thenewhumanitarian.org/news-feature/2021/2/23/fear-runs-deep-after-ethnic-violence-in-western-Ethiopia>
- Massey, D. S., Arango, J., Hugo, G., Kouaouci, A., Pellegrino, A., & Taylor, J. E. (1993). Theories of International Migration: A Review and Appraisal (Vol. 19, Issue 3). <https://about.jstor.org/terms>
- Mcgregor, E., & Siegel, M. (2013). #2013-068 Social Media and Migration Research UNU-MERIT Working Paper Series. <http://mgsog.merit.unu.edu>
- McKinney, W. (2013). Python for Data Analysis. www.it-ebooks.info
- Mehari, A. T. (2019). Determinants of Mobile Banking Usage or Transaction Frequency: Evidence from Selected Branches of Commercial Bank of Ethiopia (CBE) in Addis Ababa, Ethiopia. *Modern Management Forum*, 3(1). <https://doi.org/10.18686/mmfv3i1.1084>
- Merisalo, M., & Jauhiainen, J. S. (2020). Digital Divides among Asylum-Related Migrants: Comparing Internet Use and Smartphone Ownership. *Tijdschrift Voor Economische En Sociale Geografie*, 111(5), 689–704. <https://doi.org/10.1111/TESG.12397>
- Mohammed, R., Rawashdeh, J., & Abdullah, M. (2020). Machine Learning with Oversampling and Undersampling Techniques: Overview Study and Experimental Results. 2020 11th International Conference on Information and Communication Systems, ICICS 2020, 243–248. <https://doi.org/10.1109/ICICS49469.2020.239556>
- Molnar, P. (2019). New technologies in migration: human rights impacts. www.fmreview.org/ethics
- Nalbandian, L., & Dreher, N. (2022). Advanced Digital Technologies in Migration Management: A Review of Emerging Literature The Working Papers Series is produced jointly by the Toronto Metropolitan Centre for Immigration and Settlement (TMCIS) and the CERC in Migration and Integration www.torontomu.ca/centre-for-immigration-and-settlement www.torontomu.ca/cerc-migration
- Norwegian Center for Humanitarian Studies. (2023). New report shows conflict-related deaths have more than doubled - NCHS. <https://www.humanitarianstudies.no/news/new-report-finds-conflict-related-deaths-have-more-than-doubled-in-the-last-year/>
- Obi, C., Bartolini, F., & D’Haese, M. (2020). Digitalization and migration: the role of social media and migrant networks in migration decisions. An exploratory study in Nigeria. *Digital Policy, Regulation and Governance*, 23(1), 5–20. <https://doi.org/10.1108/DPRG-08-2020-0101>

- Opalo, K. O., & Smith, L. (2021). Ideology and succession politics in Ethiopia: autocratic leadership turnover and political instability. *Democratization*, 28(8), 1463–1482. <https://doi.org/10.1080/13510347.2021.1929178>
- Pham, K. H., & Luengo-Oroz, M. (2022). Predictive modeling of movements of refugees and internally displaced people: Towards a computational framework. <https://centre.humdata.org/catalogue-for-predictive-models-in-the-humanitarian-sector/>.
- Philippe Pellet. (2021). KÜLÜGYI ÉS KÜLGAZDASÁGI INTÉZET KKI Policy Brief | Understanding the 2020-2021 Tigray Conflict in Ethiopia - Background, Root Causes , and Consequences. <https://doi.org/10.47683/KKIElemzesek.KE-2021.39>
- Pries, L. (2015). Transnationalism. <https://www.researchgate.net/publication/280555539>
- Richard Vidgen. (2023). BEE UML PROVIDES A POWERFUL FRAMEWORK AND NOTATION FOR MODELLING BUSINESS PROCESSES.
- Rokach, L., & Maimon, O. (2006). Decision Trees. In *Data Mining and Knowledge Discovery Handbook* (pp. 165–192). Springer-Verlag. https://doi.org/10.1007/0-387-25465-x_9
- Sabir Rosidin, Muljono, Guruh Fajar Shidik, Ahmad Zainul Fanani, Farrikh Al Zam, & Purwanto. (2021). Improvement with Chi Square Selection Feature using Supervised Machine Learning Approach on Covid-19 Data.
- Sanchez, G., Hoxhaj, R., Nardin, S., Geddes, A., Achilli, L., & Sona Kalantaryan, R. (2018). A study of the communication channels used by migrants and asylum seekers in Italy, with a particular focus on online and social media.
- Shahhosseini, M., & Hu, G. (2020). Improved Weighted Random Forest for Classification Problems.
- Shen, Z., Tan, S., & Siau, K. (2018). Challenges in learning unified modeling language: From the perspective of diagrammatic representation and reasoning. *Communications of the Association for Information Systems*, 43(1), 545–565. <https://doi.org/10.17705/1CAIS.04330>
- Simon, J. L. (1990). The economic consequences of immigration.
- Spyratos, S., Vespe, M., Natale, F., Iacus, S. M., & Santamaria, C. (2020). Explaining the travelling behaviour of migrants using facebook audience estimates. *PLoS ONE*, 15(9 September). <https://doi.org/10.1371/journal.pone.0238947>
- Steel, J. L., Dunlavy, A. C., Collette, •, Harding, E., & Theorell, T. (2017). The Psychological Consequences of Pre-Emigration Trauma and Post-Migration Stress in Refugees and Immigrants from Africa. <https://doi.org/10.1007/s10903-016-0478-z>

- Stremlau, N., & Tsalapatanis, A. (2022). Social Media, Mobile Phones and Migration in Africa: A Review of the Evidence. *Progress in Development Studies*, 22, 56–71. <https://doi.org/10.1177/14649934211043615>
- Tabit, S., & Moussir, C.-E. (2017). IMPACT OF MIGRANTS' REMITTANCES ON ECONOMIC GROWTH: CASE OF MOROCCO. *International Journal of Innovation and Applied Studies*, 20(1), 226–233. <http://www.ijias.issr-journals.org/>
- Tataru, G. F. (2020). Migration – an Overview on Terminology, Causes and Effects. *Logos Universality Mentality Education Novelty: Law*, 7(2), 10–29. <https://doi.org/10.18662/lumenlaw/24>
- Telsaç, C. (2022). CAUSES OF MIGRATION AND ITS EFFECTS. <https://orcid.org/0000-0002-1063-8919>
- Tristan Claridge. (2018). Introduction to Social Capital Theory SOCAP101 Tristan Claridge. www.socialcapitalresearch.com
- Tsegay, B. (2023). Internal Displacement in Ethiopia: Towards a New Policy and Legal Framework for Durable Solutions. <https://doi.org/10.13140/RG.2.2.19420.16003>
- Tsegaye Bereket, & Gezahegne Kiya. (2023). Internal Displacement in Ethiopia: Towards a New Policy and Legal Framework for Durable Solutions.
- Turner, H., & Gülerce, H. (2021). MIGRANTS AND COMMUNICATION TECHNOLOGIES IN CHALLENGING TIMES; A DOUBLE-EDGED SWORD. <https://doi.org/10.26650/B/XX00.2021.006.11>
- UNHCR. (2021). UNHCR > ETHIOPIA SITUATION REGIONAL APPEAL.
- UNHCR. (2022). UNHCR ramps up assistance to refugees, displaced families in northern Ethiopia as peace returns | UNHCR. <https://www.unhcr.org/news/unhcr-ramps-assistance-refugees-displaced-families-northern-ethiopia-peace-returns>
- Unruh, J., Frank, E., & Pritchard, M. (2017). A digital advance for housing, land and property restitution in war-affected states: Leveraging smart migration. *Stability*, 6(1). <https://doi.org/10.5334/sta.562>
- Urbański, M. (2022). Comparing Push and Pull Factors Affecting Migration. *Economies*, 10(1). <https://doi.org/10.3390/economies10010021>
- USA for UNHCR. (2022). Ethiopia's Tigray Refugee Crisis Explained. <https://www.unrefugees.org/news/ethiopias-tigray-refugee-crisis-explained/>

- Vandana Kolisetty, V., & Singh Rajput, D. (2020). A REVIEW ON THE SIGNIFICANCE OF MACHINE LEARNING FOR DATA ANALYSIS IN BIG DATA. In *Jordanian Journal of Computers and Information Technology (JJCIT)* (Vol. 06, Issue 01).
- Vaughan, S., & Gebremichael, M. (2011). Rethinking business and politics in Ethiopia The role of EFFORT, the Endowment Fund for the Rehabilitation of Tigray. www.odi.org.uk
- Veronis, L., Tabler, Z., & Ahmed, R. (2018). Syrian Refugee Youth Use Social Media: Building Transcultural Spaces and Connections for Resettlement in Ottawa, Canada. *Canadian Ethnic Studies*, 50(2), 79–99. <https://doi.org/10.1353/ces.2018.0016>
- Vestal, T. M. (1985). Famine in Ethiopia: Crisis of Many Dimensions (Vol. 32, Issue 4). <https://www.jstor.org/stable/4186321>
- Walani, S. R. (2015). Global migration of internationally educated nurses: Experiences of employment discrimination. <https://doi.org/10.1016/j.ijans.2015.08.004>
- Waskom, M. (2021). seaborn: statistical data visualization. *Journal of Open Source Software*, 6(60), 3021. <https://doi.org/10.21105/joss.03021>
- World Economic Forum. (2024). The challenges facing Ethiopia’s digital transformation | World Economic Forum. <https://www.weforum.org/agenda/2024/03/bridging-the-digital-divide-challenges-and-opportunities-for-ethiopias-digital-transformation/>
- Xu, P., Ji, X., Li, M., & Lu, W. (2023). Small data machine learning in materials science. In *npj Computational Materials* (Vol. 9, Issue 1). Nature Research. <https://doi.org/10.1038/s41524-023-01000-z>
- Zijlstra, J., & Liempt, I. Van. (2017). Smart(phone) travelling: understanding the use and impact of mobile technology on irregular migration journeys. *International Journal of Migration and Border Studies*, 3(2/3), 174. <https://doi.org/10.1504/ijmbs.2017.083245>

List of Tables

Table 1 - The socio-demographic profile of IDPs	35
Table 2 - The Job and income-related characteristics of the IDPs	39
Table 3 - Smartphone technology usage by IDP	43
Table 4 - Social media usage among IDPs	51
Table 5 - Challenges of technology usage among migrants	52
Table 6 - Lists of actors	71
Table 7 - Lists of use cases	71

List of Figures

Figure 1 - Machine learning phases	9
Figure 2 - Machine learning approaches	9
Figure 3 - Conflict area	12
Figure 4 - Sub-Saharan countries with the highest IDPs.....	12
Figure 5 - Data analysis process.....	18
Figure 6 - Importing libraries for the data analysis	19
Figure 7 - Loading and reading the dataset	20
Figure 8 - Renaming column.....	21
Figure 9 - Correcting misspelling.....	21
Figure 10 - Splitting into separate columns.....	22
Figure 11 - Drop columns	22
Figure 12 - dropping duplicates	23
Figure 13 - Merging 2 column values	23
Figure 14 - Label encoding	23
Figure 15 - handling imbalance class	24
Figure 16 - Oversampling technique.....	24
Figure 17 - Decision tree random.....	25
Figure 18 - Random forest classifier	26
Figure 19 - Machine learning implementation procedure	26
Figure 20 - Interview language distribution in pie and percentage	28
Figure 21 - Generating Age distribution visualization using Goggle Collaboratory	29
Figure 22 - Age distribution in bar chart and percentage	29
Figure 23 - Gender distribution	30
Figure 24 - Migrants' location.	31
Figure 25 – Migrants’ current location.....	32
Figure 26 - The proportion of participants residing in the current location	32
Figure 27 - Marital Status.....	33
Figure 28 - Participants Parenthood	33
Figure 29 - Generating a table of IDPs' education level using Google Colaboratory	34
Figure 30 - Education level of respondents.....	34
Figure 31 - Generating a table of the participants’ previous jobs using Google Colaboratory	36
Figure 32 - Respondents' previous jobs.....	36
Figure 33 - Respondents' current source of income status	37
Figure 34 - Source of income.....	37
Figure 35 - Participants' bank system usage preferences	38
Figure 36 - How participants receive money from abroad.....	38
Figure 37 - Mobile brands used by participants	40
Figure 38 - Participants' phone language preference.....	40

Figure 39 - Amount of money spent on mobile per week	41
Figure 40 - Is a mobile phone helpful?.....	41
Figure 41 - Participant's reasons for using mobile phone	42
Figure 42 - Mobile usage trend of the participants	43
Figure 43 - Correlation between the trend of smartphone usage after migration and characteristics	45
Figure 44 - having social media	46
Figure 45 - Participants' social media platform preferences	46
Figure 46 - Correlation between social media platforms and participants' Age and Gender	47
Figure 47 - Facebook usage among participants	48
Figure 48 - Correlation between having social media and migrants' characteristics	49
Figure 49 - Reasons given by participants for not having social media accounts	50
Figure 50 - Reasons participants use social media.....	50
Figure 51 - Splitting the data set into X and Y.....	53
Figure 52 - Applying chi2 for feature selection	53
Figure 53 - Selected top 10 features using chi-square.....	54
Figure 54 - Initial imbalanced classes and classes after applying the oversampling technique	55
Figure 55 - Splitting into train and test sets.....	55
Figure 56 - Decision tree	56
Figure 57 - Hyperparameter tuning on decision tree.....	57
Figure 58 - Random forest prediction	57
Figure 59 - Hyperparameter tuning	58
Figure 60 - Comparison between the models.....	58
Figure 61 - Feature selection for mostly used application	59
Figure 62 - select the top 10 features	60
Figure 63 – Oversampling technique	60
Figure 64 - Random forest model	61
Figure 65 - Hyperparameter tuning on random forest model.....	61
Figure 66 - Decision tree prediction.....	62
Figure 67 - Hyperparameter tuning on decision tree.....	62
Figure 68 - Implemented models comparison (2)	63
Figure 69 - interaction between migrants, smartphones, and other stakeholders.....	72

Abbreviations

IDP:	Internally Displaced Person
UML:	Unified Modeling Language
ML:	Machine Learning
IOM:	International Organization for Migration
IDMC:	Internal Displacement Monitoring Centre
UNHCR:	United Nations High Commissioner for Refugees
ILO:	International Labor Organization
EPRDF:	Ethiopian People's Revolutionary Democratic Front
TPLF:	Tigray People Liberation Front
OLA:	Oromo Liberation Army
UAGs:	Unidentified Armed Groups
CSV:	Comma Separated Values
UN:	United Nations
EDA:	Exploratory Data Analysis

Appendix

Questionnaire

This part of Questionnaire is filled by the interviewer.

1.1 Date: ____/____/____

1.2 The interview is made in _____ language

1.3 Location of interview _____

SECTION 2: PERSONAL INFORMATION

In this part of the Questionnaire, You are asked to provide your Personal Information, Read the questions, and check the box ☒

2.1 Age (18+ at minimum): _____

2.2 Gender: _____

2.3 Education Level:

☐ Middle School/ High School

- What grade are you? _____

☐ University Student

- Level

☐ Bachelor

☐ Master's Program

☐ PhD Program

☐ Other, please Specify _____

- Mention the university name _____

- Study Program _____

- What Year are you? _____

☐ University Graduate

- Level
 - ☐ Bachelor
 - ☐ Master's Program
 - ☐ PhD Program
 - ☐ Other, please Specify _____
- Mention the university

- Study Program _____

-

☐ I don't study

☐ Other/please specify _____

2.4 What was your previous job in your home town? (More answers are possible)

☐ Private Business (Shop, café...)

☐ Teacher (KG, Middle school, high school or university)

☐ Government office worker

- Mention the Sector

☐ Farmer

☐ Just Student

☐ Craft person

☐ Other, please specify _____

2.5 Are you currently working?

☐ Yes

- Mention your Job

-
- Mention the Sector
-

☐ No, I'm not

2.6 Where do you live now? (Location)

☐ Addis Ababa

- Mention which sub city
-

☐ Outside of Addis Ababa

- Mention which Region and town
-

☐ Other/please specify

2.7 How long do you stay in your current location?

☐ < 1 Month

☐ 1 – 5 months

☐ 5 Months – 1 year

☐ > 1 year

2.8 Do you currently have a source of income?

☐ Yes, I have

- What is your source of economy? (More answers is possible)

☐ My Salary

☐ Cash Transfer from the government

☐ Self-employed

☐ Family support

☐ From aid organization

☐ Other/please specify _____

☐ No, I don't have

2.9 How do you mostly receive money from abroad?

☐ By draft money transfer

☐ Via black market

☐ The sender sends the money in cash to someone he/she knows

☐ Other, please specify _____

☐ I don't know

2.10 When you send money to someone, which way do you use the most?

☐ I use banks

☐ I go to the bank in person

☐ I use mobile banking

☐ I prefer to send cash to someone I know

☐ I use *tebibirr* application

☐ Other, please specify _____

☐ I don't know

2.11 Are you married?

☐ Yes

☐ No, I'm not

2.12 Do you have children?

☐ Yes, I do

- How many children do you have?

☐ 1 child ☐ 2 children ☐ 3 and more

☐ No, I don't have

2.13 Whom do you live with? (More answers are possible)

☐ With my family (wife/husband and/or children)

☐ With my parents

☐ With my friends

☐ I'm alone

☐ Other, please specify _____

SECTION 3: USAGE OF MOBILE DEVICES

In this part of the questionnaire, you are asked to provide information about your smart mobile device usage. Read the questions and check the box ☒

3.1 What brand of mobile device do you use?

3.2 For which Activity do you use your mobile the most? (More answers are possible)

☐ For Internet use

☐ For Gaming

☐ For playing Music

☐ Making Regular call and SMS

☐ Taking Picture

☐ Record Video

☐ Navigating places

☐ Other/ please specify _____

3.3 In which language you prefer to use your mobile device? (More answers are possible)

- ☐ Amharic
- ☐ Tigrinya Language
- ☐ English
- ☐ Other/please specify _____

3.4 Do you have a social media account?

- ☐ Yes, I have
 - Which social media do you use the most? (Check all if you use all)
 - ☐ Facebook
 - ☐ Instagram
 - ☐ TikTok
 - ☐ WhatsApp
 - ☐ Telegram
 - ☐ Other/please specify _____
- ☐ No, I don't have
 - If you don't have, which one would be the reason?
 - ☐ I can't afford the internet price
 - ☐ I don't think it is crucial for me
 - ☐ I don't know how to use it
 - ☐ Other/please specify _____

3.5 What is your main reason of using social media? (Check ALL that apply)

- ☐ To meet friends, family and relatives
- ☐ For Entertainment

- ☐ To Get Information
- ☐ Discover a better place
- ☐ Business Purpose
- ☐ Other/please specify: _____

3.6 Do you think your mobile usage has been increasing since you migrated from your region?

- ☐ Yes, it is increasing
- ☐ It is still the same
- ☐ No, it doesn't (decreasing)
- ☐ I don't know

3.7 How much money do you spend on internet service in a week?

- ☐ 5-10 ETB
- ☐ 10-20 ETB
- ☐ 20-50 ETB
- ☐ >50 ETB

3.8 Do you think using a mobile Device is helpful for your current situation?

- ☐ Yes, Fully helpful
 - How? (More answers are possible)
 - ☐ It helped me to pay bills (electric, water bill, internet bills...)
 - ☐ For the shopping payment
 - ☐ It helped me with online registration.
 - ☐ It helped me for the money transfer
 - ☐ It helped me to get online services from government offices.
 - ☐ Other/please specify: _____

☐ No, it doesn't

☐ I don't know

3.9 Do you think using a mobile device helped to move to your current place? (During your migration process?)

☐ Yes

- How? (More answers are possible)

☐ It helped me to navigate the location

☐ It helped me to discover my destination places

☐ It helped me to ask for information

☐ Other/please specify: _____

☐ No, it doesn't

☐ I don't know

3.10 How would you rate the following challenges for using smart mobile devices? put tick mark (✓)

No.	Questions	Rating scales				
		Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	The cost of the Device is not affordable					
2	Shortage of Electric power to charge the phone					
3	There is an Internet connection problem					



Usnesení Etické komise č. 36

Etická komise neshledává porušení etických standardů v rámci výzkumu Smart Migration - Digital Technologies and Migration in Northern Ethiopia jehož zpracovatelem bude Tinsae Simeon Wonjelo, z Provozně ekonomické fakulty Mendelovy univerzity v Brně.
Etická komise souhlasí s realizací výzkumu.

The Ethics Committee of Mendel University in Brno has reviewed the application to conduct the research project Smart Migration- Digital Technologies and Migration in Northern Ethiopia which will be conducted by Tinsae Simeon Wonjelo from Faculty of Business and Economics Mendel University in Brno. The Ethics Committee approves the research.

23. 6. 2023

Hana Kelblová, v. r.
Předsedkyně Etické komise

```

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder
import warnings
warnings.filterwarnings('ignore')
from google.colab import drive
drive.mount('/content/drive')
df = pd.read_csv('/content/drive/MyDrive/Smart Migration - Digital
Technologies and Migration in the Northern Ethiopia.csv')

```

Data cleaning

Renaming Columns

```

columns=['Date', 'Language', 'Location', 'Age group', 'Gender', 'Education
Level', 'Grade', 'Level', 'University', 'Program',
        'year', 'level', 'Mention University', 'ys program', 'previous
job', 'sector', 'currently working', 'current job',
        'job sector', 'current location', 'sub
city', 'region&town', 'howlongyou stay', 'income source', 'economy sr',
        'money from abroad', 'money transfer way', 'bank
use', 'married?', 'children', '# of children', 'live with',
        'phone brand', 'mobile usage reason', 'phone language', 'social
media?', 'sm most used', 'sm not usage reason',
        'main reason of social media', 'mobile usage increase?', 'money
spend for mobile', 'is MD helpful?', 'how?',
        'MD for moving', 'MD for moving How', 'Device
affordable?', 'hortage of Electric power', 'Internet connection problem'
]
df = df.set_axis(columns_new, axis=1)
df = df[columns_new]

```

Replace values

```

replace_amh=['Amharic`', 'Amharic ']
replace_tig=['Tigrigna`', 'Tigrigna ', 'Tigrigna']
df['Language'].replace(replace_amh, 'Amharic', inplace=True)
df['Language'].replace(replace_tig, 'Tigrigna', inplace=True)

```

Splitting values in column that are separated (;)

```
df['Social Media'] = df['sm most used'].astype(str).replace(['nan',
'NaN'], '')
df['Social Media List'] = df['Social Media'].apply(
    lambda x: x.split(';') if x.strip() != '' else []
)
unique_platforms = set(
    platform for sublist in df['Social Media List'] for platform in
    sublist
)
for platform in unique_platforms:
    df[platform] = df['Social Media List'].apply(
        lambda platforms: 1 if platform in platforms else 0
    )
```

Drop columns

```
df.drop(columns=['Social Media List', 'Social Media', 'sm most used'],
inplace=True)

print(df)
```

Label encoding

```
df[df.columns] = df[df.columns].apply(LabelEncoder().fit_transform)
```

Exploratory data analysis

Gender distribution

```
data = np.array([df['Gender'].value_counts().get('Male', 0),
df['Gender'].value_counts().get('Female', 0)])
myexplode = [0.2, 0]
colors = ['#519DE9', '#F4B678']
def autopct_format(values):
    total = sum(values)
    def my_autopct(pct):
        count = int(pct * total / 100.0)
        return '{:.1f}% ({} )'.format(pct, count)
    return my_autopct
plt.pie(data, labels=['Male', 'Female'], explode=myexplode,
shadow=True, autopct=autopct_format(data), colors=colors)

plt.title("Gender Distribution", pad=20)
plt.show()
```

Displaying reason for smartphone usage

```
column_totals = df[Col_reason_phone].sum()
plt.figure(figsize=(10, 6))
barplot = sns.barplot(x=column_totals.index, y=column_totals.values,
                      palette='pastel') # Create the bar chart
for p in barplot.patches:
    count = p.get_height()
    barplot.annotate(
        f'{count}',
        (p.get_x() + p.get_width() / 2, count),
        ha='center',
        va='bottom',
        fontsize=10
    )
plt.title("Bar Chart with Counts Only")
plt.xlabel("Social Media Platforms")
plt.ylabel("Total Count")
plt.xticks(rotation=45, ha='right')
plt.show()
```

Select mostly used applications

```
Col_socialmedia = ['WhatsApp', 'Youtube', 'Tiktok', 'Facebook',
                  'Instagram', 'Viber', 'Telegram', 'imo' ]
```

Display in barchart

```
column_totals = df[Col_socialmedia].sum()
plt.figure(figsize=(10, 6))
barplot = sns.barplot(x=column_totals.index, y=column_totals.values,
                      palette='pastel')
for p in barplot.patches:
    count = p.get_height()
    barplot.annotate(
        f'{count}',
        (p.get_x() + p.get_width() / 2, count),
        ha='center',
        va='bottom',
        fontsize=10
    )
plt.title("Bar Chart with Counts Only")
```

```
plt.xlabel("Social Media Platforms")
plt.ylabel("Total Count")
plt.xticks(rotation=45, ha='right')
plt.show()
```

Displaying correlations between variables

```
selected_columns5 = ['Education level', 'Where do you live now? (Location)', 'How long do you stay in your current location? ', 'Mobile Usage Trend Since Migration']

df_selected5 = df[selected_columns5]
df_encoded = pd.get_dummies(df_selected5)
print("One-Hot Encoded Data:")
print(df_encoded.head())
```

Correlation matrix

```
correlation_matrix = df_encoded.corr()
print("Correlation Matrix:")
print(correlation_matrix)
```

Displaying correlation heatmap

```
plt.figure(figsize=(10, 8)) # Adjust the size to fit all columns
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm',
fmt='.2f', linewidths=.5)
plt.title("Correlation Heatmap")
plt.show()
```

Building model for predicting “Reason for smartphone usage”

```
columns = list(df.columns)
columns.remove('mobile usage reason')
columns.append('mobile usage reason')
df = df[columns]
print(df.head())
```

Removing columns that has >40% null values

```
df = df.drop(df.loc[:, list((100 * (df.isnull().sum() / len(df.index)) > 40))].columns, axis=1)
```

Encoding labels

```
df[df.columns] = df[df.columns].apply(LabelEncoder().fit_transform)
```

Feature Selection

```
from sklearn.feature_selection import SelectKBest
from sklearn.feature_selection import chi2
```

```
X = df.loc[:, df.columns != 'mobile usage reason']
y = df.loc[:, df.columns == 'mobile usage reason']
```

```
chi_scores = chi2(X, y)
chi_values = pd.Series(chi_scores[0], index=X.columns)
chi_values.sort_values(ascending=False, inplace=True)
chi_values.plot.bar()
```

```
k = 10

chi2_selector = SelectKBest(chi2, k=k)
X_new = chi2_selector.fit_transform(X, y)

selected_features = df.columns[:-1][chi2_selector.get_support()]
print("\nSelected Features:")
print(selected_features)
```

Handling class imbalance

```
from imblearn.over_sampling import SMOTE
from imblearn.over_sampling import RandomOverSampler
```

```
oversampler = RandomOverSampler(sampling_strategy='auto',
                                random_state=42)
X_resampled, y_resampled = oversampler.fit_resample(X_new, y)
```

```
plt.figure(figsize=(10, 6))
plt.hist(y, bins=range(12), align='left', rwidth=0.8, color='blue',
         alpha=0.7)
plt.title("Histogram of Initial Classes")
plt.xlabel("Class")
plt.ylabel("Number of Instances")
plt.xticks(range(8), ['Class 0', 'Class 1', 'Class 2', 'Class 3', 'Class 4', 'Class 5', 'Class 6', 'Class 7'])
plt.show()
plt.figure(figsize=(10, 6))
```

```
plt.hist(y_resampled, bins=range(12), align='left', rwidth=0.8,
color='orange', alpha=0.7)
plt.title("Histogram of Resampled Classes (Undersampling)")
plt.xlabel("Class")
plt.ylabel("Number of Instances")
plt.xticks(range(8), ['Class 0', 'Class 1', 'Class 2', 'Class 3', 'Class
4', 'Class 5', 'Class 6', 'Class 7'])
plt.show()
```

Train Test Split

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X_resampled,
y_resampled, test_size=0.33, random_state=42)
```

Building Models

Random Forest

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
```

```
rf_classifier = RandomForestClassifier(max_depth=68, random_state=83)
rf_classifier.fit(X_train, y_train)
y_train_pred = rf_classifier.predict(X_train)
y_test_pred = rf_classifier.predict(X_test)
```

```
from sklearn.metrics import classification_report
rf_classifier = RandomForestClassifier(max_depth=68, random_state=83)
rf_classifier.fit(X_train, y_train)
y_train_pred = rf_classifier.predict(X_train)
y_test_pred = rf_classifier.predict(X_test)
train_accuracy = accuracy_score(y_train, y_train_pred)
test_accuracy_rand = accuracy_score(y_test, y_test_pred)
print("Accuracy on Train Set:", train_accuracy)
print("Accuracy on Test Set:", test_accuracy_rand)
print("\nClassification Report:")
print(classification_report(y_test, y_test_pred))
```

Hyperparameter tuning

```
from sklearn.model_selection import GridSearchCV
param_grid = {
```

```

    'n_estimators': [100, 200, 300],
    'max_depth': [10, 50, 100, None],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4],
    'max_features': ['auto', 'sqrt', 'log2']
}
rf_classifier = RandomForestClassifier(random_state=83)

grid_search = GridSearchCV(estimator=rf_classifier,
                           param_grid=param_grid,
                           cv=5,
                           scoring='accuracy',
                           n_jobs=-1,
                           verbose=2)

grid_search.fit(X_train, y_train)
best_params = grid_search.best_params_
print("Best parameters found:", best_params)
best_rf_classifier = grid_search.best_estimator_
y_test_pred = best_rf_classifier.predict(X_test)
test_accuracy1 = accuracy_score(y_test, y_test_pred)
print("Test accuracy with tuned Random Forest:", test_accuracy1)

```

Decision tree

```

from sklearn import metrics
from sklearn.metrics import accuracy_score
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report
clf = DecisionTreeClassifier()
clf = clf.fit(X_train, y_train)
# Evaluate the classifier on the test set
y_pred = clf.predict(X_test)
asc_ds = accuracy_score(y_test, y_pred)
report_ds = classification_report(y_test, y_pred)
print("Decision Tree")
print(report_ds)

```

Hyperparameter tuning

```

from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score
param_grid = {
    'clf__criterion': ['gini', 'entropy'],

```

```

        'clf__max_depth': [None, 10, 20],
        'clf__min_samples_split': [2, 5, 10],
        'clf__min_samples_leaf': [1, 2, 4],
        'clf__max_features': [None, 'sqrt', 'log2']
    }
    pipeline = Pipeline([
        ('scaler', StandardScaler()),
        ('clf', DecisionTreeClassifier(random_state=42))
    ])

    grid_search = GridSearchCV(pipeline, param_grid, cv=5,
                                scoring='accuracy', n_jobs=-1)
    grid_search.fit(X_train, y_train)
    best_params = grid_search.best_params_
    print("Best Hyperparameters:", best_params)
    best_clf = grid_search.best_estimator_
    y_pred = best_clf.predict(X_test)
    test_accuracy = accuracy_score(y_test, y_pred)
    print("Test Accuracy:", test_accuracy)

```

Comparison between models

```

model_names = ['Random Forest', 'Decision Tree']
accuracy_scores = [test_accuracy, asc_ds]

plt.figure(figsize=(7, 3))
bar_plot = sns.barplot(x=accuracy_scores, y=model_names, orient='h')

# Displaying accuracy scores on the bars
for index, value in enumerate(accuracy_scores):
    bar_plot.text(value, index, f'{value:.2f}', va='center')

plt.ylabel('Model')
plt.xlabel('Accuracy Score')
plt.title('Accuracy Comparison between Models')
plt.xlim(0, 1.0)
plt.show()

```